

SPCI110

POSTGRADUATE COURSE

M.Sc. - Cyber Forensics & Information Security

FIRST YEAR

SECOND SEMESTER

ELECTIVE PAPER - II

BFSI FRAUDS & COUNTERMEASURES



**INSTITUTE OF DISTANCE EDUCATION
UNIVERSITY OF MADRAS**

**M.Sc. Cyber Forensics & Information Security
FIRST YEAR - SECOND SEMESTER**

**ELECTIVE PAPER - II
BFSI FRAUDS
& COUNTERMEASURES**

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With best wishes from mind and heart,

DIRECTOR i/c

M.Sc. Cyber Forensics & Information Security
FIRST YEAR - SECOND SEMESTER

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& COUNTERMEASURES

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M.Sc. DEGREE COURSE
FIRST YEAR
SECOND SEMESTER
Elective Paper - II
BFSI FRAUDS & COUNTERMEASURES
SYLLABUS

Unit 1: Introduction to BFSI

Banking Concepts - Broad features of Deposit and Loan Products - Types of banks: Retail, Corporate, Investment, Development, Private, etc. - Ancillary services like Trade Finance, Remittances, etc. - Anti Money Laundering and KYC concepts.

Unit 2: Computerized operations of banks

Evolution of computerization in banks - Core Banking Solution - Infrastructure requirements - Broad software features - Various methods, options available for customizing like Setting up Chart of Accounts Parameterising Products, Interest Rates and Charges - User restrictions and transaction types - Delivery Channel Options for direct customer access to databases

Unit 3 : Basel II

Need for Basel Regulations - Three pillars - Types of risks - Operational Risk overview with focus on IT risk - Relation of Bank related cyber crimes to Operational risk

Unit 4: Vulnerable areas in CBS and their exploitation

Application related - Parameters and freedom available to users - Empowerment of users - Access to -organization -wide data - Direct access to database and records - Multiple interfaces with other applications ATM Network, Anti-Money Laundering Application

Unit 5

Money Laundering and Anti Money Laundering - Money laundering techniques and the vulnerabilities of specific financial services products - The process of money laundering - How is money laundered? - Limitations of the staged interpretation of money laundering - Vulnerabilities of specific services and products - The duties and responsibilities of the Money Laundering Reporting Officer (MLRO) - The role of the MLRO - Generating management information - Common MLRO problems - Recognition, handling and reporting transactions - The legal obligation to report - Designing an effective internal reporting system - The MLRO's evaluation process - Corruption in BFSI Sector – Types – Security Controls - Counter Measures.

M.Sc. DEGREE COURSE
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BFSI FRAUDS & COUNTERMEASURES
SCHEME OF LESSONS

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Lesson 1

Introduction to BFSI

Objectives

After reading this lesson you will be able to understand the following:

- Banking: Introduction & Concept
- Types of banks
 - o RBI, Commercial, Cooperative, Regional and Rural, Development banks, Small banks, Payment banks etc.
- Banking Regulations Act 1949 with specific reference to sec 5(c),
- Forms of banking business
- Ancillary services like Trade Finance, Remittances, etc.
- Types of deposits and accounts
 - o Demand Deposits and types
 - § Current account and Savings account
 - § Other savings account – salient features and benefits
 - § Kiddy bank, Daily Deposit Collection, Insurance linked Accounts, Jan Dhan account etc
 - o Term Deposits and types
 - § Fixed Deposits, Recurring Deposits, Cumulative Deposits & Flexi Deposits
- Loans and Advances
 - o Demand Loan
 - o Term Loan
 - o Over Drafts & cash Credits
 - o Trade Finance
 - o Bill purchase & Discount

- Core Banking Solution – Salient features, advantages and Risks
- E-Banking services like – NEFT, RTGS, IMPS, UPI, USSD, E wallets- Features and risks
- Introduction to ATM Networks and how ATM services are managed

Structure

- 1 Banking: Introduction and concept**
 - 1.1 Banking – Definition**
 - 1.2 Indian Banking System – broad Classification**
 - 1.3 Reserve Bank of India**
 - 1.4 Commercial Banks**
 - 1.5 Cooperative Banks**
 - 1.6 Regional Rural Banks**
 - 1.7 Development Banks**
 - 1.8 Investment Banks**
 - 1.9 Merchant Banks**
 - 1.10 Small Banks and payment Banks**
 - 1.11 Banking Regulations Act 1949**

1 Banking: Introduction & Concept

1.1 Banking – Definition

Banking is a system of trading money. It provides for a safe place to save the excess cash in the form of deposits. It loans out this money for the growth of businesses. It helps the consumers for the purchase of homes, vehicles and the like. Section 5(b) of the banking regulation act 1949 gives a comprehensive definition of banking as, “banking” as “accepting for the purpose of lending and investment of deposits of money from the public, repayable on demand or otherwise, and withdraw able by cheque, draft, order or otherwise”.

Banks deal in money and credit. It handles the money of the public for their advantage and to make its own profits. As such banking as a financial institution links the flow of funds from the persons who save to the prospective users. Thus it plays an important role in the economic growth of the country holding the savings of the public.

Thus a bank is a financial institution that accepts deposits from the public and creates credit. Lending activities can be performed either directly or indirectly through capital markets. Due to their importance in the financial stability of a country, banks are highly regulated in most countries. Most nations have institutionalized a system known as fractional reserve banking under which banks hold liquid assets equal to only a portion of their current liabilities. In addition to other regulations intended to ensure liquidity, banks are generally subject to minimum capital requirements based on an international set of capital standards, known as the Basel Accords.

<https://www.toppr.com/guides/general-awareness/banking/history-of-banking-in-india-and-world/>

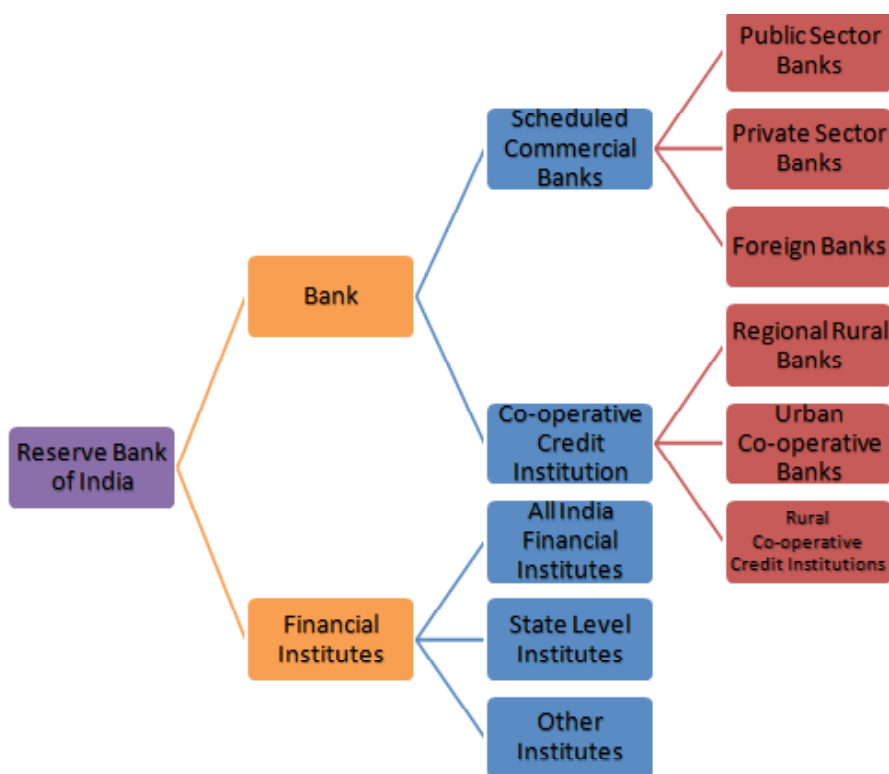


Figure 1.1 : Structure of Indian Banking System

1.2. Indian Banking system – broad classification

The structure of Banks in India is illustrated in Figure.

Broad classification of banks in India

- 1) RBI is the supreme monetary and banking authority in the country which has the responsibility to control the banking system in the country. Since it keeps the resource of all scheduled banks it is known as the reserve bank.
- 2) Public sector banks
 - a. State bank of India
 - b. Nationalized banks
 - c. Regional Rural Banks sponsored by public sector banks
- 3) Private sector banks
 - a. Old generation private banks
 - b. Foreign New generation banks
 - c. Banks in India
- 4) Co-operative Sector Banks
 - a. State co-operative banks
 - b. Central co-operative banks
 - c. Primary agricultural credit societies
 - d. Land development banks
 - e. State Land development banks
- 5) Development banks – they provide long term finance for setting up industries. They also provide short term finance for import and export activities. The following are the types of development banks:
 - a. Industrial finance Corporation of India Ltd(IFCI).
 - b. Industrial Development bank of India(IDBI)
 - c. Small industries development bank of India(SIDBI)
 - d. National bank for agriculture and rural development (NABARD)
 - e. Export-Import bank of India(ExIM)

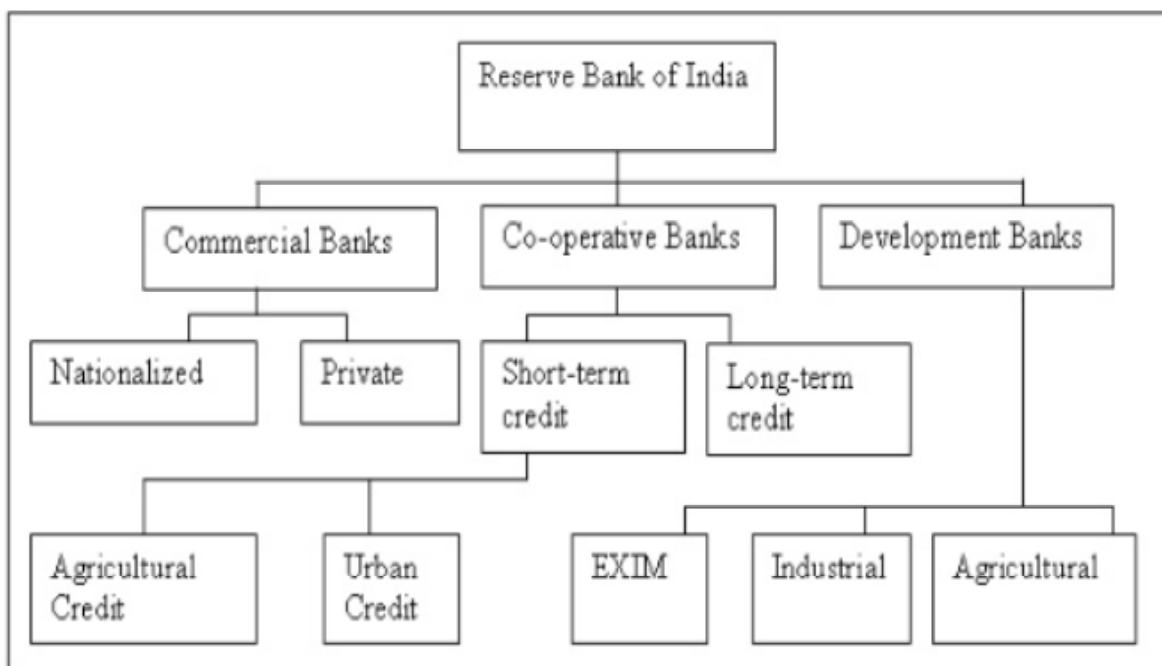


Figure 1.2: Structure of Banks in India

1.3. Reserve Bank of India (RBI)

1.3.1 History

It was established on 1st April 1935 in Kolkata with a paid up capital of 5 Crores based on the recommendations of the Hilton Young commission. It subsequently moved to Mumbai in the year 1937. It was the first bank to be nationalized in 1949. It has about 22 regional offices most of them located in state capitals. It is fully owned by Government of India.

1.3.2 Objectives of establishment of RBI

- To regulate the monetary and credit system of the country
- To maintain the internal and external value of our currency
- To achieve systematic and balanced development of banking in the country
- For regulating agricultural and industrial finance
- For proper management of public debt
- For establishment of monetary relations with the rest of the world

- For the maintenance of cash reserves of commercial banks
- To establish the money market in our country in an organized manner
- To monitor in achieving the required balance between demand and supply of currency and thus check the velocity of money
- For maintaining price stability and ensuring adequate flow of credit to productive sectors

1.3.3. Organizational structure of RBI

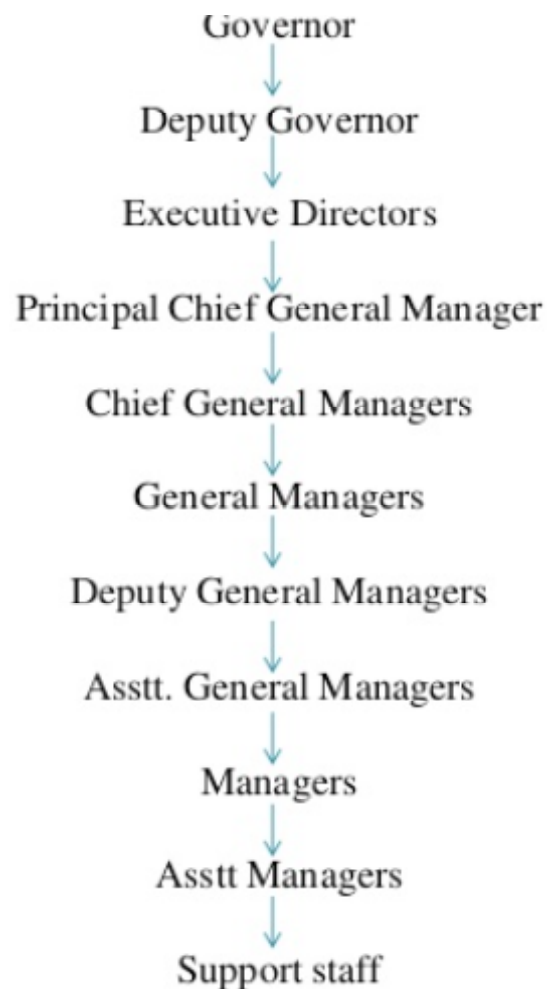


Figure 1.3: Organizational structure of RBI

1.3.4. Functions of RBI Regulator and supervisor of financial system

- o It helps formulate, implement and monitor the monetary policy
- Manager of exchange of control
 - o It helps manage the foreign exchange through foreign exchange management act, 1999
- Issuer of currency
 - o Issues and exchanges or destroys currencies and coins not fit for circulation
 - o RBI maintains currency chests for distributions of currency notes and coins
 - o Thus ensures supply of adequate quantity of currency notes and coins in good condition
- Banker to Government
 - o It performs merchant banking functions for the central and state governments by acting as their banker
- Banker to bank
 - o It maintains banking accounts of all scheduled banks
 - o It supervises and controls various types of banks in our country
 - o It has a development role by performing a wide range of promotional functions in supporting the national objectives
 - o Lender of last resort – it acts as a lender of last resort through rediscounting of bills
 - § Commercial bills
 - § Bill for financing agricultural operations
 - § Bill for financing cottage and small scale industries
 - § Bill for holding or trading in government securities
 - § Foreign bill
 - o Loans and advances

1.3.5. Instruments of Credit Control



Figure 1.4 : Instruments of Credit control

1.3.6 Subsidiaries of RBI

The following are the subsidiaries which are fully owned RBI

- Fully owned
- National Housing Bank (NHB)
- Deposit Insurance and Credit Guaranty Corporation of India(DICGC)
- Bharatiya Reserve Bank Note Mudran Pvt Ltd. (BRBNMPL)
- Major Stakes
- National Bank for Agriculture and Rural Development (NABARD)
- The RBI had recently divested stake in State Bank of India to the Government of India

1.3.7 Related Functions

- It regulates the opening/installation of ATM
- It regulates the opening of branches by banks
- It ensures that all the non banking finance companies to follow the KYC guidelines
- It supplies fresh currency notes for ATM
- It regulates the trade of gold

1.4 Commercial banks

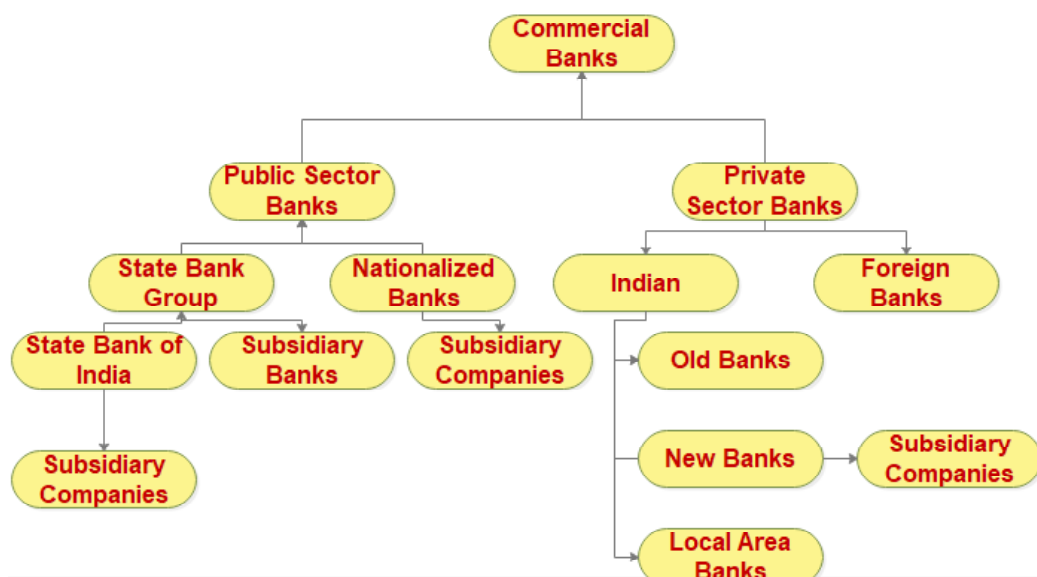


Figure 1.5 : Structure of Commercial Banks

Figure depicts the structure of commercial banks. They are classified into public sector banks and private sector banks. Public sector banks have the state bank group with state bank of India and its subsidiary companies. Nationalized banks together with its subsidiary companies also form part of public sector banks. Further the private sector banks are classified into two, namely Indian and foreign banks respectively. Indian Banks have been classified into old banks, new banks with their subsidiary companies and local area banks.

The commercial banks in India operate as the chief most component in the Indian financial system in fostering the financial processing as such.

1.4.1 Functions of Commercial Banks

The following are the functions of commercial banks:

- Accepting deposits from the general public
- Granting of loans and advances
- Services offered to depositors and borrowers
 - o Collection of cheques, demand drafts, bills of exchange, promissory notes, hundies, foreign documentary and clean bills
 - o Purchase of local and foreign currency documentary/clean bills, negotiation of bills: inland and foreign letters of credit (LOC) and also advising of inland and foreign letters of credit (LOC)
 - o Executing the standing instructions of their customers for payments of insurance premium, taxes and licenses, remittances and so on.
 - o Ancillary services - banks do perform ancillary services for the public like the following for a specific charge or fees:
 - § Issue of financial guarantees and performance guarantees on behalf of their customers
 - § Keeping in safe custody the documents and securities of customers
 - § Hiring safe deposit lockers to the public
 - § Purchase and sale of securities
 - § Collections of interest on securities/ debentures
 - § Collection of pensions
 - § Remittance of funds through bank drafts, mail transfer and telegraphic transfers and such other electronic media
 - § Issue of credit cards
- Credit creation
- Financing foreign trade

1.4.1 New business undertaken by bank

- Banks and insurance business
- As participant of securities depositories
- Sale of mutual fund products
- Advisory services for wealth management and investments

1.4.2 Subsidiaries of Banks

- Through their subsidiaries, Indian commercial banks have started doing the following:
 - o Equipment leasing
 - o Venture capital
 - o Housing finance
 - o Mutual funds
 - o Factoring services
 - o Leasing and merchant banking subsidiaries
 - § Managing capital issues
 - § Arrangement of finance from financial institutions
 - § Formulation of projects
 - § Consultancy service
 - § Trusteeship
 - § Management of portfolios investment management
 - § Equipment leasing
 - § Mergers
 - § Amalgamations

1.5 Cooperative Banks

Co-operative banks were established under the co-operative society's act of different states. While commercial banks is based on branch banking structure; co-operative banks are setup with state co-operative bank at apex level, central/district co-operative bank at district

level and primary co-operative societies at rural level. It is interesting to note while all the sections of banking regulation act are fully applicable to commercial banks only some sections of the same are applicable to co-operative banks resulting in partial control of the RBI of co-operative banks. Co-operative banks in fact centers around the principle of co-operation.

- Members of cooperative banks are themselves the owners and customers.
- These banks are created by persons belonging to the same local or professional community with a common interest
- They encourage and promote thrift and mutual help for the development of agriculturists, artisans and the like.
- They helped in reducing the exploitation of money lenders

1.5.1 Features of Cooperative Banks

- Customer themselves owners
- Control of members using democratic principles
- Allocation of profits

1.5.2 Structure of Cooperative Banks

They consist of the following segments:

- Agricultural segment and Non-agricultural segment
 - o Primary agricultural credit society at village level
 - § They form the basis of the co-operative credit structure as such
 - § So much so, it is an association of borrowers and non-borrowers residing in a specific locality.
 - § The funds of the Society are derived from the share capital and deposits from its members and the loans from central Co-operative Banks.
 - § It is interesting to note that the borrowings component is the most important element of their working Capital. However one should know that in this regard the purpose of borrowing is different in each and every state.
 - § The Loans are being accorded to the members for the purchase of Cattle, fodder, fertilizers; Pesticides and so on.

- o Central co-operative bank at the District level
 - § They are federations of primary credit societies in a district
 - § Funds for these banks consists of share capital, deposits, loans and over drafts from state cooperative banks and joint stocks
 - § They conduct all the business of a joint stock bank
- o State co-operative bank at the highest level
 - § The state cooperative bank is a federation of central co-operative bank and acts as a watchdog of the co-operative banking structure in the state
 - § It funds are obtained from share capital, deposits, loans and overdrafts from the Reserve Bank of India
 - § They lend money to central co-operative banks and primary societies and not directly to farmers
- o Land development banks
 - § They are organized in state, central and primary level to meet the long term credit requirements of farmers
 - § They oversee the primary land development banks
 - § They are governed by both state government and RBI
 - § National Bank for Agriculture and Rural Development (NABARD) supervises these banks
- o Urban co-operative banks
 - § These are primary co-operative banks situated in urban and semi-urban areas
 - § They lend to small borrowers and business

1.6 Regional Rural Banks (RRBs)

They cater to the needs of weaker sections of the rural population such as small farmers, marginal farmers, agricultural labourers, artisans and small entrepreneurs.

1.6.1 Features of RRBs

- They are more suitable for rural development works and hence the branches are usually opened in rural areas
- The eligible business of commercial banks in rural areas are being transferred to these banks
- Shareholders and equity capital are being raised to meet the initial losses of RRBs
- Facilities provided by sponsor banks are continued for ten years
- Refinance by RBI are aimed at by RRBs
- They take the banking service to the door step of rural users
- They help mobilize rural savings

1.7 Development Banks

It is a financial institution concerned with providing all types of financial assistance to business units in the form of loans, underwriting, investment and guaranty operations and promotional activities. It is basically a term lending institution. It is a multipurpose financial institution with a broad development outlook – economic development in general and industrial development in particular.

1.7.1 Functions of development banks

The following are the functions of development banks

- It helps in the process of fostering industrial growth
- It provides for long term financial assistance
- It helps in achieving balanced development
- It helps in providing promotional services
- It breeds in building infrastructure
- It helps in achieving entrepreneur development
- It helps in fulfilling socio economic objectives

1.8 Investment banks

These banks are Financial Intermediaries that acquire the Savings of People and direct these funds in to the business enterprises which essentially seek for the acquisition for the plant and acquisition and for the holding of Inventories.

1.8.1 Features of Investment Banking:

- Security
- Long term Financing
- Merchandiser
- Insurer
- Underwriter
- Security Middlemen

1.8.2 Functions of Investment Banks:

- Underwriting
- Capital Formation
- Purchase and Selling of Securities.
- Acting as a Dealer.
- Advisory Services.

1.9 Merchant Banks: They render the following services

- Managing the Customer's Securities;
- Portfolio Management;
- Counselling;
- Insurance.

1.9.1 Functions of Merchant Banks: The following are the Functions of Merchant Banks

- Sponsoring Issues;
- Helps in Mergers and Acquisition;

- Arrangement of Fixed Deposits;
- Portfolio management

1.10 Small Banks and Payment Banks

1.10.1 Small Banks

- The purpose of the small banks will be to provide a whole suite of basic banking products such as deposits and supply of credit, but in a limited area of operation.
- The objective for these Small Banks is to increase financial inclusion by provision of savings vehicles to under-served and unserved sections of the population, supply of credit to small farmers, micro and small industries, and other unorganized sector entities through high technology-low cost operations.
- Resident individuals with 10 years of experience in banking and finance, companies and Societies will be eligible as promoters to set up small banks. NFBCs, micro finance institutions (MFIs), and Local Area Banks (LABs) can convert their operations into those of a small bank. Local focus and ability to serve smaller customers will be a key criterion in licensing such banks.
- Branch expansion: For the initial three years, prior approval will be required.
- The area of operations would normally be restricted to contiguous districts in a homogenous cluster of states or union territories so that the Small Bank has a 'local feel' and culture. However, if necessary, it would be allowed to expand its area of operations beyond contiguous districts in one or more states with reasonable geographical proximity.
- The bank shall primarily undertake basic banking activities of accepting deposits and lending to small farmers, small businesses, micro and small industries, and unorganized sector entities. It cannot set up subsidiaries to undertake non-banking financial services activities. After the initial stabilization period of five years, and after a review, the RBI may liberalize the scope of activities for Small Banks.
- The promoters' other financial and non-financial services activities, if any, should be distinctly ring-fenced and not co-mingled with banking business.
- A robust risk management framework is required and the banks would be subject to all prudential norms and RBI regulations that apply to existing commercial banks, including maintenance of CRR and SLR.

- In view of concentration of area of operations, the Small Bank would need a diversified portfolio of loans, spread over its area of operations.
- The maximum loan size and investment limit exposure to single/group borrowers/ issuers would be restricted to 15 per cent of capital funds.
- Loans and advances of up to Rs 25 lakhs, primarily to micro enterprises, should constitute at least 50 per cent of the loan portfolio.
- For the first three years, 25 per cent of branches should be in unbanked rural areas.

1.10.2 Payments Banks

- Objective of payments banks is to increase financial inclusion by providing small savings accounts, payment/remittance services to migrant labour, low income households, small businesses, other unorganized sector entities and other users by enabling high volume-low value transactions in deposits and payments/remittance services in a secured technology-driven environment.
- Those who can promote payments banks can be a non-bank PPIs, NBFCs, corporate, mobile telephone companies, super market chains, real sector cooperatives companies and public sector entities. Even banks can take equity in Payments Banks
- Payments Banks can accept demand deposits (only current account and savings accounts). They would initially be restricted to holding a maximum balance of Rs 100,000 per customer. Based on performance, the RBI could enhance this limit.
- The banks can offer payments and remittance services, issuance of prepaid payment instruments, internet banking, functioning as business correspondent for other banks.
- Payments Banks cannot set up subsidiaries to undertake NBFC business.
- As in the case of Small Banks, other financial and non-financial services activities of the promoters should be ring-fenced.
- The Payments Banks would be required to use the word 'Payments' in its name to differentiate it from other banks.
- No credit lending is allowed for Payments Banks.
- The float funds can be parked only in less than one year G-Secs

These are new type of banks which are expected to reach customers mainly through their mobile phones. However, they are allowed to undertake only certain banking functions that are being allowed by the banking Regulation Act 1949.

RBI says in its guidelines “the objectives of setting up of payment banks will be to further financial inclusion by providing

- 1) Small accounts
- 2) Payment/remittance services

to migrant labour work force, low income households, small business, other unorganized sector entities and other users”.

In fact on 19th august 2015 the RBI gave permission “in-principle” licenses to the following for launching payment bank namely:

1. National Securities Depository Limited (NSDL)
2. Reliance Industries
3. Aditya Birla Nuvo
4. Airtel Mcommerce
5. Department of Posts
6. Fino PayTech
7. Tech Mahindra
8. Vodafone M-Pesa
9. Cholamandalam Distribution Services
10. Paytm
11. Sun Pharma

1.10.2.1 Structure of payment banks

The following are the aspects of payment banks.

- 1) Rs. 100 crores is the minimum capital
- 2) 75% should be maintained as deposits

- 3) Maintain 25% of deposits on other banks
- 4) Should have at least 26% investment by Indians
- 5) Can accept CASA deposit
- 6) Liability should not exceed 20 times of its net worth
- 7) Have 25% branches in unbanked areas
- 8) Be fully networked and technology driven
- 9) They can offer FOREX services
- 10) They cannot sanction loans or accept fixed deposits
- 11) They can offer services such as automatic payment of bills, purchases in cashless, chequeless transactions through phone.

1.10.2.2 Benefits of payment banks

- 1) It will ensure more money coming into banking system
- 2) The customer will not be forced to maintain a minimum balance in their account
- 3) It may make even the handling of cash easier
- 4) Service charges will come down
- 5) RBI rate cut for these banks

1.11 Banking Regulations Act 1949 with specific reference to 5(c)

Definition of Banking

“Accepting for the purpose of lending or investment of deposits of money from the public, repayable on demand or otherwise and withdrawal of cheques, draft, order or otherwise (Section 5c). The objective of this is to ensure sound banking through regulation covering the opening of branches and maintenance of liquid asset. It also provides for safety in the interest of deposits and also prevents misuse of powers by managers of banks.

Section 5(c) of the act defines banking as “a company which transacts the business of banking in India”

It means that any company which is engaged in the manufacture of goods or carries any trade and which accepts the deposits of money from public merely for the purpose of financing

its business as such manufacturer or trader shall not be deemed to transact the business of banking within the meaning of this clause.

1.11.1 Forms of Banking Business

The following are the other forms of banking business. Innovative services apart from the regular services are offered by Banks.

1) ATM Services - It is enabled through an electronic telecommunication device. It helps enable the clients of banks to perform financial transactions to have anytime money.

2) Debit Card and Credit Card facility - It can be used in ATM, Point of Sale terminals, electronic commerce sites online debit card; offline debit card; prepaid debit card; electronic pulse card, cards for mail, telephone and internet use only debit card. The types of credit cards are secured credit cards, prepaid cards, balance transfer credit cards, low interest credit cards.

3) Telebanking

Telephone banking is a service being provided by a bank which enables customers to perform their financial transactions through phone. They need not visit the bank or ATM.

4) Internet Banking

Online banking/Internet Banking/Electronic Banking has been introduced by the banks as a facility of transacting banking for its customers. It allows the customers to transact their banking transactions through website in a secured manner. It helps to check the balances, effective transfer of money, to do online shopping, pay bills online and so on.

5) Assurance

It helps the delivery of insurance products through banking channels.

6) RTGS - Real time gross settlement system is a fund transfer of money takes place from one bank to another bank. It is however a large volume of transaction.

7) NEFT - National Electronic Funds Transfer Facilitates transfer of to other bank accounts into other bank branches accross the country. It is a simple, secure, safe, faster way to transfer funds.

8) Point of Sale (POS) - It is the time and place where a retail transaction completes. It is the point at which a customer makes a payment to the merchant in exchange of goods or after the provision of service. For this electronic device point of sale terminals kept at the merchant outlets help as tool in accepting debit/credit cards as the case may be.

9) Mobile Banking - smart phones or other cellular devices are being used to perform online banking tasks. It helps to monitor account balances, transfer of funds between accounts, payment of bills and also helps in locating the ATM in use.

10) Smart cards - It is a microchip enabled card which is used for purchases without the need of keying the pin. It is a very powerful card which carries the functions of debit/credit cards and ATM card.

Thus a bank with the newly evolved services aims at achieving the following aspects of operations:

- 1) Banks and financial services will be revolving around the customer choices
- 2) Banks of the future will be on mobiles
- 3) There will be the emergence of robot advisors that will help the customers in arriving at the real and right time.
- 4) The behaviour of banks data will be monitored by powerful algorithms and thus ease out the tension from the external security threats.
- 5) Banks would be replaced by platforms run by algorithms and robots which will help know customers, and products and merchants
- 6) Blockchain technology will be used to distribute, verify and record a wide range of financial services.
- 7) Eventually bank account will become like a cell phone number which will help the customer from one bank to another.
- 8) Lending, borrowing and trading on social network platforms will get widespread.
- 9) Decentralized and crowd sourced loans, mortgages and risk management products will become the norm.

1.11.2 Ancillary services like trade finance, remittances etc

The following are the secondary services which involve risk free profit making services that are provided by the bank known as ancillary services.

1) Remittance services: - Fund transfer within the same bank of different branches or to other banks involves former charge which is the profit to the service providing bank like Banker Cheque, Demand Draft NEFT, RTGS.

2) Custodial services: - safe deposit lockers in the bank to protect the assets

3) FOREX services - Banking services which involves foreign currency exchange when a customer travels to other countries

4) Card services -credit card and debit card services for a charge..

5) E-Banking services-online banking and mobile banking

6) Gold Sale - certain bank provide this facility of gold sale in the form of a coin or a 100gm biscuit/Bar

7) Investment service - helps the customers in investing in mutual fund run by the banks

8) Insurance Sale - Banks offer a range of insurance products covering the risk of life, health, assets like vehicle, credit and debit cards, travel and the like.

1.11.3 Types of deposits

In India banks have two major types of deposit accounts namely

1) Demand deposits 2) Term deposit

1.11.3.1. Demand Deposits: If the funds deposited can be withdrawn by the customer (depositor / account holder) at any time without any advanced notice to banks; it is called demand deposit. One can withdraw the funds from these accounts any time by issuing cheque, using ATM or withdrawal forms at the bank branches. The money as demand deposit is liquid and can be encashed at any time. The ownership of demand deposits can be transferred from one person to another via cheques or electronic transfers. There is no fixed term to maturity for Demand Deposits. The demand deposits may or may not pay interest to the depositor. For

example, while we get an interest on savings accounts; no interest is paid on current accounts.

Demand Deposits have two major types of account namely:

- 1) Current account
- 2) Savings bank account

A) Current account

Current account- these are usually for businessman to carryout their daily transactions. it does not given any interest. it has overdraft faility and other facilities such as collection of cheques, transfer of money and renders agency and general utility services. it attracts interest on overdraft on a day to day basis. minimum balance should be maintained in such accounts. Depending upon the banks the charges for non compliance with respect to the minimum balance will attract charges.

Benefits of current account

- 1) It has a easy, fast access to your funds
 - 2) It has overdraft facilities
 - 3) It has electronic payment facilities'
 - 4) It helps in management of regular payments
 - 5) It gives a cheque book facility
- b) Savings bank account

B) Savings bank account

These are the most popular type of individual accounts. They help in promoting the savings habit in individuals. it provides cheque facility for withdrawal of funds. The savings bank account overdraft facility will not be given. Cheques collections are given in this account. It fulfills the KYC norms. It has facility of operation such as either or survivor and former or survivor.

Benefits of savings bank account

- 1) It helps develop savings bank habit amongst wage earners and others who have fixed income
- 2) It gives periodical interest

- 3) It provides cheque facility for making payments
- 4) It helps in fund transfer directly
- 5) It acts as a residential proof of identity of a person
- 6) It helps in doing online shopping
- 7) It provides funds immediately through ATM
- 8) It shows the aggregate salary of a person over a period of time

1.11.3.2 Term deposits

Time Depositor Term deposits:

When money is deposited with a “tenure”, it cannot be withdrawn before its maturity fixed at a particular time. Such deposits are called “Time deposits” or “Term deposits”. The most common example of Time deposits is “Fixed Deposit”. All time deposits are eligible for interest payments. Interest rate depends upon the tenure and amount of deposit. This rate varies from bank to bank. The interest rate is generally higher for time deposits of longer tenure. On the basis of their nature, time deposits may be of three types as follows:

- Fixed deposits
- Re-investment deposits
- Recurring deposits

a) Fixed Deposit

It is a type of deposit where a sum of money is given to the bank. it gives interest which varies accordingly from bank to bank. the investor receives back the amount of deposit at the time of maturity. It has nomination facility. It has facilities such as former or survivor.

Benefits of fixed deposits

- 1) it encourages savings habit for a longer period of time
- 2) it enables to earn a higher interest rate
- 3) it ensures safety for the future
- 4) it is risk free
- 5) loan on fixed deposit is allowed

b) Re-investment deposits:

Interest is compounded quarterly and paid on maturity, along with the principal amount of the deposit. In the Flexi Deposits amount in savings deposit accounts beyond a fixed limit is automatically converted into term-deposits.

Benefits of re-investment plan

1) The bank adds the regular interest to the principle and compound interest is calculated on the whole amount and paid thereon on maturity.

2) Loan on reinvestment deposit is available.

c) Recurring deposits

They are down on monthly basis for specific number of months. it is for a period from six months to one year. usually no withdrawals are allowed before the completion of stipulated time. Interest is given based on the number of months.

Benefits of recurring deposits

- 1) It encourages regular monthly savings habit
- 2) Loan on RD is allowed
- 3) minimum deposited amount is less
- 4) TDS not applicable
- 5) higher interest rates than savings bank account
- 6) low risk instrument

there are other forms of deposits like

- o 2-in-1 deposits
- o smart deposits
- o power saving deposits
- o automatic sweep deposits

d) Other savings accounts

There are other types of savings account offered by banks namely

1) Kiddy bank - this scheme is meant for the children upto the age of 18 years. minors who have completed 10 years of age are eligible to open this account. at the time of opening the account an attractive will be given free of cost to each account holder for putting their savings inside the doll. Dolls contain a secret lock which can be opened only at the branch where the account is held. the minimum balance is Rs.100/-.

2) Daily Deposit Collection

3) Insurance linked accounts

4) Jhan Dhan Account

1.11.4 Deposit schemes to Non-Resident Indian

1) NRO - Non-Resident ordinary accounts - NRI can open an NRO deposit account for collecting their funds from local bonafide transactions. when a resident becomes an NRI his/her existing rupee accounts are designated as NRO accounts. these accounts can be maintained as savings, current, recurring or term deposits.

2) Non Resident (External) Rupee Accounts (NRE - Account) - These accounts are opened with funds remitted to India through a bank channel from abroad. In the case of Bangladesh and Pakistan citizens, prior approval of RBI is necessary. These accounts are governed by non-resident (external) Account Rules 1970. The interest rates on NRE accounts shall be usually at the rates applicable to domestic savings deposit accounts. accounts can be opened singly or jointly. The account holder can give the power of attorney or other authority to a resident Indian to operate the account. Immediately on return to India, the account holder should inform the bank and convert his NRE deposit account into ordinary Rupee account. The account holder can also avail of loans abroad against the security of term deposits in India.

3) Foreign currency non - resident (FCNR) account

These deposits are accepted from non-resident individuals of Indian nationality or origin (NRIs) and are maintained in US-dollar, Pound, Sterling, Euro, Canadian dollars and Australian dollars. Only term deposits are accepted under this category. The depositors are protected

from exchange risk and the foreign exchange risk is borne by bank. The minimum maturity period is 1 year and maximum maturity period is 5 years.

Insurance linked account

Banks offer insurance products or insurance benefit to the bank customers. Banks do offer clients the option to purchase health insurance policies through the health insurance companies. Banks sell life insurance also. Banks earn additional revenue by selling insurance products.

Summary

Banking is a system of trading money. It provides for a safe place to save the excess cash in the form of deposits. It loans out this money for the growth of businesses. It helps the consumers for the purchase of homes, vehicles and the like. Section 5(b) of the banking regulation act 1949 gives a comprehensive definition of banking as, “banking” as “accepting for the purpose of lending and investment of deposits of money from the public, repayable on demand or otherwise, and withdraw able by cheque, draft, order or otherwise”.

Banks deals in money and credit. It handles the money of the public for their advantage and to make its own profits. As such banking as a financial institution links the flow of funds from the persons who save to the prospective users. Thus it plays an important role in the economic growth of the country holding the savings of the public.

They are classified into public sector banks and private sector banks. Public sector banks have the state bank group with state bank of India and its subsidiary companies. Nationalized banks together with its subsidiary companies also form part of public sector banks. Further the private sector banks are classified into two, namely Indian and foreign banks respectively. Indian Banks have been classified into old banks, new banks with their subsidiary companies and local area banks.

Co-operative banks were established under the co-operative society's act of different states. While commercial banks is based on branch banking structure; co-operative banks are setup with state co-operative bank at apex level, central/district co-operative bank at district level and primary co-operative societies at rural level. It is interesting to note while all the sections of banking regulation act are fully applicable to commercial banks only some sections of the same are applicable to co-operative banks resulting in partial control of the RBI of co-operative banks. Co-operative banks in fact centers around the principle of co-operation.

They cater to the needs of weaker sections of the rural population such as small farmers, marginal farmers, agricultural labourers, artisans and small entrepreneurs.

It is a financial institution concerned with providing all types of financial assistance to business units in the form of loans, underwriting, investment and guaranty operations and promotional activities. It is basically a term lending institution. It is a multipurpose financial institution with a broad development outlook – economic development in general and industrial development in particular.

These banks are Financial Intermediaries that acquire the Savings of People and direct these funds in to the business enterprises which essentially seek for the acquisition for the plant and acquisition and for the holding of Inventories.

The purpose of the small banks will be to provide a whole suite of basic banking products such as deposits and supply of credit, but in a limited area of operation.

Objective of payments banks is to increase financial inclusion by providing small savings accounts, payment/remittance services to migrant labour, low income households, small businesses, other unorganized sector entities and other users by enabling high volume-low value transactions in deposits and payments/remittance services in a secured technology-driven environment.

1) ATM Services - It is enabled through an electronic telecommunication device. It helps enable the clients of banks to perform financial transactions to have anytime money.

2) Debit Card and Credit Card facility - It can be used in ATM, Point of Sale terminals, electronic commerce sites online debit card; offline debit card; prepaid debit card; electronic pulse card, cards for mail, telephone and internet use only debit card. The types of credit cards are secured credit cards, prepaid cards, balance transfer credit cards, low interest credit cards.

3) Telebanking

Telephone banking is a service being provided by a bank which enables customers to perform their financial transactions through phone. They need not visit the bank or ATM.

4) Internet Banking

Online banking/Internet Banking/Electronic Banking has been introduced by the banks as a facility of transacting banking for its customers. It allows the customers to transact their banking transactions through website in a secured manner. It helps to check the balances, effective transfer of money, to do online shopping, pay bills online and so on.

5) Assurance

It helps the delivery of insurance products through banking channels.

6) RTGS - Real time gross settlement system is a fund transfer of money takes place from one bank to another bank. It is however a large volume of transaction.

7) NEFT - National Electronic Funds Transfer Facilitates transfer of to other bank accounts into other bank branches across the country. It is a simple, secure, safe, faster way to transfer funds.

8) Point of Sale (POS) - It is the time and place where a retail transaction completes. it is the point at which a customer makes a payment to the merchant in exchange of goods or after the provision of service. For this electronic device point of sale terminals kept at the merchant outlets help as tool in accepting debit/credit cards as the case may be.

9) Mobile Banking - smart phones or other cellular devices are being used to perform online banking tasks. It helps to monitor account balances, transfer of funds between accounts, payment of bills and also helps in locating the ATM in use.

10) Smart cards - It is a microchip enabled card which is used for purchases without the need of keying the pin. It is a very powerful card which carries the functions of debit/credit cards and ATM card.

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It is a type of deposit where a sum of money is given to the bank. it gives interest which varies accordingly from bank to bank. the investor receives back the amount of deposit at the time of maturity. It has nomination facility. It has facilities such as former or survivor.

Interest is compounded quarterly and paid on maturity, along with the principal amount of the deposit. In the Flexi Deposits amount in savings deposit accounts beyond a fixed limit is automatically converted into term-deposits.

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Banks offer insurance products or insurance benefit to the bank customers. Banks do offer clients the option to purchase health insurance policies through the health insurance companies. Banks sell life insurance also. Banks earn additional revenue by selling insurance products.

Check your Answers

1. What is Banking?
2. Describe broad classification of Banking.
3. Explain the Roles of RBI and Organizational Structure of RBI.
4. Explain the structure of Commercial Banks.
5. What are the features of Co-operative Banks?

Lesson - 2

Computerized Information System for Banks (BIS)

Objectives

After reading this lesson you will be able to understand the following:

- Overview
- SDLC
- MIS for Banks
- LAN Model
- Architecture
- Banking Software
- Core Banking System (CBS)
- CBS Security and control

Structure

- 2 Introductions**
- 2.1 Systems Development Life Cycle**
- 2.2 MIS for Banks**
- 2.3 Accounting Information Systems for Banks.**
- 2.4 Architecture of bank branch computerization**
- 2.5 The Model of Mini-computer with Multi-Termini Systems**
- 2.6 Minicomputer systems with a Multi terminal – software Architecture**
- 2.7 LAN Model**
- 2.8 Banking software**
- 2.9 Core Banking Systems.**

2. Introduction

The Banking in India have cross three eras – pre-nationalization era, post nationalization era and post liberalization era. During the pre-nationalization era till 1996, banks were focused more on basic principles of banking- accepting deposits for the purpose of lending. Customers approached bankers for giving deposits which banker accepted for the purpose of lending and making profits.

During the post-nationalist ion era till 1994, several banks were brought under Government control and directed to towards a major thrust to rural lending and poverty alleviations programs. Employment generation was the focus. Banks were re-oriented themselves to open and manage branches in the remotest corners of the country to meet the region wise, sector wise lending targets. Thousands of employees were recruited. Lakhs of borrowers benefited. Cores of rupees were mobilized as deposits and were also lent with focus on increasing agricultural production. Developing small scale sectors, self employment programs and so on.

The post liberalization era started in the late 1990s and continuing even now fuelled by the emergence of IT sector as major export earner for the country. Computerization is a positive step to bank growth. With the aid of computers, the bank work can be done faster. Not only the present work load can be reduced to a great extent but also the bank can expand its working area with the same manpower. Reserve bank of India and State Bank of India installed computer systems for processing and reconciliation of inter branch transactions, processing statistical data and for research purpose.

“Based on the norms worked out by Rangarajan Committee (II), 7827 branches of the public sector banks were identified for full branch computerization up to March 2000 of which around 4620 were computerized as on March 99. Meanwhile, the networking of the already-computerized branches also assumed urgency and some of the Banks have started inter-connecting their computerized branches using made software like railway reservation system, banking information system, library management system etc. Apart from this software there is a need of Database Management System, File Management system, software for run the computer network etc.

2.1 System Development Life Cycle

Like human beings, systems too have a similar life cycle. After a system has been in operation for a number of years, it gradually declines and becomes less and less effective

because of the changing environment. For some period it is possible to overcome the problems by amendments (changes) and minor modification to the system, but it will not work for a long period of time, and at one point of time the user have to buy or develop a new system to fulfill his requirement. Same thing is applicable in Information system.

2.2 MIS for Banks

One of the major features of computerization in banking is designing and development of “Management Information System”. It is very convenient and user friendly. “The importance of effective MIS for control operations and of maintaining customer and business/industry data base for strategic planning has also surfaced; while Banks are looking at Data warehousing, Data mining, Business Restructuring etc. as most essential things to have as early as possible, they are taking urgent steps to computerized the operations in their administrative and controlling machinery, so as to evolve an effective MIS. In this phase, the new communication revolution sweeping the nation and the world has come in extremely handy, as the communication infrastructure has improved significantly and the Internet technologies are available to network branches at a relatively low and affordable cost with a high degree of reliability. The present level of MIS covers, basically, information needed for control, performance monitoring, decision making etc. and encompasses most activities in administrative offices like processing of statutory returns under Reserve Bank of India Act, monthly/quarterly performance reports from branches, credit information/BSR, inter-branch transactions, personnel inventory, provident fund accounting, profit and loss accounts, cash and investment management, stationery stock accounting, and branch housekeeping etc.”

2.3 Accounting Information System for Banks

Information technology, globalization, and related developments have made business increasingly dependent on high-quality information for decision making. Business decision makers need the relevant and reliable information, which should be available at the right time and at reasonable cost. In earlier days, customers obtained the information through manually data processing. It takes lot of time and with the possibility of error. Today is the age of Information Technology. Data is processed by computer system and generate the useful information. There is an information generating software which provides information to the user at required time. For Example, Accounting Information System, is computer based software or system with process the raw data and generate the information for the user or for the bank.

Being an information system, accounting information must have a target (objectives) system. This system will perform banking accounting operations as well as non-accounting operations such as Human resources information system, Production planning and scheduling system, Strategic planning system and so on.

The target system for an accounting system has to do with the aspects of business operations that have to do with accountability for assets/liabilities of the enterprise, the determination of the results of operations that ultimately leads to the computation of comprehensive income, and the financial reporting aspects of business operations.

AIS and related disciplines: Accounting Information System (AIS) may be considered a discipline as well as a collection of system. AIS study the structuring and operation of planning and control processes. Its main objectives are:

- a) Providing information for decision making and accountability to internal and external user
- b) Providing the right conditions for sound decision making
- c) Ensuring that no assets illegitimately exit the organization.

AIS relates to several more elementary disciplines of which Accounting and Administrative Organization (AAO), Internal Control (IC), Managerial Information Provision (MIP), and Information Systems (IS) are main ones shown in figure below:

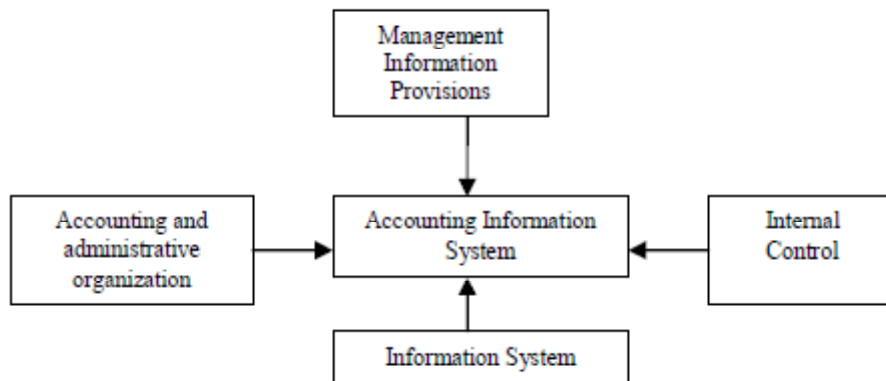


Figure 2.1 : Accounting Information System and related disciplines

**Source: Information system for Banks, Indian Institute of Banking and Finance
Taxman Publication (P) Ltd. 2005**

a) Accounting and Administrative Organization: The system of organizational measures with respect to data processing aimed information provision for management entity functionality and accounting purposes.

b) Managerial Information Provision: The system gathering, recording and processing of data aimed at the provision of information for management decisions.

c) Internal Control: Control of judgments and activities of others in so far as control is conducted for the management of the organization by or on behalf of that management.

d) Information system : An information system is an asset of interrelated components working together to collect, retrieve, process, store and disseminate information for the purpose of facilitating planning, control, coordination, and decision making in businesses and other organizations.

2.4 Architecture of bank branch computerization

a) General approaches: One of the important aspects of branch automation is to determine the size and type of the computer required at the branch. The computer size or the configuration is specified in terms of memory, disk space, number of terminals, printers etc. All capacity planning is directed towards optimizing throughout during peak loads and maximizing its use during other periods. The hardware architecture will define the parameters to be used in determining the size (capacity). "In banks computerization used hardware architecture consist of:

- a) Mini –computers at each branch and multiple terminals
- b) LAN at the branch with work station and
- c) ALPMs at the branch"

b) Database design: The database design can be classified as follows:

- i) Common databases and
- ii) Individual functional area databases
- iii) Common Data bases:

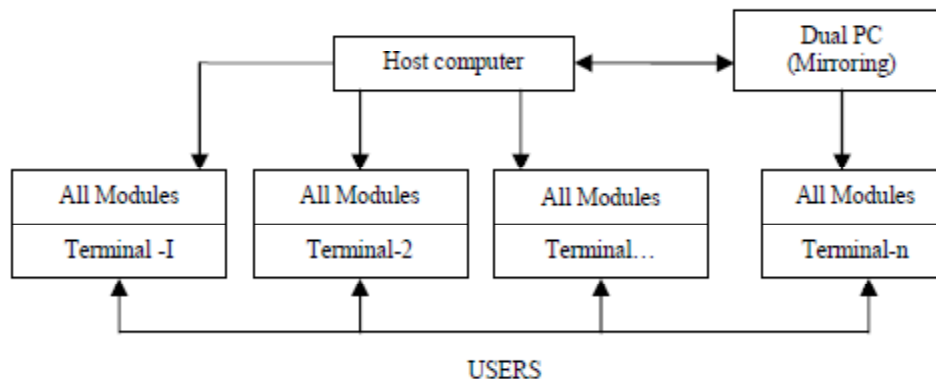
This type of data base contain particular information which is frequently required for both transaction posting as well as other services like query, Management Information System (MIS)

reports, etc. This is derived from all the functional areas where the information is common in nature for the for the areas such as transaction, cheques issue, stop payment, standing instructions, balances attachments, etc.

ii) Individual Functional Area Data bases: This data base has detailed information about each functional area. They have information which is not required frequently and also it contain master information about the functional area. If the transaction is not validated at common database then the transaction may be required to go through the functional areas for validation.

2.5 The Model of Mini-computer with Multi-Terminal System

“In the minicomputer with multiple terminals at branch where a customer will be served at any terminal with any type of transactions and other services. The module uses single window concepts, i.e. all modules or functional areas can be operated through one window. The configuration required for this model is: N number of terminals depending upon the work-load distribution (i.e., number of terminates at counters, back-office, authorization, development, etc.) connected to a main host computer. The intelligent terminals connected may have a hard disk for off-loading the data at the end of the day for back-up purposes.” The operator can access all functional areas of the branch from any terminals. Any type of service can be obtained by the customer from any terminal. This eliminates the formation of queues at the counters.



(Figure 2.2: Multi-Terminal System)

(Source: IT, MIS & Productivity – S.M. Padwal)

The operator can access all functional areas of the branch from any terminals. Any type of service can be obtained by the customer from any terminal. This eliminates the formation of queues at the counters.

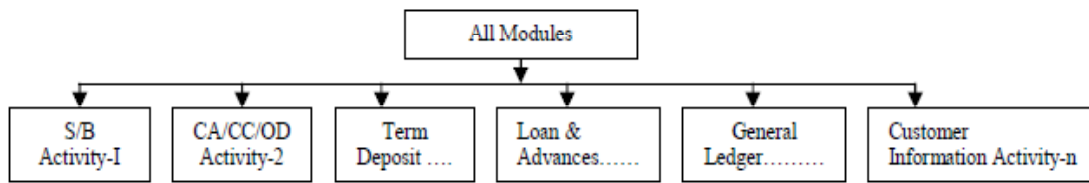


Figure 2.3 : Module Functions Details

(Source: IT, MIS & Productivity – S.M. Padwal)

2.6 Mini Computer System with a Multi-Terminal – Software Architecture

The development of software depends upon the approach selected. Accordingly the software may be favourable or unfavourable to the customers. Customer services are divided in to mainly two types- transactions and other services. Transaction type services are validated and posted to common data base with authentication by the offices/ manager as per the requirements. If more details are required for posting the transactions then the control will go to individual functional area's database, and it refers to additional information required and then the transaction will be posted. For any other type of customer service i.e., query, updation, changing details, then it will be routed directly to the individual functional area's databases and operation will be carried out. "In this design internal processing functions (main functions) have been identified, and such high-level functions are decomposed into sub-functions like issuing a cheque book, stop payments, loans, advance bills, term loan agricultural loan, bill purchases, bill receivable, etc. the data storage is based on these sub-functions."

2.7 LAN Model

The LAN models are designed on the basis of two approaches:

- Each LAN having a number of dedicated terminals for each depending upon the work- load. Those terminals will be networked at back-office to the host system.
- The second approach has separate LAN for a particular module depending upon the dominating or crowded modules and remaining modules can have a separate
- LAN or individual ALPMs. "LAN for some modules and LAN nodes or ALPMs

- for the remaining will be networked with the host computer, i.e., we will be using a LAN in a LAN.”

Therefore in both the approaches a customer will be served at dedicated terminals for particular modules and one window concept need not be used. Each operator will have access to only that module for which his terminal is dedicated. But the officer/manager can have access to all modules from his terminal.

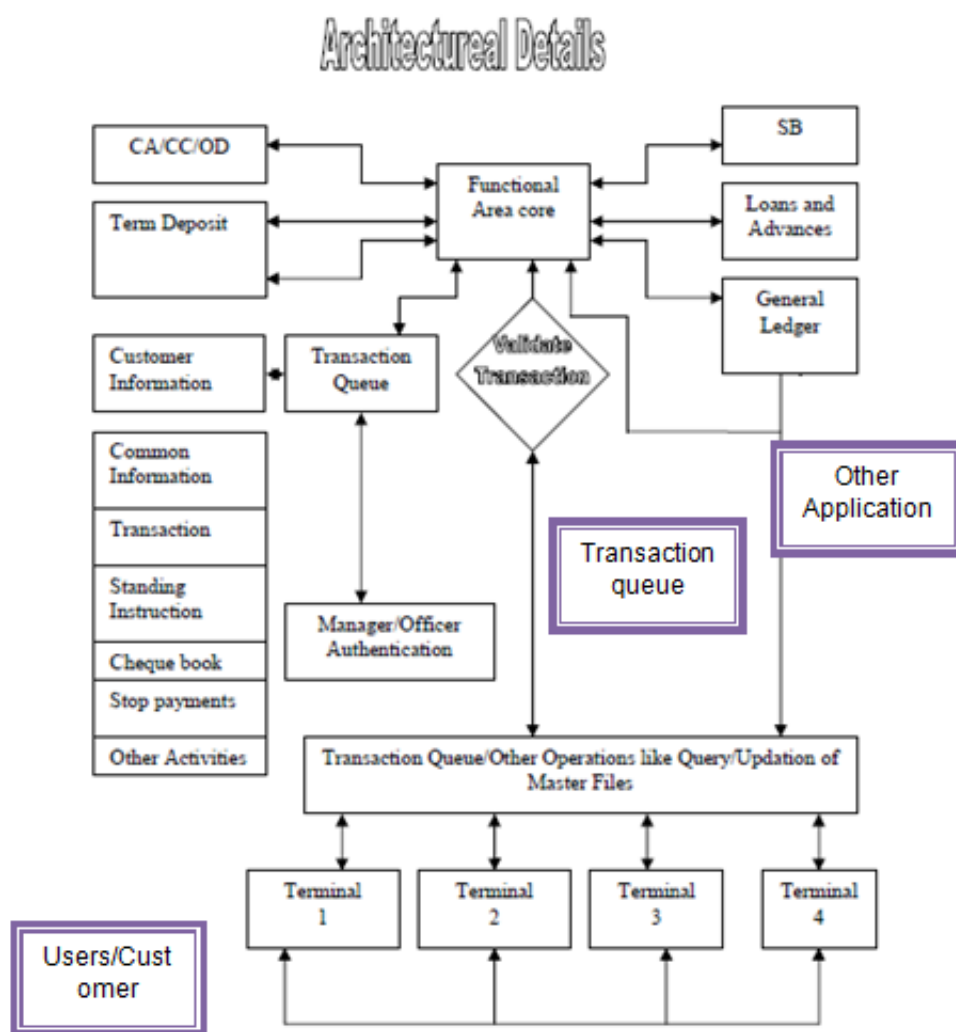


Figure 2.4 : LAN Model

LAN Model/ Approach- Software Architecture: “Architecture details for LAN model/ approach is also the same as multi-terminal approach. The difference is that access restrictions to each functional module are incorporated. As the architecture is same as the multi-terminals mode, there is flexibility for allocating and reallocating of terminals for different modules as per the work-load requirements.” ALPMs are simply the stand-alone PCs. Because of restrictive practices of the staff union regarding technology related matters, banks use ALPMs. The cost effectiveness of this approach is limited due to permissible number of accounts, and also the type applications that are taken up for mechanization. But this approach is slowly converting itself into LAN approach.

2.8 Banking Software

The Software Packages for Banking Applications in India had their beginnings in the middle of 80s, when the Banks, spurred on by RBI and the Rangarajan Committee Report, started computersing the branches in a limited manner. The Committee Report, started computersing the branches in a limited manner. The approach was to empanel a few hardware vendors who will also develop the software as per bank’s specifications and also help to install at the branches. This was a multi-vendor approach to foster competition and to assess the relative vendor approach was to empanel a few hardware vendors who will also develop the software as per bank’s specifications and also help to install at the branches. This was a multi-vendor approach to foster competition and to assess the relative vendor capabilities. These packages were written usually in fox-pro or c language and were capabilities. These packages were written usually in fox-pro or c language and were Dos-based-and rarely Unix-based. The early 90s witnessed a reduction in the hardware prices due to which high powered PCs and Servers were available in the affordable price. This led to what is called the Total Branch Automation (TBA) package. The solutions offered by the vendors were architecturally different. Some has centralized solutions and some vendors offered distributed processing models. The platforms used ranged from simple UNIX-C to powerful RDBMS like Oracle etc. The middle and late 90s there was rapid revolution in communication technologies and evolution of novel concept of ‘Convergence’ of computer and communication technology, like Internet, mobile/cell phones etc. The arrival of foreign and private banks with their superior technology-based services pushed Indian Banks also to follow suit by going in for the latest technologies so as to meet the threat of competition and retain customer base. This also brought in revolutionary products and services to which, the Indian Software Industry has significantly contributed.

2.8.1 Categories of packages

“The banking software present by available in India can be classified into six types:

- i) Stand-alone branch-level package
- ii) Multi-branch solutions
- iii) Foreign Packages
- iv) Packages for specialized niche areas
- v) Service Branch / high-volume transactions processing packages
- vi) IT Services”

- **Stand-alone branch-level package:** These are usually in FoxPro, C language or Dbase and are capable of handling specific functions at branches; these are sometimes networked on a LAN to simulate a TBA (Total Branch Automation) environment. AT the same time the high-end packages with a central server (which can be a Pentium PC or NT or a MINI or even a Main Frame, supported by multiple dumb or intelligent) terminals are also used. Some of them use sophisticated RDBM like ORACLE as back-end and provide user-friendly front-end with windows GUI (graphical user interface).
- **ii) Multi-branch solutions:** These solutions are used to network a cluster of branches in a city; the account maintenance can be central or can be distributed, networking being achieved through Wide Area Network (WAN) or terrestrial lines / high speed lines/ satellite networks – and now even wireless.
- **Foreign Packages:** The Foreign banking software solutions which are commonly used are Bank Master, Kappiti, and Sanchez etc. This need to be suitable modified to make Indian requirements- but their strength in their proven capabilities in developing and offering modern/ global banking products/ services that India is just ushering in.
- **Packages for specialized niche areas:** Like Assets Liability Management (ALM), Treasury Management, Trading / Dealing Room activities, Custodial Services / Depository Participant etc. These are high-end packages with sophisticated analytical and decision tools. v) **Service Branch / high-volume transactions processing packages:** These solutions include clearing, drafts issue/payments/reconciliation (Remittances), Bills (payments/collection/purchases), Dividend warrant processing,

inter-branch reconciliation etc. Mostly the service providers themselves implement these packages.

- **IT Services:** These are the sundry services provided by the software dealers. They include services like disaster recovery, virus protection, security handling /networking multiple legacy system between themselves of two new platforms or to new delivery channels like ATMs etc.

2.9 Core Banking System

The one of the biggest achievement in this period is development and implantation of “Core banking Solution (System)”. In the Core banking, all the data base of a particular bank is centralized and a bank can access any information and customer can get the services of that bank in any of its branch. It is a combination of Management Information system and small network banking system. Core banking is a general term used to describe the services provided by a group of networked bank branches. Bank customers may access their funds and other simple transactions from any of the member branch offices.

2.9.1 Definition of core banking

Core Banking is normally defined as “the business conducted by a banking institution with its retail and small business customers. Many banks treat the retail customers as their core banking customers, and have a separate line of business to manage small businesses. Larger businesses are managed via the Corporate banking division of the institution. Core banking basically is depositing and lending of money.” Normally core banking functions include deposit accounts, loans, mortgages and payments. Banks today know better than anyone the opportunities and the risks they face in an ever-changing competitive environment. The Old Management Information system is turned to words Core banking System (solution). The banks were traditionally grouped into one of the several business areas such as commercial, wholesale or Retail Banking to indicate the thrust they are giving to customer acquisition strategies, the “core” of their banking operations continue to be Deposits, Loans, Money Transfers and MIS. The aim of any core banking solution will have to meet these core-banking requirements. The role of technology is to make available core banking solutions that can be deployed centrally to take advantages of both the central administration and distribution operations. The core banking solution fulfills the requirements of wholesale banking by providing robust and industry strength Trade Finance and forex modules. And For Retail banking, such a solution should address

various types consumer lending from housing loans to commodity loans etc. The core banking operations will be done through all branches, wholesale banking will be confined to selected branches and much of the retail banking will be done in metros. Hence the core banking solution should be not only cost effective in meeting this varied requirement but also efficient and consistent when accessed centrally. The new private sector banks have brought in new shift in banking operations by meshing core banking with subsidiary banking which generally cover Investment banking, Insurance, Mutual funds and so on and have set expectations that core banking solution should address these requirement as well. They want to do “Universal Banking” through a small network of brick and mortar branches and large network of ATMs and expect that the technology should address this effectively. The emergence of Internet technology has reduced the cost of information delivery substantially and is also readily “available” for commercial exploitation to reduce the cost of online banking operations.

2.9.2. Banking Technology Models of Core Banking

The trends world-over are towards adoption of core banking system as the new IT system. “There are basically four models:

Model Stage	Model-I	Model-II	Model-III	Model-IV
I	Take Sys and PROs as given	Take CBS as given	Try to map existing Sys and PROs	Identify critical Sys and PROs
II	Develop SW package	Implement CBS	Customize CBS	Do BPR and Customize
III	Implement SW package	Develop Sys and PROs based CBS	Implement CBS	Implement CBS and BPR
IV	No MIS/DSS Inefficient, ad-hoc, outdated IT stand alone solution	Effective IT solution, but useful for start-off banks	Good solution but with limited benefits	Most efficient Solution Justifying ROI in IT

(Table - 2.1 Banking Technology Models)
(Source: Transformation of Indian Banks with Information Technology – Prof. S.M. Padwal & Dr.V.T. Godse)

Model-I: Take the given legacy systems and procedures of banking and try to map them on IT products and services without attempting to undertake business process re-engineering for whatever the reasons such as prevailing Industrial Relations, or for associated cost or for

want of time. The effect of such model has been witnessed by most of them existing banks in Indian banking in 1990s.

Model-II: In this model the management takes core banking solution (CBS) and starts a bank with systems and procedures as dictated by the CBS. The era of starting new technology driven private banks in India is a good illustration of this model. Limitation of this model is, it is good for a new start-off bank where there is no legacy or any system in existence prior to starting of the bank.

Model-III: In this model, the management selects a CBS and tries to map existing systems and procedures on the CBS. The result would be similar but little better than in the first model.

Model-IV: This model makes a departure in basic approach from first three models. In this case CBS is chosen and the systems and procedures are classified as critical (must have), semi-critical (should have), and non-critical (nice to have). For critical systems and procedures the customization is undertaken by the solution provider in consultation with user acceptance test (UAT) in place, the Business Process reengineering exercise is also initiated for others.

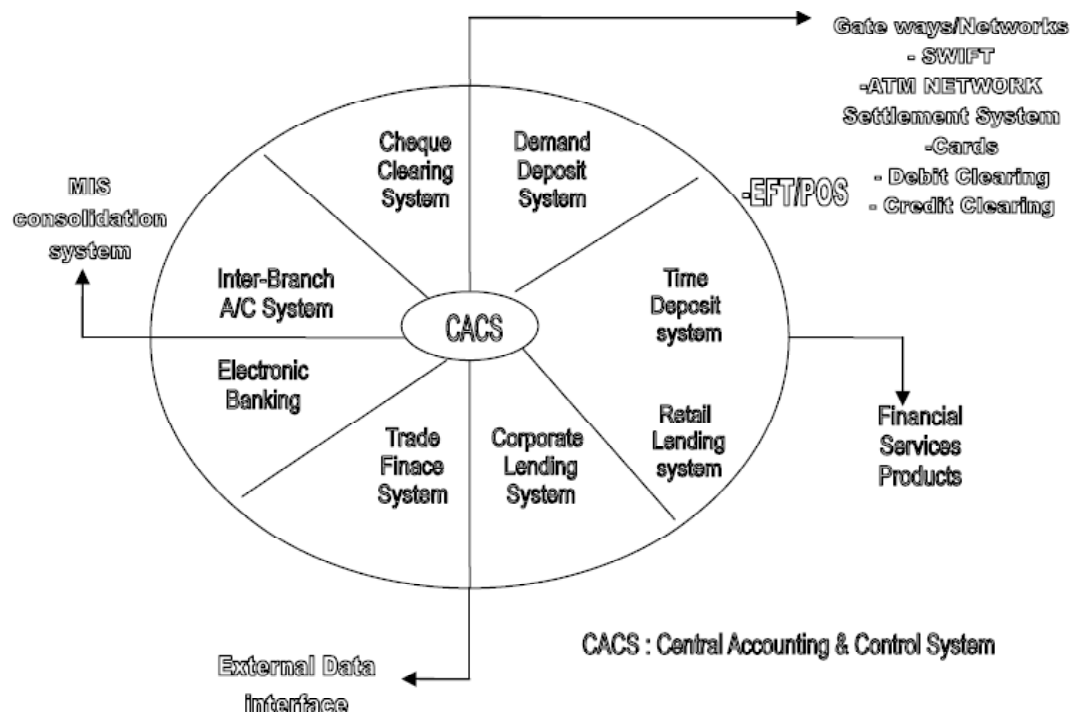
2.9.3 Core Banking System-selection process

In the present scenario, in India, both public and old private sector banks face intense competition and are struggling for customer retention. They have legacy problems of outdated corporate MIS, Outnumbered staff, outstretched branch spread outstanding dues; outclass products and out-of-line customer service. Many of the smaller banks with branch network less than 300 have become target for acquisition by the new private sector banks who want to expand their branch network to meet the social obligations as well as mobilize low cost deposits. In this backdrop, there is urgency on the part of the banks to embrace technology quickly not only to thwart competition but also to stall (their) acquisition itself. Therefore, the availability of centralized core banking solution and the tele-communication infrastructure making the post merger system integration quick and easy for these banks that the getting a little bit adventurous for acquisition by the day.

2.9.4 Steps of selection process

The following steps to acquire an online core banking solution and deploy it effectively:

- Analysis: The first step is to analyse the present competition of the banking sector and also to determine the requirements of the customer. This is the key parameters for selection of core banking solution.
- Perspective plan: Draw the perspective plan to bring online core banking in the main ranches, which contribute to 75% to 80% of banks business.
- Network plan: Draw the optimistic plan to provide core banking, to other remaining branches through proper network.
- Basic solution: Develop the laundry list of requirements for selection of a suitable core banking solution that will have to meet the basic business functionality and network requirements show in fig.



(Figure 2.5: Core banking solution)

(Source: INDIAN INSTITUTE OF BANKING & FINANCE, "Information System for Banks, Taxmann Publication (P.) Ltd)

- **Back-office MIS:** In head office of the banks, the operations are focused on administration and supervision as well as policy making. But, data and information from the branches will have to flow in time to make these operations effective and efficient. Core banking solutions, installed centrally and administered distributed fashion, will have to have these functions supported. The major users of these functions will be top management of the bank and the operating staff in Head Office (HO) who support them.
- **Flexibility:** The solution should be flexible to meet new requirements such as multi-currency, Internet banking, integrated Treasury management, integrated Forex management, anywhere banking etc.
- **Government Banking:** banks in India also handle accounts and financial transactions on behalf Central Government Departments, State Government departments, Public sector undertaking providing utility services. The core banking solution should be flexible to fulfill these requirements.
- **E- channels:** The core banking solution support the network branches and also support the multi delivery channels Like ATM, Debit/Credit cards, telephone banking etc.
- **Response Time:** It should be an industrial strength product capable of handling present and future transaction volumes in networked environment with acceptable response time.
- **Ease of use:** It should be user friendly. That everyone should be able to easily operate and understand.
- **Security:** Banks are custodian of public funds and hence a core banking solution should be secured enough to insure against financial loss caused by frauds, misappropriation and embezzlement. It should have highest-level access controls to safeguard against intrusions and should be well-defined and time-tested and must have the control procedures.

2.9.5 Core Banking Systems- Products and Vendors: There are many vendors and different products are available for core banking solution in India and the same is listed in the table :

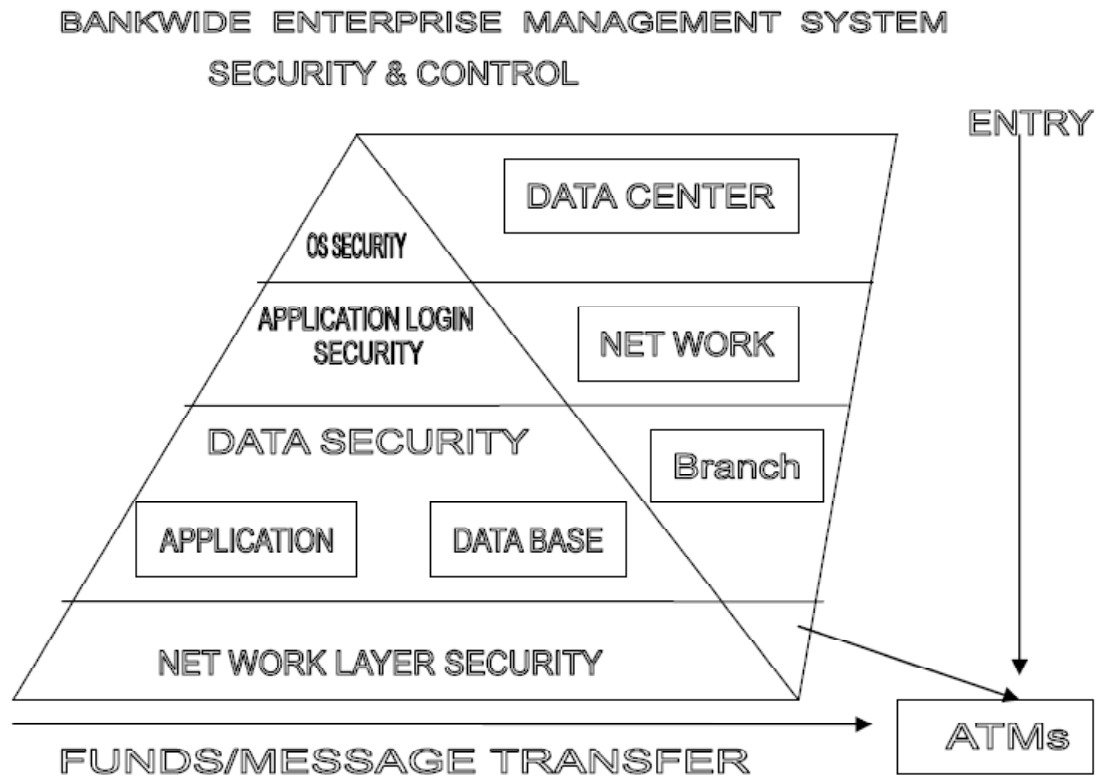
Product Name	Vendor Name	Leading Banking Customer	Largest Transaction volumes/ Branches
Finacle 6.x	Infosys Technologies Ltd.	• ICICI Bank • UTI Bank • IDBI Bank • ABN Amro Bank • Punjab National Bank • Union Bank of India • Tamilnadu Mercantile Bank • Karnataka Bank	Millions of Transactions; Several Hundreds of branches; Thousands of ATMs;
Flexcube	I-Flex solutions Ltd.	• Syndicate Bank • Bharath Overseas bank • Karur Vysya Bank • HDFC Bank	Lakha of Transactions; Hundred of Branches Several Hundred ATMs;
Sanchez	IBM	• Vysya Bank	More than 100 branches
Equation	ACT Kindle	• IndusInd Bank • Centurion Bank	More than 100 branches
FNS	TCS	• SBI	Under implementation
TC/4	TCS/CMC	• United Western Bank	Under implementation
Newton	ICICI Info Tech	• Lakshmi Vilas Bank	Under implementation
Quartz	TCS	• City Union Bank	Under implementation

(Table 2.2: Core Banking Systems- Products and Vendors)

(Source: INDIAN INSTITUTE OF BANKING & FINANCE, “Information System for Banks”, Taxmann Publication (P.) Ltd.)

2.9.6. Core banking system-security & control: There are several dimensions to security and control aspects in a core banking solution environment implemented in wide area network. “The main too many dimensions are:

- **Solution Dimension:** Covering the application, operating system controlling the application, network (both physical as well as software) and the database, which keep the data and information.
- **Topology:** The topology in which the solution is implemented covering the data center, the network connecting the branches and the brick and mortar branches form where the system is accessed and e-channel such as ATMs as shown in Figure 2.6



(Figure 2.6 : Bank wide Enterprise Management System security & Control)

(Source: INDIAN INSTITUTE OF BANKING & FINANCE, “Information System for Banks”, Taxmann Publication (P.) Ltd.)

Therefore, the objective of security and control are presented as below:

The objective of Security is to

- i) Prevent unauthorized users from using system
- ii) Allow authorized users to use the system
- iii) Safeguard assets from intentional as well as accidental damages hardware, software and data. On the other hand the objectives of control are to ensure that, authorized users use the application and adapt for the intent and purpose for which they are designed and made available by the business. When banks go for selection of core banking solution, they have to also concurrently consider installing a security and

control system to administer the security and control policy and procedures. For example:

- appoint a chief security officer (CSO) - He should be an authorized person for to take care the security.
- Appoint Risk Analysis Team – to analyze the risks and develop counter measures
- for implementation
- Develop Safeguard manuals – Administrative safeguard, physical safeguard and technical safeguard.

Summary

Information technology, globalization, and related developments have made business increasingly dependent on high-quality information for decision making. Business decision makers need the relevant and reliable information, which should be available at the right time and at reasonable cost. In earlier days, customers obtained the information through manually data processing. It takes lot of time and with the possibility of error.

Architecture of bank branch computerization

a) General approaches: One of the important aspects of branch automation is to determine the size and type of the computer required at the branch. The computer size or the configuration is specified in terms of memory, disk space, number of terminals, printers etc. All capacity planning is directed towards optimizing throughout during peak loads and maximizing its use during other periods.

Individual Functional Area Data bases: This data base has detailed information about each functional area. They have information which is not required frequently and also it contain master information about the functional area. If the transaction is not validated at common database then the transaction may be required to go through the functional areas for validation.

- **Stand-alone branch-level package:** These are usually in FoxPro, C language or Dbase and are capable of handling specific functions at branches; these are sometimes networked on a LAN to simulate a TBA (Total Branch Automation) environment. AT the same time the high-end packages with a central server (which can be a Pentium PC or NT or a MINI or even a Main Frame, supported by multiple dumb or intelligent) terminals are also used. Some of them use sophisticated RDBM

like ORACLE as back-end and provide user-friendly front-end with windows GUI (graphical user interface).

- **ii) Multi-branch solutions:** These solutions are used to network a cluster of branches in a city; the account maintenance can be central or can be distributed, networking being achieved through Wide Area Network (WAN) or terrestrial lines / high speed lines/ satellite networks – and now even wireless.
- **Foreign Packages:** The Foreign banking software solutions which are commonly used are Bank Master, Kappiti, and Sanchez etc. This need to be suitable modified to make Indian requirements- but their strength in their proven capabilities in developing and offering modern/ global banking products/ services that India is just ushering in.
- **Packages for specialized niche areas:** Like Assets Liability Management (ALM), Treasury Management, Trading / Dealing Room activities, Custodial Services / Depository Participant etc. These are high-end packages with sophisticated analytical and decision tools. v) Service Branch / high-volume transactions processing packages: These solutions include clearing, drafts issue/payments/reconciliation (Remittances), Bills (payments/collection/purchases), Dividend warrant processing, inter-branch reconciliation etc. Mostly the service provides themselves implement these packages.
- **IT Services:** These are the sundry services provided by the software dealers. They include services like disaster recovery, virus protection, security handling /networking multiple legacy system between themselves of two new platforms or to new delivery channels like ATMs etc.

The one of the biggest achievement in this period is development and implantation of “Core banking Solution (System)”. In the Core banking, all the data base of a particular bank is centralized and a bank can access any information and customer can get the services of that bank in any of its branch. It is a combination of Management Information system and small network banking system. Core banking is a general term used to describe the services provided by a group of networked bank branches. Bank customers may access their funds and other simple transactions from any of the member branch offices.

Core Banking is normally defined as “the business conducted by a banking institution with its retail and small business customers. Many banks treat the retail customers as their core

banking customers, and have a separate line of business to manage small businesses. Larger businesses are managed via the Corporate banking division of the institution. Core banking basically is depositing and lending of money.”

In the present scenario, in India, both public and old private sector banks face intense competition and are struggling for customer retention. They have legacy problems of outdated corporate MIS, Outnumbered staff, outstretched branch spread outstanding dues; outclass products and out-of-line customer service.

Check your Answers

1. Explain the need for MIS in banking?
2. Explain in detail architecture of bank branch computerized.
3. Describe LAN Model with a neat diagram.
4. What is CBS ?

Lesson - 3

Banking Regulations and Basel Accord

Objectives

After reading this lesson you will be able to understand the following:

- Need for Banking Regulations
- Basel Committee
- Basel Capital Accord (BASEL I) - Salient features and short comings
- Basel II Accord
 - o Three Pillar Concepts
 - o Core Capital & Supplementary Capital
- Risks and categories of risk
 - o Credit Risk
 - o Market Risk
 - o Operational Risk
 - o Capital Adequacy Risk
 - o Risk Adjusted Asset
 - o Risk Weighted Asset

Structure

- 3.1 Need for Banking Regulations**
- 3.2 Basel Committee**
- 3.3 Risks and categories of Risks.**

3.1 Need for banking regulations

The banking regulations act 1949 is a legislation in India. The Act provides a frame work using which Banking in India is supervised and regulated. In 1965 the act was amended in 1965 the act was amended to include cooperative banks. It gives power to the RBI to regulate the operations of Banks as such.

3.2 Basel Committee

A committee of central banks of G10 countries was formed to stabilize the international banking system and regulate them as well. This is called as Basel committee on banking supervision (BCBS). The Basel committee published a set of minimum capital requirements for banks and this is called as 1988 Basel accord, which was enforced by law in G10 Nations in 1992.

Basel 1: The main objective of the Basel accord was to reinforce the capital base of the world's major banks, which were being eroded due to severe competition among the banks. This is also recognized as Basel 1.

as per the 1988 Accord, the Banks were advised to maintain capital equal to a minimum 8% of a basket of assets measured based on this risk. Banks were advised to maintain two tiers of capital viz:

Tier 1 covering additional internal and external resources available to the bank.

The twin objective of Basel were:

- 1) to ensure an adequate level of capital in the international banking system
- 2) to create more level of playing field in the competitive environment

Shortcomings of Basel 1

- 1) limited differentiation of credit risk
- 2) static measure of default risk
- 3) No recognition of term - structure of credit risk

Basel II

In January 2001, the Basel committee on banking supervision issued a new proposal for a Basel capital Accord to replace the 1988 accord. The new Basel capital accord focused on three pillars namely:

1) Pillar I - Minimum Capital Requirement

2) Pillar II - Supervisory Review

3) Pillar III - Market Discipline

1) Pillar I - Minimum Capital Requirement

The committee on Banking supervision recommended the target standard ratio of capital to Risk weighted assets should be at least 8% (of which the core capital element would be at least 4%) The minimum capital adequacy ratio of 8% was prescribed taking into account the credit risk

- > however in India the RBI has prescribed the minimum capital adequacy ratio of 9% of weighted risks

Pillar II

The supervisory review should be carried in the following manner

- > Banks should have a process for assessing their overall capital adequacy
- > Supervisors should review banks assessments
- > Banks are expected to operate above minimum
- > Supervisors intervention if capital is not sufficient

Pillar III

Market Discipline

- > Role of the market in evaluating the adequacy of bank capital
- > Streamlined catalogue of disclosure requirements
- > close coordination with international accounting standards
- > in principle, disclosure of data on semi annual basis

Core capital and supplementary capital

Under Basel accord a bank has to maintain a certain level of cash or liquid assets as a ratio of its weighted assets. Tier I capital is banks core capital, where as tier 2 capital is a banks supplementary capital. A banks capital is calculated by adding its tier 1 and tier 2 capital together.

Basel III

Basel III is an extension of Basel II framework and introduces new capital and liquidity standards to strengthen the regulation, supervision and risk management of the whole of the banking and finance sector. It was agreed upon by the members of the Basel committee on banking supervision in 2010 - 2011 and was scheduled to be introduced from 2013 until 2015. The changes made from 2013 April extended implementation up to 31st march 2018. The Basel 111 requirements were in response to the deficiencies in financial regulation that is revaluated by the financial crisis seen during 2000. Basel III was intended to strengthen the capital requirements of banks by increasing the banks liability and decreasing bank leverages.

The global capital framework and new capital buffers require financial institutions to hold more capital and higher quality of capital than under the Basel II rules. The new liquidity ratio ensure the adequate funding is maintained in case there are other severe banking crisis.

Basel III has the following three pillars

- 1) Pillar I : minimum capital requirements
- 2) Pillar II : supervisory review process
- 3) Pillar III: Enhanced disclosure(Discipline of market)

Basel III rule introduced the following measures

A) Capital requirements

i) Capital conservation buffer is designed to absorb losses during periods of financial and economical stress. Financial institutions will be required to hold a capital conservation buffer of 2.5% to withstand future period of stress, bringing the total common equity requirement to 7% (4.5% common requirement and 2.5% capital conservation buffer). The capital conservation buffer must be met exclusively with common equity. Financial institutions that do not maintain the capital conservation buffer faces restrictions on payouts of dividends, share buy backs, share buy backs and bonuses.

Counter cyclic capital buffer

It is counter cyclical buffer within a range of 0% to 2.5% of common equity or other fully loss absorbing capital is implemented according to national circumstances. The buffer serves as an extension to the capital conservation buffer.

-> higher common equity Tier I (CETI)

constitutes an increase an increase from 2% to 4.5%

The ratio is set at:

-> 3.5% from Jan 2013

-> 4% from 1 Jan 2014

-> 4.5% from 1 Jan 2015

-> Minimum Total capital ratio: remains at 8%. The addition of the capital conservation buffer increases the total amount of the capital, a financial institution must hold 10.5% of risk weighted assets of which or which 8.5% must be tier 1 capital. Tier 2 capital instruments are harmonized and tier 3 capital is abolished.

3) Leverage Ratio

Basel III introduced a minimum "leverage ratio". The leverage ratio was calculate by dividing tier 1 capital by the banks were expected to maintain a leverage ratio in excess of 3% under Basel III. In July 2013, the US Federal Reserve announced that minimum Basel III leverage ratio would be of 6% for 8SIFI Leverages ratio would be 6% for *SIFI Banks and 5% for their holding companies.

C) Liquidity requirements

Basel III introduced two required liquidity ratios.

-> Liquidity coverage ratio (LCR): ensures that sufficient levels of high quality liquid assets are available for one month survival in a severe stress scenarios.

-> Net stable funding Ratio (NSFR): Promotes resilience over long term time horizons by creating more incentives for financial institutions to fund their activities with more stable sources of funding on an ongoing structural basis.

D) Changes to counter party credit risk (CCR)

Basel III introduced capital requirements to cover credit value adjustments (CVR) risk and higher capital requirements for securitization products.

3.3 Risks and categories of Risks

Banks do face complex environment from which array of risks that came on account of new type of players, new technologies, ever growing complexities of national and international regulations ever changing consumer behaviour and the like require measures to address the financial and other related risks naturally occurring as a result of these changes.

Risk is an uncertainty of an event to occur. So much so in banking context risk is the exposure to the uncertainty of an outcome where exposure could be defined as the position a bank takes in the market.

a) Credit risk

The Basel committee on banking supervision defines credit risk as the potential that a bank borrower or counter party will fail to meet its payment obligations regarding the terms agreed with the bank. It includes both uncertainty involved in repayment of the bank's dues and repayment of dues on time.

Credit risk could occur owing to

- > Inadequate income of borrowers
- > Inadequate underwriting frameworks
- > Business failure of the borrowers
- > The unwillingness of the borrowers to repay

b) Market risk

Basel Committee on banking supervision defines market risk as the risks of losses in on -or - off balance sheet positions that arise from movement in market prices

Important components of market risks are:

- > Interest risks: potential loss due to change in interest rates
- > It requires banking asset/liability management

c) Equity risks: Potential losses due to change in stock prices as banks accept equity against disbursing loans

d) Commodity risk: Potential losses due to change in commodity (agricultural industrial, energy, prices)

Massive fluctuations occur in these prices due to continuous variations in demand and supply. Banks may hold them as part of their investment and hence face losses.

d) foreign exchange risk: potential loss due to change in the value of the banks assets or liabilities resulting from exchange rate fluctuations as banks transact with their customer/ other stake holders in multiple currencies.

e) Operational Risk

Basel committee on banking supervision defines operational risk as the risk of people and systems or external events. All their departments including treasury, credit, investment, information technology.

> Operational risks occurs due to

a) Human intervention

b) Human error

c) Failure of information technology

d) Failure of external software

e) Failure of internal processes to transmit data and information accurately

Banking Regulations and its Needs

Bank regulation is a form of government regulation which subjects banks to certain requirements, restrictions and guidelines, designed to create market transparency between banking institutions and the individuals and corporations with whom they conduct business, among other things.

Needs of Banking Regulations:

- Prudential Purpose: to reduce the level of risk to which bank creditors are exposed (i.e. to protect depositors)
- for systemic risk reduction: to reduce the risk of disruption resulting from adverse trading conditions for banks causing multiple or major bank failures

- to avoid misuse of banks: to reduce the risk of banks being used for criminal purposes, e.g. laundering the proceeds of crime
- to protect banking confidentiality
- Credit allocation: to direct credit to favoured sector.

Basel Norms:

Basel is a city in Switzerland which is also the headquarters of Bureau of International Settlement (BIS).

- o BIS fosters co-operation among central banks with a common goal of financial stability and common standards of banking regulations.
- o The Bank for International Settlements (BIS) established on 17 May 1930, is the world's oldest international financial organisation. There are two representative offices in the Hong Kong and in Mexico City. In total BIS has 60 member countries from all over the world and covers approx. 95% of the world GDP.
- o The set of the agreement by the BCBS (BASEL COMMITTEE ON BANKING SUPERVISION), which mainly focuses on risks to banks and the financial system are called Basel accord.
- o The purpose of the accord is to ensure that financial institutions have enough capital on account to meet the obligations and absorb unexpected losses.
- o India has accepted Basel accords for the banking system.
- o "BASEL ACCORD" has given us three BASEL NORMS which are BASEL 1, 2 and 3.

1. BASEL I:

- In 1988, The Basel Committee on Banking Supervision (BCBS) introduced capital measurement system called Basel capital accord, also called as Basel 1.
- It focused almost entirely on credit risk, it defined capital and structure of risk weights for banks.
- The minimum capital requirement was fixed at 8% of risk-weighted assets (RWA).
- India adopted Basel I in the year 1999.

2. BASEL II:

In 2004, Basel II guidelines were published by BCBS, which were considered to be the refined and reformed versions of Basel I accord. The guidelines were based on three parameters which are as follows

- Banks should maintain a minimum capital adequacy requirement of 8% of risk assets.
- Banks were needed to develop and use better risk management techniques in monitoring and managing all the three types of risks that is credit and increased disclosure requirements.
- The three types of risk are- operational risk, market risk, capital risk.
- Banks need to mandatory disclose their risk exposure, etc. to the central bank.
- Basel II norms in India and overseas are yet to be fully implemented.

THREE PILLARS OF BASEL NORM II:

1. Pillar I - Minimum Capital Requirements
2. Pillar II - Supervisory Review Process
3. Pillar III - Market Discipline

3. BASEL III:

- In 2010, Basel III guidelines were released. These guidelines were introduced in response to the financial crisis of 2008.
- In 2008, Lehman Brothers collapsed in September 2008, the need for a fundamental strengthening of the Basel II framework had become apparent.
- Basel III norms aim at making most banking activities such as their trading book activities more capital-intensive.
- The guidelines aim to promote a more resilient banking system by focusing on four vital banking parameters viz. capital, leverage, funding and liquidity.
- Presently Indian banking system follows Basel II norms.

The Reserve Bank of India has extended the timeline for full implementation of the Basel III capital regulations by a year to March 31, 2019.

RISK & CATEGORIES OF RISK:

i) Credit Risk:

Credit risk is the probable risk of loss resulting from a borrower's failure to repay a loan or meet contractual obligations. Traditionally, it refers to the risk that a lender may not receive the owed principal and interest, which results in an interruption of cash flows and increased costs for collection.

Although it is impossible to know exactly who will default on obligations, properly assessing and managing credit risk can lessen the severity of loss. Interest payments from the borrower or issuer of a debt obligation are a lender's or investor's reward for assuming credit risk

ii)Market Risk:

Market risk is the possibility of an investor experiencing losses due to factors that affect the overall performance of the financial markets in which he or she is involved. Market risk, also called "systematic risk," cannot be eliminated through diversification, though it can be hedged against. Sources of market risk include recessions, political turmoil, and changes in interest rates, natural disasters and terrorist attacks.

iii) Operational Risk:

Operational risk summarizes the risks a company undertakes when it attempts to operate within a given field or industry. Operational risk is the risk not inherent in financial, systematic or market-wide risk. It is the risk remaining after determining financing and systematic risk, and includes risks resulting from breakdowns in internal procedures, people and systems

iv) CAR:

The capital adequacy ratio (CAR) is a measurement of a bank's available capital expressed as a percentage of a bank's risk-weighted credit exposures. The capital adequacy ratio, also known as capital-to-risk weighted assets ratio (CRAR), is used to protect depositors and promote the stability and efficiency of financial systems around the world.

Two types of capital are measured: tier-1 capital, which can absorb losses without a bank being required to cease trading, and tier-2 capital, which can absorb losses in the event of a winding-up and so provides a lesser degree of protection to depositors.

v) Risk-Weighted Assets:

Risk-weighted assets are used to determine the minimum amount of capital that must be held by banks and other institutions to reduce the risk of insolvency. The capital requirement is based on a risk assessment for each type of bank asset. For example, a loan that is secured by a letter of credit is considered to be riskier and requires more capital than a mortgage loan that is secured with collateral.

vi) Risk-Adjusted Asset:

A risk-adjusted capital ratio is a financial institution's total adjusted capital divided by its risk-weighted assets (RWA). There are variations of risk-adjusted capital ratios whose use is dependent on how analysts define capital. Risk-adjusted capital ratios assess the adequate capital needs of a financial institution, and its ability to withstand a downturn in the economy.

Summary

A committee of central banks of G10 countries was formed to stabilize the international banking system and regulate them as well. This is called as Basel committee on banking supervision (BCBS). The Basel committee published a set of minimum capital requirements for banks and this is called as 1988 Basel accord, which was enforced by law in G10 Nations in 1992.

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-> Liquidity coverage ratio (LCR): ensures that sufficient levels of high quality liquid assets are available for one month survival in a severe stress scenarios.

-> Net stable funding Ratio (NSFR): Promotes resilience over long term time horizons by creating more incentives for financial institutions to fund their activities with more stable sources of funding on an ongoing structural basis.

The Basel committee on banking supervision defines credit risk as the potential tha a bank borrower or counter party will fail to meet its payment obligations regardig the terms agreed with the bank. It includes both uncertainty involved in repayment of the banks dues and repayment of dues on time.

Basel Committee on banking supervision defines market risk as the risks of losses in an on -or - off balance sheet positions that arise from movement in market prices

Basel committee on banking supervision defines operational risk as the risk of people and systems or external events. All their departments including treasury, credit, investment, information technology.

Bank regulation is a form of government regulationwhich subjects banks to certain requirements, restrictions and guidelines, designed to create market transparency between banking institutions and the individuals and corporations with whom they conduct business, among other things.

Check your Answers

1. Explain the following:
 - a) Basel Committee
 - b) Need for banking Regulations
 - c) Credit Risk
 - d) Risk weighted asset
 - e) Liquidity requirement

Lesson - 4

Frauds in BFSI

Objectives

- Definition of a fraud in banking and financial sector
- Categories of fraud
 - o Account related fraud
 - o Cheque related fraud
 - o Advance related fraud
 - o Alternate channel frauds
 - § Internet Banking related - Social Engineering, Phishing tactics
 - § Some common frauds with ATMs
 - o Frauds by banking employees
 - o Account related frauds - Input & output manipulation
 - o User rights escalation methods
- Reporting of frauds
- Frauds in insurance sector
 - o Policy holder & claims fraud
 - o Intermediary fraud
 - o Assessment related frauds
- IRDA – Introduction, functions and responsibilities

Banks are an essential part of the Indian economy.

- While the primary responsibility for preventing frauds lies with banks themselves.
- Banks dealing with public's money: due care and diligence
- The RBI advisory to banks for prevention of frauds

Structures

- 4 frauds Definitions**
 - 4.1 Categories of Banking Frauds**
 - 4.2 Insurance Frauds.**
 - 4.3 The Insurance Regulatory and Development Authority of India (IRDAI)**
 - 4.4 Cyber Frauds / crime and Fraud Management.**
 - 4.5 Online Transactions.**
 - 4.6 E-Commerce.**
 - 4.7 Funds Transfer.**
 - 4.8 Computer Insecurity.**
 - 4.9 User failures and causes.**
 - 4.10 Vulnerability in Banking Environment**
 - 4.11 Case studies**

4. Fraud Definition

Fraud can loosely be defined as “any behavior by which one person intends to gain a dishonest advantage over another “fraud, under section 17 of the Indian contract act, 1872,

RBI has defined the term “fraud” in its guidelines on frauds which reads as under.

“A deliberate act of omission or commission by any person, carried out in the course of a banking transaction or in the books of accounts maintained manually or under computer system in banks, resulting into wrongful gain to any person for a temporary period or otherwise, with or without any monetary loss to the bank”.

4.1 Categories of Banking Frauds

- § Account opening fraud: this involves a deposit and cashing of fraudulent cheques.
- § Cheque kiting: is a method where by a depositor utilizes the time required for cheques to clear to obtain an unauthorized loan without any interest charge.

- § Cheque fraud: most common cases of this kind of fraud are through stolen cheques and forged signatures.
- § Counterfeit securities: documents, securities, bonds and certificate could be forged, duplicated, adjusted or altered and presented for loan collection.
- § Computer fraud: hacking, tampering with a diskette to gain access to unauthorized areas and give credit to an account for which the funds were not originally intended.
- § Loan fraud: when funds are lent to a non-borrowing customer or a borrowing customer that has exceeded his credit limit.
- § Money laundering fraud: this is a means to conceal the existence, source or use of illegal obtained money by converting the cash into untraceable transactions in banks.
- § Letters of Credit: Most common in international trading, these are instruments used across borders and can be forged, altered, adjusted and take longer to identify.
- § Advanced Fees Fraud: Popularly known as „419“, advanced fees fraud may involve agent with an offer of a business proposition which would lead to access often for a long term.

4.2 Insurance Fraud

Insurance fraud is an illegal act on the part of either the buyer or seller of an insurance contract. Insurance fraud from the issuer (seller) includes selling policies from non-existent companies, failing to submit premiums and churning policies to create more commissions.

4.2 Categories of Insurance Fraud

- § Minor Car Damage Scheme: The most common type of auto insurance fraud is when an individual reports minor damages to their vehicle, collects the money, and then never has any work performed on the car.
- § Engineered Home Fires: A homeowner will remove all priceless items from the home (like photo albums, collections, etc.) and then stage a home fire in order to collect some quick cash.
- § Reporting False Thefts: In most cases of theft fraud, the owner exaggerates as to the value of the items that were stolen. However, some people go as far as to report stolen items that were not really stolen.

- § Fraudulent Stolen Car Claims: Faking a stolen car is a popular form of fraud that can be carried out in a number of different ways. Some individuals will simply dump the car somewhere and hope that it's never found. Others go as far as to sell the car to a chop shop and have it disassembled and sold for parts.
- § Car Accident Scam: An individual uses a current accident to claim damage that happened in the past. Sometimes several people will conspire to stage an accident just to get a claim.
- § Natural Disaster Damage Fraud: In the wake of a natural disaster like a major storm, tornado, or earthquake, insurance companies get overrun with claims. Some people will take advantage of this and enhance the damage to their property in order to get a bigger payday.
- § Health Insurance Fraud: There are times when a health care provider will bill an insurance company extremely high fees for low cost procedures.
- § Unnecessary Medical Procedures: Doctors will sometimes force patients to go through unnecessary testing just to increase the amount they can bill an insurance company.
- § Renter's Insurance Fraud: Renter's often take out a renter's insurance policy because it's so cheap. When they decide to move, they will sell their stuff and then report it stolen.
- § Faked Death: Criminals sometimes take out life insurance claims and then fake their own death. They need an accomplice to carry out this type of insurance fraud

4.3 The Insurance Regulatory and Development Authority of India (IRDAI)

The Insurance Regulatory and Development Authority of India (IRDAI) is an autonomous, statutory body tasked with regulating and promoting the insurance and re-insurance industries in India. It was constituted by the Insurance Regulatory and Development Authority Act, 1999, an Act of Parliament passed by the Government of India. The agency's headquarters are in Hyderabad, Telungana, where it moved from Delhi in 2001.

The functions of the IRDAI are defined in Section 14 of the IRDAI Act, 1999, and include:

- § Issuing, renewing, modifying, withdrawing, suspending or cancelling registrations
- § Protecting policyholder interests
- § Specifying qualifications, the code of conduct and training for intermediaries and agents
- § Specifying the code of conduct for surveyors and loss assessors
- § Promoting efficiency in the conduct of insurance businesses
- § Promoting and regulating professional organisations connected with the insurance and re-insurance industry
- § Levying fees and other charges
- § Inspecting and investigating insurers, intermediaries and other relevant organisations
- § Regulating rates, advantages, terms and conditions which may be offered by insurers not covered by the Tariff Advisory Committee under section 64U of the Insurance Act, 1938 (4 of 1938)
- § Specifying how books should be kept
- § Regulating company investment of funds
- § Regulating a margin of solvency
- § Adjudicating disputes between insurers and intermediaries or insurance intermediaries
- § Supervising the Tariff Advisory Committee
- § Specifying the percentage of premium income to finance schemes for promoting and regulating professional organisations
- § Specifying the percentage of life- and general-insurance business undertaken in the rural or social sector. Specifying the form and the manner in which books of accounts shall be maintained, and statement of accounts shall be rendered by insurers and other insurer intermediaries.

4.4 Cyber Frauds/ Crime and Fraud Management

This section deals at length on the growth of e-commerce on account of the enhanced technological advancements. The various types of on line transactions and their inherent characteristics open up avenues for different type of Cyber Crimes.

The last decade has seen a phenomenal rise in the usage of World Wide Web and the Internet. This has led to the evolution of a concept called e-Commerce, which has broken down all physical and political barriers with regard to banking, shopping and other payment systems. Today, businesses are being increasingly transacted through the Internet E-Commerce has come to establish itself with businesses shifting from the real time market to the digital market.

This shift has been continuous over the past few years and has now become a part of online.

4.5 Online Transactions

Online transaction can be broadly classified under the heads:

1. E-Commerce Transaction
2. Banking Transactions

Example of e-commerce transactions:

- To buy books, we need not drive to the nearest shop. We can check the Internet for their rates, reviews and purchase
- To buy property, the best option would be to check for such dealers and their properties on the net and form and idea of the option available.

Examples fo banking transactions:

- To transfer funds, a physical visit to the bank is not necessary. We can transfer them electronically through facilities that the bank offers.
- To pay our utility bills, we do not anymore stand in queues. They get paid automatically on a simple instruction to the bank that the same be debited to our accounts

4.5.1 What are Online Transactions?

The term online” became popular in the late 1980s. Initially, it referred to the use of a computer with an access to a banking system. Of late, the use of this term has widened,

including the use of Internet for all kinds of transactions like banking, purchasing, investments and insurance.

Online, a user can now perform a complete business transaction within a few seconds, facilitated by an interaction within computers. Since, these include critical operations and address urgent requirements of a consumer; high levels of performance, integrity and data security are required, apart from continuous availability of information.

Today, businesses cannot function without having to rely on data. And, because online transactions are on the rise, processing such data requires specific software supports, which are widely spread over a 'large network, including outside national boundaries. Such online transactions are processed through Online Transaction Processing Systems (OLTPs), examples of which include the Customer Information Control System (CICS).

4.5.2 Process of Making an Online Transaction

During an online transaction, for example, a merchant payment system is directly connected to the bank or the credit card issuer of the consumer. It provides the user access to a large amount of data pertaining to the merchant. Usually, the interaction between the user and the merchant is very brief. Most transaction are executed within short periods.

An online transaction processing system collects, modifies and stores all transactions that a user executes during his interaction with an organisation. The main requirements of OLTPs are:

- Availability of transaction interface to the user at all times
- Quick response
- Data security and integrity

To facilitate processing of large updates to a database in an OLTP, sophisticated transaction management software are used.

4.5.3 Features of an Online Transaction

A system has to pass an ACID test for entry into a transaction processing system. This test holds good for an OLTP as well. "ACID" here is an acronym for the four properties of a transaction processing system. Their details are:

1. Atomicity

If any single operation fails, then the whole operation fails. For example, in electronic payments, the system has to withdraw from one account and credit it to another. The transaction cannot be deemed complete if only one of the actions has taken place. Either both must occur or neither. Any partially executed transaction must be rolled back by the OLTP system.

2. Consistency

The essence of an OLTP is that changes in the data before and after a transaction should be consistent. The actions taking part in a transaction should not violate any integrity constraints associated with the state.

3. Isolation

An OLTP has multiple users accessing the database simultaneously. It should ensure that nobody is able to manipulate the database. Neither should a user be able to change a piece of data before another is done with it. For example, in an airline ticketing system, if one user is reserving the last available seat in a flight, another user who is using the system simultaneously should not get an impression that the seat is available for booking. In short, concurrent transactions should not overlap each other.

4. Durability

A completed transaction should always be on record regardless of any failures within the system network, like network unavailability or crash of a hard disk.

4.6 E-Commerce

4.6.1 What is E-Commerce?

Online transactions include e-commerce. It is a process in which an online shop or website is used by the purchaser to select the items he/she wants to purchase, before he/she places an order. The seller then fulfils this order. The sale and payment for such goods and services may

be completed either online or any offline method like “Cash on Delivery”. For example, if the online order is for the purchase of a software or an e-book, the seller of the software can give access to the purchaser who can then download the same online. The payment for such transactions is completed online. If the purchase is for physical goods, such as purchase of flowers and gifts, the delivery takes place offline. Payment for such physical items may be either online using credit/debit cards or the online banking system. At times, payments for this kind of products can also be made through the “Cash on Deliver” method.

Electronic commerce that takes place between businesses is referred to as business-to-business or B2B. B2B can be open to all interested parties, for example in case of a commodity exchange. Or, it can be limited to specific, pre-qualified participants, referred to as a private electronic market. Electronic commerce that takes place between businesses and consumers, on the other hand, is referred to as business-to-consumer or B2C.

Payment through credit cards has become the most common form of payment for purchase of goods and services. An online supplier/retailer cannot survive without a support system for credit cards.

The service that automates the payment for transactions between the shoppers and merchants is called a ‘Payment Gateway’. For all online B2B and B2C transactions, payment gateways play a very important role.

4.6.2. Payment Gateways

In e-commerce, the service that automates the payment transaction between the shopper and the merchant is called the ‘payment Gateway’. It is a third-party service that verifies, accepts or declines credit card transactions on behalf of the merchants. It is equivalent to a physical point-of sale terminal available in retail outlets.

Payment gateways ensure that information is passed securely between the customer and the merchant and also between merchant and the payment processor. This is done by way of encrypting sensitive information, such as credit card numbers, and transmitting them through a secure Internet connection.

Transcure Pvt.Ltd., Pay Seal (ICICI), Elitral payment Gateway payment Gateway

India .com, Team VII Payment Gateway, Citibank and HDFC are some of the payment gateways in India

Here diagrams

When a customer orders a product online from a payment gateway-enabled merchant. There are three vital things that the gateway does:

1 Authorisation: It check whether the credit/debit card is acceptable. It affords a secure link between the customer, merchant and credit card issuer. The credit card issuer authorises the purchase through the payment gateway.

2 Settling: At the end of the day, the payment gateway groups all transactions of a merchant and sends them to the bank in a single batch. The bank settles the merchant. This process is called settling.

3 Reporting: It records all transactions, which can be viewed using the payment gateway report facilities.

The steps involved in an online purchase transaction are:

1. Customer selects the goods or services that he/she has to buy online. On clicking the "Buy "button, a message is sent to the website.
2. The Web Store's server sends over a secured network a "Digital Order "by adding its digital certificate, the consumers IP address and the transaction amount to the payment Gateway. This data is encrypted before transmission to ensure its security.
3. Based on the digital certificate, the Payment Gateway authenticates the Web Store.
4. The buyer is presented with several payment options on the screen.
5. The chosen payment option is transmitted to the payment Gateway securely.
6. The Payment Gateway sends the details to the acquiring bank, which, in turn, sends the information to the issuing bank.
7. The issuing bank either accepts or rejects the transaction based on the credit limit and the payment instructions 'validity.
8. The acquiring bank sends the confirmation to the Payment Gateway.
9. The Payment Gateway sends digital receipts to the shopping site and the buyer.
10. The Web Store ships the goods to the buyer.

4.7 Online Banking Systems

Online transactions also include online banking. The days of walking to the bank branch every time you want to withdraw cash or make a transaction are gone. Banking can now be completed from within the comfort of your home. All banking services are available at the click of a button. All one needs is a computer terminal with an Internet connection.

Online banking includes the following services:

4.7.1 Funds Transfer

Funds can be transferred online regardless of geographical locations. Money is electronically transferred online from one bank/financial institution to another based on the instructions given by a user. Most common examples of fund transfers include direct credit of salaries to employees, wire transfers and so on.

4.7.2 Bill Payment

All bill payments including those of utilities like electricity and telephones can be paid online through electronic banking systems. This is subject to the respective service providers making arrangements for collecting such payments using the online channels. The user has to have a fair idea of his regular billers and their billing cycle. Online registration of such billers along with appropriate instructions as per the bank's requirements can take care of such bills.

4.7.3 Debit/Credit Card Payments

Internet banking also allows payments to credit/debit cardholders through a simple instruction or request by the user.

4.7.4 Ticket Booking

Booking of airline and train tickets as also cinema ticket can be done through online banking. On confirmation of such a transaction, the amount gets debited into the bank account and payment is made to the concerned organisation.

4.7.4 Financial Planning

Online banking also has options to help a user integrate his financial investments. Investing in the stock market and other investment vehicles like derivatives and mutual funds can now be

done directly through banks.

4.7.5 Online Loans

Nowadays, loans from banks are also available online. Features like applications, sanction, calculation and payment of EMI can be done using e-banking solutions.

4.7.6 Real-Time Gross Settlement System

Real – Time Gross settlement (RTGS) plays a very important role in online funds transfer between banks.

RTGS is a fund transfer system, where settlement between banks takes place on a “real time” and “gross” basis. “Real time” means payment is not subject to any waiting period. “Gross settlement” is processing of instructions on a one-to-one basis and there is no bunching or netting with any other transaction. Once processed, the payments are final and irrevocable.

4.8 Computer Insecurity

The way things have gone on, nobody would raise any objection if 2011 is to be observed as the ‘Year of The Breaches’. Innumerable cyber attacks and hackings have taken place across organisations during year. Several websites were hacked including Sony’s picture website in Canada, Sega’s gaming site this bearer home the website of Indian Railways. There was data theft in Google. The list is endless. Anonymous and LulZSec were the notable hackers on the news. The 2011 attackers were not from outside organisations alone; culprits were there within too. Lockheed Martin was a victim of a “significant and tenacious” cyber attack on its computer systems. This was most likely related to the security flaw in RSA Secure ID tokens, used for two-factor authentication by some of the Lockheed employees.

These numerous attacks have only proved that many of the computer systems and websites of several organizations are insecure giving ways to perpetrators. Nowadays, top managements emphasize on highly secure websites and systems, particularly because considerable business is being done online. The role of top management should also include defending cyber attacks. They need to be on their tiptoes to counter any cyber attacks that may occur at anytime from anywhere.

4.8.1 Types of Computer Attacks

In this Internet era, all businesses are excessively dependent on computers. All the data relating to the customers are stored in computers. These data are the assets of the subject organizations. The non availability or loss of such data at any given point of time would lead to irreparable loss to the organization way of legal or reputational consequences. Hence data security becomes the top agenda. Their focus is on protection of database from destructive forces and the unauthorized S of the users. Here comes the need for computer security.

Computer security refers to ensuring of appropriate levels of assurance of the s with regard to data or information stored in the computers.

The general definition of computer security refers to ensuring of appropriate levels of assurance of the following elements with regard to data or information stored in the computer.

- Confidentiality
- Availability
- Integrity
- Auditability
- Non-repudiability

However, there is no computer system, which does not have a loophole. Everyday, new vulnerabilities arise followed by attacks. As per the CVE site, computer systems have more than 5000 vulnerabilities and the list keeps increasing.

Therefore information security management system should address the different may creep into the system and take appropriate countermeasures to prevent or der Breach in security is interchangeably termed as insecurity/vulnerability/exposure of 1 and the like. Security failure is a costly affair as it results in:

- Financial/business loss
- Cost of recovering from the attack
- Cost of fixing and securing the system again
- Cost of measures to be adopted to avoid further failure

The following table gives the different classification of computer attacks.

4.8.2 Type of Attack and Description

Internal Attacks - These are done normally by insiders, such as disgruntled employees to employees or employees doing it for financial gain.

External attacks - These are done by different types of hackers, such as Black hat hackers and Hactivists.

Active attack - An active attack is one, when it attempts to alter system resources or affect their operations. Due to active attack, integrity or availability of information assets can be compromised.

Passive attack - A passive attack is an attempt to extract information from a system without affecting the system resources. In other words, such an attack compromises the confidentiality of information assets

4.8.3 Meaning of Computer Vulnerability

According to NIST 800-30, vulnerability is a flaw or weakness in system security procedures, design, implementation or internal controls that could be exercised (accidentally triggered or intentionally exploited) to result in a security breach or violation of the system's security policy.

According to ISO 27005 defines vulnerability as a "weakness of an asset or a group of assets that can be exploited by one or more attempts. These assets can be anything that can add value to the organisation, its business operations and their continuity, including information resources that support the organization's mission."

Vulnerability is a flaw that can exist in any aspect of a system, not necessarily from only the perspective. It can be at the policy level or in a set of procedures followed by a system administrator help desk personnel for resetting a password. It can also be an inadequate review of a log. and audit logs not reviewed regularly can lead to vulnerability. For example, a review of audit trail can identify any kind of password attack, like brute force. This can even be tracked since the ser have a number of failed logins within a short period of time.

The ISO/IEC 27005 gives examples of Threats - Exploiting - Vulnerabilities. Given here are a few of the examples from this list.

Hardware

- Examples of Vulnerabilities
- Insufficient maintenance/faulty installation of storage media
- Lack of efficient configuration change control
- Unprotected storage

Examples of Threat

- Breach of information system maintainability
- Error in use
- Theft of media or documents

Software

- Examples of Vulnerabilities
- No or insufficient software testing Lack of documentation
- Lack of documentation
- Unclear or incomplete specification for developer

Examples of Threat

- Abuse of rights
- Error in use
- Software malfunction

Network

- Examples of Vulnerabilities
- Unprotected sensitive traffic
- Insecure network architecture
- Poor joint cabling

Examples of Threat

- Eavesdropping
- Remote spying

- Failure of telecommunication equipment
- Personnel
- Examples of Vulnerabilities
- Lack of security awareness
- Unsupervised work by outside or cleaning staff
- Lack of policies for the correct use of telecommunications media and messaging

Examples of Threat

- Error in use
- Theft of media or documents
- Unauthorized use of equipment

Site

Examples of vulnerabilities

- Inadequate or careless use of physical access control to buildings and rooms
- Lack of physical protection for the building, doors and windows
- Examples of Threat
- Destruction of equipment or media
- Theft of equipment
- Theft of equipment
- Organisation

Examples of Vulnerabilities

- Lack or insufficient Service Level Agreements
- Lack of email usage policy
- Lack of control over off-premises assets
- Examples of Threat
- Breach of information system maintainability
- Error in use
- Theft of equipment

Computer insecurity, vulnerability and attacks should not be viewed from a standalone point of view. They can lead to cyber and Internet crimes and frauds, which can result in huge financial loss, compromise of security, sometimes with disastrous effects.

4.8.4 Internet Crime and Internet Fraud

Recall our discussion of the various types of cyber crimes. Of them, crimes committed through email or any online Internet media are Internet crimes and frauds.

a) Internet crime

It is defined as any illegal activity involving one or more components of the Internet, such as websites, chatrooms and/or email. Internet crime involves the use of Internet to communicate false or fraudulent information to consumers. Such a crime may include, but is not limited to, advance fee schemes, non-delivery of goods or services, computer hacking or fraudulent employment/business opportunity schemes. In other words, Internet crime is:

“A crime in which the perpetrator develops a scheme using one or more elements of the Internet to deprive a person of property or any interest, estate, or right by a false representation of a matter of whether by providing misleading information or by concealment of information.” For example, money laundering is normally done using fake/benami accounts and transmit an online over different countries and tax havens involving numerous banks. The entire exercise is to ensure that the actual source of funds remains hidden.

b) Internet Fraud

Internet fraud according to a U.S. Justice Department website devoted to the topic, refers to any type of scheme in which one or more Internet elements are employed in order to put forth statements to prospective victims, to conduct fraudulent transactions, or to transmit the proceeds of fraud to financial institutions or to others connected with the scheme.” As pointed out in a report prepared by the National White-Collar Crime Centre and the Federal Bureau of Investigation (FBI) in 2001, major categories of Internet frauds include, but are not limited to, auction or retail fraud, securities fraud, and Identity theft.

Internet fraud is any type of intentional deception that uses the Internet. This includes fraud that occurs in chat rooms, message boards, website and through emails. It occurs in the form of deceitful solicitations and fraudulent transactions.

Difference between Internet Crime and Internet Fraud The difference between Internet crime and Internet fraud is that Internet crime includes all kinds of traditional crimes, from abuse to sabotage to monetary crimes through Internet. Internet fraud is committing a fraud through Internet, which will mostly cause financial or economic loss to an individual or a group of individuals or an organization. It can be stated that Internet Fraud is a subset of Internet Crime.

Types of Internet Frauds and Crimes In Internet crime, Internet is used as a media to perform illegal activities. Internet gives a criminal a wide scope, a public domain to connect with a huge set of people without boundaries and commit the crime.

Let us look at the various types of Internet crimes and frauds. They include:

- Spam (Unsolicited emails)
- Phishing (Email or website spoofing/masquerading)
- Chain letter scam (Hoax mails)
- Purchase scams (Less delivery or non-delivery of the purchase)
- Ponzi or Pyramid schemes (Misleading investors)
- Unethical hacking (Denial of service, password cracking, dumpster diving or salami attack)
- Credit and debit card fraud (Swindle money by gaining access to card details)
- Check fraud (Forgery of the check)

c) Countermeasures

To counter such frauds and crime, certain steps need to be taken. These include

User awareness about:

- Kinds of crimes being committed online
- Not replying to unsolicited mails
- Taking care while browsing or chatting
- Protecting the computer systems (for example, installing anti-virus)
- Online banking and other transactions

- Understanding the importance of digital signatures and certificates
- Protection of computer and network systems
- Restriction on the usage of cyber cafes and equipment's in public places

According to the Federal Bureau Investigation, the most reported offense Internet auction fraud, which consists of 34.5% of complaints. This usually occurs when people advertise merchandise on auction site and then either fail to deliver or deliver an item of lesser value than was described in the auction.

The second most common type of Internet fraud is non-delivery of merchandise or payment, which accounts for about 19 to 25% of all Internet frauds.

Hence, it is inevitable that with Internet crimes rising, there should also be a significant increase in awareness about Internet crimes and simultaneous strengthening of system controls and individual precautions.

4.9 User Failures and Causes

Let's look at the vulnerabilities arising from human failures.

The human element is the most important link when it comes to security. It is humans who create computer systems, write programs and build security systems. Ironically though, humans are also the weakest link when it comes to computer security.

Application passwords and Login IDs and passwords should be kept very confidential by the respective users; else, it might lead to serious threat to security. The users may discuss casually the login IDs and passwords in a social gathering. Or they may write down the log in details in a piece of paper and keep it on their own tables fearing they would forget the same as there may be a number of such user IDs for different applications. In other cases, the users may simplify the passwords which could be easily guessed by others. Such practices give scope for others to get access to their computers and in turn get access to the sensitive information.

This kind of information gathering is called Eavesdropping, which falls under the social engineering technique of cyber crimes.

4.9.1 Causes for Human Failure

Human failures are generally due to:

- Internal employees
- Lack of awareness
- Lack of training
- 4 Errors and omission
- Password sharing
- Social engineering

let us look at each of these in detail,

a) Internal Employees

Under the internal employee category, the following contribute to human failure:

Improper screening of employees at the time of recruitment Unhappy/disgruntled employees and/or former employees

b) Improper Screening

In these employees, contractors or third-party users are hired without proper background and character verification checks. This can result in potential threats to the organization.

For example, a competitor can send his employee to join the firm to get insider information from the organization. Another example could be that a criminal, with intent to commit a financial fraud, joins a bank giving false previous employment details and other personal information.

Countermeasures

The HR should not only have a strong screening policy, it should also properly and strictly implement and monitor this policy.

c) Unhappy/Disgruntled Employee and/or Former Employee Disgruntled present or former employees are a bigger risk than external unethical hackers. Since they have or have had access to information as authorized users, they are well versed with the systems and applications of the organization. News 5th April 2007

Example:**Former Morgan Stanley Employee Arrested on Data Theft Charges**

A former Morgan Stanley client service representative was arrested and charged with stealing proprietary information relating to the brokerage firm's hedge fund clients. Ronald Peteka, 41, and co-conspirators accessed Morgan Stanley's computers and obtained confidential information on the firm's prime brokerage business, including its client rate lists, the U.S. Attorney in Manhattan alleged in a complaint. He is accused of using the information to set up his own financial consulting

Business with another unidentified Morgan Stanley employee. Peteka was arrested on charges of conspiracy, theft and unauthorized computer access in July 2006. He pled guilty to the charges in February magistrate judge set the bail for Peteka at \$35,000 at his hearing in the federal court.

Survey indicates that the main causes for employees becoming disgruntled are poor managers and inadequate pay.

Countermeasures

Performance appraisals should be the only yardstick for promotion and increments. Competent people should rise in the organisation. Policy should be stringent for email content monitoring, removable media access employees and access removal.

d) Privileged Users

When you call the helpdesk to reset your password, the helpdesk engineer does it in few minutes. Engineer has a special access ID, which enables him to create, modify, delete or reset IDs u helpdesk engineer is a privileged user as he can reset any ID.

Privileged users are hence users with special access/privileges who generally have higher-level access. These user IDs are also called as super user IDs or special privileged IDs. They include the IDs system administrators, security administrators and network administrators as also server IDs. A break at these levels can cause a tremendous financial loss.

Countermeasures

- Proper segregation of duties
- Review of administrator activity log
- Supervision of the violations committed by these users
- Vigilant screening policy in their hiring process

e) Lack of Awareness

Ignorance is not bliss but fatal when it comes to information security. Most employees are not aware nor understand why they cannot leave a confidential paper in their unlocked drawers or leave their laptops on their desk without cable locks. But these very people, when they leave their homes, remember to lock their houses and keep their money and jewellery under lock and key. It is just that they lack the sense of ownership when it comes to office property.

Unlocked property always attracts criminals. Similarly, unlocked confidential data or laptops attracts fraudsters. You may end up losing business, money or reputation for such negligence. Also, any theft through your ID implies a theft by you. This is because you are responsible for the activity take place under your ID. This underlines the importance that you should attach to keeping you passwords and other sensitive information secure.

In Latin “Ignorantia juris non excusat”, means “ignorance of the law does not excuse”.

From information security perspective this can mean, “ignorance of security does not excuse”.

Countermeasures

Regular/periodic information security awareness programmes for all (employees, contractors and, third parties wherever relevant)

f) Lack of Training

In most organization when the question “What is Information Security is asked, many among the staff This itself is a vulnerability and the attacker can use social engineering techniques confidential and sensitive information. Lack of training can lead to inadvertent sharing of critical information, which can lead to crimes and frauds.

For example, when waiting for a flight, a marketing executive of a financial company happens to lose his sensitive reports he had planned to carry for a customer meet. Due to lack of awareness, he doesn't call his office helpdesk to disable his ID. This results in compromise of data. Actually, a rival company employee to gain competitive advantage at the forthcoming meeting has stolen the PDA.

Countermeasures

There should be mandatory basic Information Security (IS) and Awareness training programmes related to job responsibilities. Company personnel need to be told what they need to do when they realize that sensitive information has got compromised.

g) Errors and Omissions

A simple example of an error of this type would be a bank employee from the Operations Department transferring by mistake Rs. 10,00,000 (10 lakhs) instead of Rs. 1,00,000 (one lakh). The employee does not realise this till the reconciliation. Meanwhile, the customer to whose account the transfer has taken place realises the bonanza. He withdraws the full amount and disappears. A simple example of omission would be a security guard in a high-security area checking the bag of an individual but forgetting to physically check the individual or vice versa.

Countermeasures

Monitoring of operations and inputs well-documented processes and procedures that should be followed without fail.

h) Password Sharing

Let us look at a scenario for a better understanding of this aspect.

Ankita has fallen sick with conjunctivitis and cannot make it to office. She has been asked to complete a corporate report on an urgent basis. She calls her colleague Nisha to go to her (Ankita's) terminal and login with her (Ankita's) ID and password. Once Nisha had logged in with Ankita's ID and password, Ankita through phone requests Nisha to update the report and forward it to her boss.

When you look at this scenario, you may feel nothing is wrong and it was a genuine request in the interests of the company. The answer is it is a policy violation. No passwords can be shared. What would happen if at the same time Nisha had passed on some sensitive information on Ankita's computer, to a rival company?

Countermeasures

- Strict disciplinary action against violation of the password sharing policy
- Proper training and awareness with regard to implementation of IS policies and practices

i) Social Engineering

Social engineering is an effective way/process of deceiving and luring a victim into giving a confidential and sensitive information or data. It is also called the art of manipulating people.

Humans are gullible. They take things at face value. They fail to see the larger picture of things. They become victims of social engineering and end up parting with sensitive information.

Let us look at some common terminologies used with regard to social engineering

- Eavesdropping: A technique where the intruder overhears a private conversation or peeps into a chat on computers between two people to gain confidential information. The conversation between Ramnath and Sashi as given earlier is an example of eavesdropping
- Shoulder Surfing: A technique wherein the attacker is looking over someone's shoulder and trying to get information. For example, looking at the monitor of the target when he/she is entering the ID and password while logging into the system
- Piggy Backing/Tailgating: A technique in which the intruder gains access into a secured premises by walking in immediately behind an authorized user
- Dumpster Diving: A technique where discarded trash or computer junk is searched for valuable information. For example, a hacker may sift through the trash bin of a big corporate where the documents and computer junk are dumped. These may reveal some sensitive information
- Phishing: These are techniques where the intruder masquerades as authorized or legitimate user person and tries to trick the victim into giving confidential information like ID and password, or credit card details. For example, a fraudster impersonates as a security administrator and call user. He then tells the user that there is a problem with his ID, which needs to be fixed. To fix the user has to provide the password. The user thinks it is a genuine call and gives the details becoming a victim himself.

Countermeasure

- Have a good and effective Awareness and Training Programme detailing the various social engine techniques
- Conduct awareness campaigns and put up posters and warnings as reminders

4.10 Vulnerability in Banking Environment

Criminals and hack commit fraudulent acts involving both individuals and corporates including banks. Their concentration is more on banks because they are likely to result in immediate financial gains. For example, when they hack the credit card details of individuals from a banking site, they can use the information immediately for financial gain. If they hack confidential data from a corporate, unless they can sell it to, they may not be able to make any financial gain.

What is a Bank Failure?

A bank failure is when the bank in question fails to safeguard the sensitive information belonging to its customers or itself.

Bank's losses can be broadly classified as:

- Reputation Loss - When a bank's data has been compromised and the same gets highlighted in the media, the image of the organisation takes a beating. It loses its reputation and credibility
- Financial Loss - Fraudsters gaining access to credit or debit card information can lead to enormous losses. For example, Citigroup lost \$ 2.7 million to unethical hackers
- Customer Loss - When attacks on a bank keep increasing, customers start fearing for the safety of their funds and sensitive personal documents or possessions. They may decide not to use the services of such a bank. The flow of new customers also gets affected under such circumstances
- Legal Losses - When a bank is hacked and customers end up losing funds and data, the aggrieved customers can launch legal proceedings against the bank. The costs of such legal action can turn out to be enormous. For example, Comerica Bank ended up paying its customers losses amounting to US \$560,000 following a cyber-attack.

The following news items give an idea of the losses suffered by respective banks:

News 25th June 2011: Citigroup suffered about US\$2.7 million in losses after paying post-fraudulent charges when hackers found a way to steal credit card number from its website

16th June 2011: A Michigan court has ruled that Comerica bank is liable for a US\$560,000 cyber heist, saying the bank should have done a better job to spot millions of dollars in fraudulent transaction after one of the bank's customer was tricked in a phishing attack two years ago.

4.10.1 Causes of Bank Failure

Let us look at the various factors that cause bank failures along with countermeasures to be adopted by banks.

1. People

As explained earlier in this chapter, people working for the bank are a critical factor. Their lapses can lead to the bank's failure. Their responsibility is critical considering the types of attacks that banks nowadays face.

2. Policies and Procedures

The bank's IS policies and procedures play an important part in failures. IS Policy is the standard based on which the IS baseline is given to all employees and related staff to strictly follow and implement.

Inadequate and inappropriate IS policies and procedures may lead to major flaws in control, lead into fraudsters exploiting the vulnerabilities.

For example, a bank's IS policy may mention revoking of access rights to all bank staff pending issuance of new IDs and passwords. Because of this, access to contractors and third parties continues. This is a major control flaw as intruders can still take advantage of these IDs.

Countermeasure

An adequate and appropriate IS Policy and procedure should be formulated, implemented and duly monitored for continued suitability.

Physical Security

The perimeter and movement of people and goods in a bank should be secured and monitored. If not there can be external threats, which may compromise the information assets.

For example, social engineering like piggy backing gives access to an unauthorised user into the premises, which can lead to a fraud.

- Stringent physical security should be there based on the classification of the area,

For example, in a highly restricted area, the security guard checking people should also time in and on out the people. Biometrics is a good method of restricting access to genuine people. While this access to pantry or canteen can be given to all employees possessing access cards

3. System Security

A bank's information assets are used, stored, transmitted, retrieved and disposed off using system. Hence, the IT infrastructure itself should have inbuilt security measures with people and process level controls.

Some of the flaws in the system security are:

- Networks - Unencrypted data packet, patches not updated, absence of Intrusion Detection System (IDS), absence or insufficient firewalls, open ports, ineffective network architecture and access management
- Operating system - Security lapses like patches not updated
- Application services - Improper configuration of application, improper version updates, outdated software, software errors, absence of anti-virus
- Inbuilt applications - Flaws in the design, development, testing, implementation and review
- Servers - Improper backups, no proper restriction to server rooms, server malfunctioning, inadequate review of the activities of system administrators
- Terminals - No anti-virus, improper configuration of applications

Countermeasure

- Security controls based on the required standards should be in place
- There should be a continuous upgrade of the system with latest technology and controls There should be effective monitoring of existing controls and vulnerabilities

4. Internal Controls

Internal controls should not be merely audit-based, taken as just a paper filing activity. They should be meticulous depending on the controls required with regard to both information assets and people.

5. Third-Party Services

Service-Level Agreements (SLAS) should be regularly reviewed and updated in tune with the changing times and threats. The access to third parties should be limited only to the areas that they are commissioned to serve.

6. Threat and Risk Management

There should be regular monitoring of threats and risks and adequate steps taken to counter them.

Example of a Bank Becoming a Phishing Victim

Bank Phishing scam attempts are very common. How do they work?

A scammer sends out thousands or millions of phishing emails. The email claims that the recipient's account has been suspended, and that in order to return the account to normal standing, they need to verify some information. The unsuspecting user clicks a link and submits the information sought. The scammer then compromises the users account.

Countermeasure

Do not reply to any unsolicited or strange email; do not click any link claiming to be from a bank. This is because such scams have become a common occurrence. It is better to ignore such mails or enquire with the respective bank. Whenever you come across an unauthorized activity in your bank account, contact your bank as soon as possible.

HSBC

HSBC Bank

Customer ID: 000-5432-654386-PSI

Dear Valued HSBC Customer

This e-mail is to inform you that your account will be suspended within 48 hours due to your Account Inactivity. You will have to confirm certain Account information in order to continue your account subscription:

<https://securityalertHSBC.co.uk/12/>

HSBC Bank Plc Security Advice HSBC Bank PLC

Source: HSBC Bank

4.11 Case studies

The Morris Worm, 1988

The Morris worm, named after its creator Robert Tappan Morris, then a student at the Cornell University, emerged even before most people knew there was an Internet. It was created not to cause any disruption but to measure for size of the Internet by propagating itself across the network. One of its side effects was that it slowed computers down to the point of becoming redundant. It affected about 6,000 computers, which were around 1-per cent of all the computers using the Internet in those days. This was quite a high number for that time.

The worm was created by Robert Morris in 1988. Morris was studying at the Massachusetts Institute of Technology (MIT) at that time and although this program was not intended to be a virus, it halted the working of all the computers it went into. He is now an associate professor in the same university. He sent the virus making it look as if it was sent by the MIT. The virus exploited weakness in passwords, in Unix and 'finger' and took advantage of the exploits in Unix's send mail, fingerd, rsh/rexec and so on. The 'finger' program was a utility that allowed users to obtain information about other users, while the sendmail program was a mailer designed to route mail in a heterogeneous internet network. The actions of the Internet worm exposed some specific security flaws in standard services provided by Berkeley Software

Distribution 9BSD-derived versions of Unix, and only affected Digital Equipment Corporation's (DEC'S) 'VAX' (virtual Address Extension) and Sun Microsystems's 'Sun3'systems. The worm has a design flaw. The critical error that transformed it from a potentially harmless intellectual exercise into a virulent denial-of-service attack was its spreading mechanism. This incident

VirtualAddress eXtension (VAX) was an instruction set architecture (ISA) developed by Digital Equipment Corporation (DEC) in the mid-1970s. however, motivated the world to think seriously about information security. During the Morris appeal process, the United States Court of Appeals estimated the cost of removing the virus from each installation to be in the range of \$200-\$53,000 (Hafner and Mark off 1991).

The Melissa Virus, 1999

A 30-year-old man launched the email computer virus known as 'Melissa 'on 26March 1999, a Friday. Named after a topless dancer from Florida, the virus spread rapidly around the world over the weekend, swamping the e-mail systems of thousands of computers including those at government agencies. Melissa targeted users of Microsoft Outlook and spread over the Internet like fir. The virus came in the form of an e-mail, usually containing the subject line important Message, 'and appeared to be from a friend or colleague. The body of the e-mail message said, 'Here is that document you asked for ... don't show it to anyone else.' It had a winking smiley face formed by punctuation marks. Attached to the message was a Microsoft Word document file. If the user opened that file, the virus dug into the user's address book and sent infected documents to the first 50 addresses.

Investigators traced the origin of the virus to a New Jersey resident, David Smith (Beer 1999). The virus was reported to have caused 80 million dollars' worth of damage in North America alone and about \$1.1 billion worldwide. Some estimates say at least 100,000 computers were infected.

Script kiddie refers to an inexperienced malicious actor who uses programs developed by other to attack computer systems and deface websites. It is generally assumed that script kiddies and kids who lack the ability to write sophisticated hacking programs on their own and that their objective is to try to impress their friends or gain credit in underground cracker communities (OECD 2008a).

Organizations reported infections by it. As per the United Sates Department of Defence (DoD) FUND AND Computer Emergency Response Team (CERT) claims, 233 organizations

and 81,285 computers has Melissa infections and one site reported receiving 32,000 copies of mail messages containing Melissa on its systems within 45 minutes. Several antivirus software makers posted patches on their websites that detected and rejected the Melissa virus (Richard 1999).

The Trojan Hydraq,2010

Trojan Hydraq, also known as 'Áurora', was first discovered on 11 January 2010, when it was used as part of a targeted attack, likely in an attempt to gain access to a corporate network and steal confidential information. Hydraq entered computers via e-mail attachments or was downloaded via malicious websites. It allowed attackers to compromise systems by exploiting a zero-day vulnerability in the client-side software. Once executed, the Trojan installed a back door that allowed an attacker to control the computer and perform a variety of compromising actions which included modifying, executing and deleting files; executing malicious files; and most importantly, gaining access to the compromised corporation's network_ which then opened up the target to additional attacks. As with other common Trojans, Hydraq would attempt to remotely contact its command and control server via a number of uniform resource locators (URLs) in order to receive updates and further instructions (Symantec Corporation 2011c).

The motivation of the virus was to take control of corporate networks and steal confidential and sensitive information from corporations. The number of infections from Hydraq was limited due to the nature of the targeted attacks—only a small number of corporations were targeted since attackers preferred to avoid attracting attention by limiting their attacks and, thus, remained concealed in a small volume of computers for as long as possible, rather than risk exposing themselves by compromising too many systems at once. The Trojan Hydraq thwarted antivirus sensors because it contained an obfuscation technique called 'spaghetti code' in which blocks of codes of the Trojan are rear-ranged to avoid detection. However, once a targeted attack is discovered, it becomes less effective due to increased awareness and the adoption of increased security measures, such as updating patches and sanitizing infected machines. As such, the life of Hydraq was short-lived, peaking in January 2010 and subsequently tapering off after February 2010.

Root Kit in E-mail Attachment

Let us take another example. A user's friend sends her a funny video. When she opens it he gets a security warning, but nevertheless she wants to see it. So she proceeds to get past

the warning. However, nothing happens and the user thinks nothing of it. Later when she meets her friend, she comes to know that her friend did not send her any such video. What actually happened in the background when she clicked on the video was that malware was installed. There is no way one can find out the intent of such attacks. One may not even notice anything. The computer could be used as a botnet drone to attack websites or other computers.

Spyware is the name given to a specific type of malware that can be installed on computer. It collects little bits of information about users at any given time without their knowledge. Spyware programs can collect various types of personal information, such as Internet-surfing habits, including sites that have been visited. It can also interfere with a user's control of the computer, such as installing additional software and redirecting web-browser activity.

In case of spyware, when one clicks on a link in search results and click on the download, one immediately starts getting pop-ups. One may close the pages but still get weird errors. One may think nothing harmful can come of it since one simply 'drove by' the website and never installed anything. However, the computer might have had a software flaw that let the website install spyware without the owner's permission. There was no warning because of the flaw in the programming of the web browser. There is a possibility that spyware is now resident on the system. Information on what one types in web forms, login pages, chat and the sites one visits could all be sent to the hacker's website;

Cyber crooks manipulate search results from search engines such as Google, Bing and Yahoo to lure victims to their malicious pages. Once a victim is lured to the desired webpage, they are redirected to rogue or poisoned sites. Through these sites, cybercriminals infect users' machines with malware or push fake goods and services to users while attempting to steal personal information.

The Zeus Family

The rapidly growing Zeus family is a classic example of one of the biggest increases in malicious URLs and websites. Zeus' ease of use and prevalence among cybercriminals has proved that there has been a distinct shift to truly malicious servers using automated domain registration practices. Once a Zeus machine is tracked it is easy to find dozens more. In one particular case, a Zeus command server which was identified yielded another 160 malicious domain scarring on everything from social networking to media sharing infections and other credential phishing activities. According to McAfee Threats Report: First Quarter 2010

(McAfee2010), the Zeus URLs are found primarily in the United States, with around 40 per cent in Europe alone.

The Zeus malware kit is primarily designed to steal users' online banking login credentials. It is made up of many, many small botnets linked by their use of the same crimeware. Zeus has proliferated partly due to the availability of Zeus toolkits, which allow cybercriminals to rapidly create Zeus variants in a matter of minutes. It is the handiwork of organized criminals from eastern Europe that has now entered the underworld cybercriminal market as a commodity (Trend Micro 2010b). Zeus may arrive as an attachment or link in a spammed message or be unknowingly downloaded via compromised web sites. Most Zeus botnets target bank-related websites.

The Slammer Worm, 2003

The Slammer worm, more commonly known as the SQL Slammer worm, was infamous for its denial-of-service (DoS) attack on various Internet hosts. The attack, which occurred on 25 January 2003 at 5:30 pm, infected more than 75,000 machines within 10 minutes. Despite the name, the Slammer worm did not use the SQL language as its method of exploitation. Instead it exploited a buffer overflow condition in the Microsoft-branded structured query language (SQL) server and other database products.

The Internet Storm Center (The Slammer Worm, <http://www.spamlaws.com/slammer-worm.html>) and other sites monitoring Internet traffic reported significant performance issues throughout the globe, similar to the impact of the Code Red Worm that struck in 2001. Yonhap (<http://www.spamlaws.com/slammer-worm.html>), a news agency in South Korea, reported that several Internet services worldwide were shut down for some time on 25 January 2001/ The Slammer worm was also detected throughout most of North America, Asia and Europe. The overall impact was somewhat mitigated by the fact that the worm struck over the weekend.

The exploitation of the Microsoft SQL server desktop engine (MSDE) substantially increased the number of infected systems. This, combined with the fact that many home PC users were unaware of MSDE's presence, essentially worsened the impact.

According to several analysts the propagation of the worm followed an exponential path with a doubling rate of 8.5 seconds in the earlier stages of the attack. This was contained only because of the failure of several networks due to the DoS attack caused by Slammer's traffic.

A router is designed to delay or temporarily halt traffic when it becomes too much to handle. The Slammer worm caused these routers to crash instead, forcing neighbouring routers to router, causing the flooding of multiple routing tables, which eventually caused other routers to fail. The routers were soon restarted, announcing their status and sparking another wave of updates in various routing tables. Shortly thereafter, large portions of Internet bandwidth were consumed as the routers were in constant communication with one another, trying to update their tables. Since the Slammer worm was small in size, it was able to get through the network, bringing the Internet to a standstill.

The Slammer worm was more of a network scare than a threat to personal users. Home computers typically were not vulnerable to infection unless they had MSDE installed on their system. The Slammer worm was so small that it contained no code allowing it to be written to a hard disk. This meant that it had to stay resident in the memory, making the infection fairly easy to remove. Symantec Corporation and several other security vendors offered free utilities that effectively remove the worm. Slammer has spread across the globe in 30 minutes. While nothing was actually destroyed by this attack, the revenues lost were estimated to be around one Billion dollars (Moore et al. 2005). South Korea was the hardest hit, where 27 million people lost Internet and cell-phone usage. In Australia the American Express website had to be closed for several hours and Bank of America customers were unable to use 13,000 automated teller machines. The United States had the most traffic and several businesses and government agencies were negatively affected.

The growing use of Internet infrastructure and broadband has led to a significant increase in malicious activities across all countries. The easy availability of resources, highly skilled professional and prosperous society is other factors for most cybercrimes. It is for these reasons that developed countries such as the United States and European nations show high instances of cybercrime both as a source and a destination. Countries which mastermind cybercrimes use highly sophisticated techniques like SQL injection, phishing and Trojans for financial gains. Growing economies such as India are used as a medium because of the vast pool of computers. Botnets through these countries are also the origin of minor cybercriminal incidents such as identity thefts for individual financial gains.

The frequency with which a link is clicked is an indication of the success of using shortened URLs that lead to malicious websites. These shortened URLs are observed more on social networking sites and lead to attack toolkits when clicked. A favoured method to distribute and

attack from a compromised profile is to post links to malicious websites from that profile so that they appear in the news feeds of the victim's friends. The use of shortened URLs for this purpose specifically helps because the actual destination of the link is obscured from the user/

According to a Trend Micro report, most malicious websites are seen to be in the adult entertainment category. Education is another industry that has been hit hard by infections. This has probably been due to a large number of students using old and out-of-date software and security, and possibly visiting suspect websites (Trend Micro 2010b).

The Koobface Worm, 2008

A malware that uses social networking sites to infect users in a concerted attack is a threat. Current variants of the Koobface worm can not only send direct messages from an infected user's account on a site to all of that user's friends in the network but are also capable of updating status messages or adding text to profile pages (Koobface Facebook Trojan on the March Again, <http://www.webologist.co.uk/interent-security/koobface-facebook-trojan-on-the-march-again>). Moreover, in addition to possibly giving attackers access to an infected user's social networking site account, some threats could also infect the user's computer. The Koobface worm attempts to download fake antivirus applications onto the compromised computer. These threats are a concern for network administrators because many users access their social networks from computers at workplaces.

Koobface cybercriminals have also been known to encrypt their command and control communications to avoid being monitored and taken down by security researchers and the authorities.

The first version of the Koobface worm was detected in December 2008 and a more potent version appeared in March 2009. The main intention of the koobface worm was to gather login information for File Transfer Protocol (FTP) SITES, Facebook and other social media platforms, but not any sensitive financial data; It used compromised computers to build a peer-to-peer botnet. A compromised computer would then contact other compromised computers to receive commands in a peer-to-peer fashion. The botnet was used to install additional 'pay-per-install' malware on the compromised computer, as well as hijack search queries to display advertisements. Its peer-to-peer topology was also used in showing fake messages to other users with the purpose of expanding the botnet.

The Stuxnet Worm, 2010

Another interesting case to examine is that of the Stuxnet worm. First spotted in June 2010. Stuxnet was a new kind of worm aimed at industrial control computers running software from the Germany company Siemens (Business Week 2010; Matrosov, Rodionov et al. 2011). Its robust capabilities and a high concentration of infections in Iran led to speculation by security researchers in Germany and the United States that it had been created by a nation state—the United States or Israel being the most likely candidates –and was aimed at sabotaging Iran's nuclear programme.

The worm initially spread via Microsoft Windows, and targeted Siemens' industrial software and equipment. While it was not the first time that hackers had targeted industrial systems, it was the first discovered malware that spied on and subverted industrial systems, as well as the first to include a programmable logic controller (PLC) root kit.

The worm initially spreads indiscriminately, but includes a highly specialized malware payload that is designed to target only Siemens supervisory control and data acquisition (SCADA) systems that are configured to control and monitor specific industrial processes.

Variants of /stuxnet targeted five Iranian organizations, with the probable target widely suspected to be the uranium enrichment infrastructure in Iran. Symantec noted in August 2010 that 60 per cent of the infected computers worldwide were in Iran (Fields2010; Symantec 2010). Siemens stated on 29 November 2010 that the worm had not caused any damage to its customers, but the Iran nuclear programme, which uses embargoed Siemens equipment procured clandestinely. Had been damaged by Stuxnet (CBS News 2010; 'An Overview of the Industrial worm called StuxNet', http://embeddedsdsw.net/doc/Stuxnet_white_paper.html).

The Conficker Worm, 2008

Conficker, also known as 'Down up', 'Downadup' and 'Kido', was a computer worm that targeted the Microsoft Windows operating system (Microsoft Corporation 2012a). First detected in November 2008, it made use of the flaws in the Windows software and launched dictionary attacks on administrator passwords to propagate, while forming a botnet. It was unusually difficult to counter this worm because of its combined use of many advanced malware techniques. At its peak, the Conficker had infected an estimated seven million governments, business and home computers spread across more than 200 countries, making it a large worm infection.

The motivation behind it was to mobilize a large number of compromised or infected computers into a kind of peer-to-peer network. This would allow the perpetrators to run malicious applications on them. This conficker worm was also known to piggyback with anti-virus products offered free for download at some websites/ However, in reality, the antivirus carried malware and spyware with it and once the user installed the anti-virus carried malware and spyware with it and became a part of the infected network.

Five variants of the Conficker virus have been known to exist and have been dubbed Conficker A, B, C, D, and E. They were discovered on 21 November 2008, 29 December 2008, 20 February 2009, 4 March 2009 and April 2009 respectively. The Conficker Working Group (CWG) uses the names A, B, B++, C and E for the same variants respectively.

Recent estimates of the number of infected computers have been notably difficult because the virus has changed its propagation and update strategy from version to version (Neild 2009). The estimated number of infected computers ranged from almost 9 million to 15 million (UPI 2009).

Avalanche, 2008

Avalanche is the name given to the world's most prolific phishing gang and to the botnet infrastructure it uses to host phishing sites. This criminal enterprise perfected a system for deploying mass produced phishing sites, and for distributing malware that gave the gang additional capabilities for theft (APWG 2010).

During the second half of 2010, Avalanche reduced its phishing activities significantly and completely disappeared. However, reports (APWG 2010) suggest that the people behind Avalanche switched to distributing the notorious Zeus Trojan. Zeus is a sophisticated piece of malware used by many different gangs of cybercriminals. The Avalanche gang began incorporating Zeus into its phishing and spamming campaigns in 2009. Zeus is crimeware or malware designed specifically to automate identity thefts and facilitate unauthorized transactions.

The Avalanche infrastructure is still in existence as of this writing, but its activities are stealthier, more focused on malware distribution and control, money mule recruitment, and other illicit activities/Recent law enforcement actions and continued takedown efforts by industry do seem to have had an effect on the operation (APWG 2010).

Panera

Number of victims: 37 million

Who was targeted?

All PaneraBread.com customer accounts were targeted.

What data was exposed: Names, email and physical addresses, birthdays, and the last four digits of the customers' credit card numbers.

Timeframe: Disclosed April 2018

What happened: Despite being warned by a cyber security expert in August 2017 that their website was leaking data, the Panera IT team failed to act until 8 months later when it announced the leak and took the site down for security maintenance.

Newegg

Number of victims: 50 million

Who was targeted?

Newegg online shoppers.

What data was exposed: Credit card information

Timeframe: August 14, 2018 - September 18, 2018

What happened: The online retailer was hacked by cybergangMagecart, who injected a credit card skimming code into the Newegg website. Whenever a customer bought something online, that payment info went straight to Magecart's C&C (command and control server).

Elasticsearch

Number of victims: 82 million (57M consumers, 26M businesses)

Who was targeted: Users and online businesses across the internet

What data was exposed: From individual users — names, email and physical addresses, phone numbers, IP addresses, employers, and job titles. From businesses — names, company

details, zip codes, carrier routes, latitudes/longitudes, census tracts, phone numbers, web addresses, email addresses, employee count, revenue numbers, NAICS codes, SIC codes, and more.

Timeframe: Discovered November 14, 2018

What happened: This is one of those cases we mentioned above where a regular security audit led to a researcher stumbling upon over 80 million records of sensitive, aggregated data. It is unknown how long the databases were sitting unguarded and who, if anyone, has had the opportunity to copy and steal all the data. Cybersecurity experts believe they have tracked down the source of the unguarded datasets to a data management company that has since closed its doors, but it is still officially unknown.

Facebook

Number of victims: 87 million

Who was targeted: Facebook users

What data was exposed: Profile info, political beliefs, friend networks, private messages

Timeframe: Disclosed September 2018

What happened: This is the notorious Cambridge Analytica scandal where the data-collecting firm illegally harvested users' info without their permission. The secret operation was politically motivated—namely, to influence the 2016 US presidential campaign. And though the breach occurred a couple years ago, it's only this year that investigatory conclusions have come out, giving us a clearer picture of what happened.

MyHeritage

Number of victims: 92 million

Who was targeted: MyHeritage users

What data was exposed: email addresses and hashed passwords

Timeframe: Alerted June 2018

What happened: Cybersecurity researchers alerted the genealogy site in June 2018 that

an outside server had been discovered with sensitive MyHeritage info. The company confirmed the info was legitimate and alerted its users that any account holders who signed up earlier than October 26, 2017 were at risk and should change their passwords.

Quora

Number of victims: 100 million

Who was targeted?

Quora users

What data was exposed: Names, email addresses, hashed passwords, profile data, public and non-public actions

Timeframe: Discovered December 3, 2018

What happened: Many questions still surround the details of this breach, but the question-and-answer site reported to its users that a third party had gained unauthorized access to one of their systems, expounding no further.

Under Armour

Number of victims: 150 million

Who was targeted?

MyFitnessPal users

What data was exposed?

User names, email addresses, hashed passwords

Timeframe: Late February 2018

What happened: The company's food and nutrition app was hacked, opening up the above info to the attackers, but not, thankfully, any payment info, which the company processes through a separate discovery

Exactis

Number of victims: 340 million (230M consumers, 110M businesses)

Who was targeted?

Users and businesses across the internet.

What data was exposed: Over 400 categories of detail, such as phone numbers, email and physical addresses, interests, ages, religions, pet ownership, etc.

Timeframe: June 2018

What happened: Data collection firm Exactis somehow had 2 terabytes of data relocated to a public site for all to see. It's unknown who or how many people accessed the info before it was discovery purpose.

Starwood

Number of victims: 500 million

Who was targeted?

Starwood guests.

What data was exposed: Names, email and physical addresses, phone numbers, passport numbers, account info, birth dates, gender, travel info, and accommodation information. Some of the breached info also included hashed credit card info.

Timeframe: Discovered September 10, 2018, but could have stretched as far back as 2014

What happened: Like many of the other official breach statements, the Marriott-owned hotel chain issued a statement that its servers had suffered "unauthorized access," but recent discoveries from the investigation indicate the breach may have been caused by the Chinese government for political purposes.

Aadhaar

Number of victims: 1.1 billion

Who was targeted: Indian citizens

What data was exposed: Aadhaar numbers, names, email and physical addresses, phone numbers, and photos

Timeframe: August 2017 - January 2018

What happened: Anonymous sellers over WhatsApp charged Rs 500 and lower for a portal into India's Unique Identification Authority where the records of virtually every citizen was at the payer's fingertips.

IFAFIFA

Football Leaks organisation leaked around 3.4 terabytes of data and 70 million documents which possessed a good number of corruption allegations. They got leaked to a newspaper in German magazine Der Spiegel. These 3.4 terabytes breach managed to overshadow the 2.6 terabyte Panama Papers, which is known as "the biggest whistle-blower leak in history" and the biggest mystery revealed and ever reported by investigative journalists.

Der Spiegel claimed that they acquired the information from whistle-blower whose name was "John" to bring corruptions in sports world into light. John told Spiegel that no hacker was engaged in this activity and many sources shared this information with him -not a single person. The attack took place a few months later when the Russian hacking group opened another internal Pandora box from FIA which had detailed reports of unsuccessful drug tests by footballers.

Google+

At the beginning of 2018, Google identified a vulnerability in an API. It noticed an API for Google's social networking effort Google+ gave third-party app developers the access to data from the friends of the app users.

As per the documents reviewed by The Wall Street Journal, Google not only compromised this data but kept it a secret due to the fear of reputational damage. So, as a response to the situation, the parent company of Google+ called Alphabet has decided and announced that it will shut down Google+ entirely for the good of everyone.

The investigators have found out that a bug within the website was sharing the access to the accounts and data from Google+ profile between 2015 till March 2018 until it got fixed. The

management of Google was naturally afraid of defamation. Hence disclosing the potential breach was not appropriate at any point. Even though, there were comparisons with Facebook in terms of legal penalties and how Google+ would have gained all the attention for all the right reasons for which it might have to pay. The repercussions were too high that temporary public attention looked an understatement.

Uber

Even though, Uber had already faced allegations and was infamous for compromising user data back in 2016. They even paid £133m to settle the legal penalisation owing to the cyber-attack which happened to expose 57 million customers and driver data. Uber, the ride-hailing company, also tried to keep it concealed, however, following the numerous allegations from the public, they did make that public in a smart manner though. In November 2017, they released the information saying that Uber paid \$100,000 (£761, 71) to hackers for deleting the acquired data from their systems.

British Airways

This year, British Airways also had to face the cybersecurity breach which affected around 380,000 transactions. This catered the stolen personal and financial data, nevertheless, the passport and flight details were safe. The data remained unsafe and insecure for 2 weeks during the period of 21st August to 5th September when the company's website and apps were under a "sophisticated" attack.

Alex Cruz, CEO of BA apologised in the BBC's Today programme saying "We're extremely sorry that it is causing concern to some of our customers, predominantly the ones who that made transactions over BA.com and app".

Currently, Rufus Grig, CTO at Maintel, is investigating the case and it is predicted that they might have to pay fines for compensating the data loss and insecurity. Customers are also suggested to check with their bank and credit card provider for comprehending how to cope up with the data breach.

T-Mobile

Around 2 million T-Mobile customers who were based in the US had their account details breached in which their names, email IDs, account numbers, billing details and encrypted

passwords. Their UK based acquired remained unaffected though, as per their statement to The Registrar.

T-Mobile announced it saying “unauthorised capture of some information.

Summary

Fraud can loosely be defined as “any behavior by which one person intends to gain a dishonest advantage over another “fraud, under section 17 of the Indian contract act, 1872,

“A deliberate act of omission or commission by any person, carried out in the course of a banking transaction or in the books of accounts maintained manually or under computer system in banks, resulting into wrongful gain to any person for a temporary period or otherwise, with or without any monetary loss to the bank”.

Insurance fraud is an illegal act on the part of either the buyer or seller of an insurance contract. Insurance fraud from the issuer (seller) includes selling policies from non-existent companies, failing to submit premiums and churning policies to create more commissions.

The Insurance Regulatory and Development Authority of India (IRDAI) is an autonomous, statutory body tasked with regulating and promoting the insurance and re-insurance industries in India. It was constituted by the Insurance Regulatory and Development Authority Act, 1999, an Act of Parliament passed by the Government of India. The agency's headquarters are in Hyderabad, Telungana, where it moved from Delhi in 2001.

An online transaction processing system collects, modifies and stores all transactions that a user executes during his interaction with an organisation.

1. Atomicity

If any single operation fails, then the whole operation fails. For example, in electronic payments, the system has to withdraw from one account and credit it to another. The transaction cannot be deemed complete if only one of the actions has taken place. Either both must occur or neither. Any partially executed transaction must be rolled back by the OLTP system.

2. Consistency

The essence of an OLTP is that changes in the data before and after a transaction should be consistent. The actions taking part in a transaction should not violate any integrity constraints associated with the state.

3. Isolation

An OLTP has multiple users accessing the database simultaneously. It should ensure that nobody is able to manipulate the database. Neither should a user be able to change a piece of data before another is done with it. For example, in an airline ticketing system, if one user is reserving the last available seat in a flight, another user who is using the system simultaneously should not get an impression that the seat is available for booking. In short, concurrent transactions should not overlap each other.

4. Durability

A completed transaction should always be on record regardless on any failures within the system network, like network unavailability or crash of a hard disk.

Online transactions include e-commerce. It is a process in which an online shop or website is used by the purchaser to select the items he/she wants to purchase, before he/she places an order. The seller then fulfils this order. The sale and payment for such goods and services may be completed either online or any offline method like “Cash on Delivery”.

In e-commerce, the service that automates the payment transaction between the shopper and the merchant is called the ‘payment Gateway’. It is a third-party service that verifies, accepts or declines credit card transactions on behalf of the merchants. It is equivalent to a physical point-of sale terminal available in retail outlets.

Online Banking Systems

Online transactions also include online banking. The days of walking to the bank branch every time you want to withdraw cash or make a transaction are gone. Banking can now be completed from within the comfort of your home. All banking services are available at the click of a button. All one needs is a computer terminal with an Internet connection.

RTGS is a fund transfer system, where settlement between banks takes place on a “real time” and “gross” basis. “Real time” means payment is not subject to any waiting period. “Gross settlement” is processing of instructions on a one-to-one basis and there is no bunching or netting with any other transaction. Once processed, the payments are final and irrevocable.

Computer security refers to ensuring of appropriate levels of assurance of the s with regard to data or information stored in the computers.

Internal Attacks - These are done normally by insiders, such as disgruntled employees to employees or employees doing it for financial gain.

External attacks - These are done by different types of hackers, such as Black hat hackers and Hactivists.

Vulnerability is a flaw that can exist in any aspect of a system, not necessarily from only the perspective. It can be at the policy level or in a set of procedures followed by a system administrator help desk personnel for resetting a password. It can also be an inadequate review of a log. and audit logs not reviewed regularly can lead to vulnerability. For example, a review of audit trail can identify any kind of password attack, like brute force. This can even be tracked since the server have a number of failed logins within a short period of time.

Internet crime

“A crime in which the perpetrator develops a scheme using one or more elements of the Internet to deprive a person of property or any interest, estate, or right by a false representation of a matter of whether by providing misleading information or by concealment of information.” For example, money laundering is normally done using fake/benami accounts and transmit an online over different countries and tax havens involving numerous banks. The entire exercise is to ensure that the actual source of funds remains hidden.

Internet fraud is any type of intentional deception that uses the Internet. This includes fraud that occurs in chat rooms, message boards, website and through emails. It occurs in the form of deceitful solicitations and fraudulent transactions.

The human element is the most important link when it comes to security. It is humans who create computer systems, write programs and build security systems. Ironically though, humans are also the weakest link when it comes to computer security.

Improper Screening

In these employees, contractors or third-party users are hired without proper background and character verification checks. This can result in potential threats to the organization.

Privileged users are hence users with special access/privileges who generally have higher-level access. These user IDs are also called as super user IDs or special privileged IDs. They include the IDs system administrators, security administrators and network administrators as also server IDs. A break at these levels can cause a tremendous financial loss.

Lack of Awareness

Ignorance is not bliss but fatal when it comes to information security. Most employees are not aware nor understand why they cannot leave a confidential paper in their unlocked drawers or leave their laptops on their desk without cable locks. But these very people, when they leave their homes, remember to lock their houses and keep their money and jewellery under lock and key. It is just that they lack the sense of ownership when it comes to office property.

Social engineering is an effective way/process of deceiving and luring a victim into giving a confidential and sensitive information or data. It is also called the art of manipulating people.

Check your Answers

Explain the following:

- a) Categories of Frauds
- b) IRDA
- c) Cyber Frauds
- d) Vulnerability

Lesson - 5

Vulnerable areas in CBS and their exploitation

Objectives

After reading this lesson you will be able to understand the following:

- Core Banking Concepts
- Application related parameters and freedom available to users - Empowerment of users
- Access to organization wide data
- Direct access to database and records
- Multiple interfaces with other applications ATM network
- Anti money laundering applications

Structure

- 5 Core Banking System**
 - 5.1 Banking Overview**
 - 5.2 Conventional Bank Branch- setup – Traditional**
 - 5.3 Computerisation and Indian Banks**
 - 5.4 Drivers and options**
 - 5.5 TBA architecture computerisation and Indian Banks – options.**
 - 5.6 Physical design CBS-Ovarian**
 - 5.7 General Ledger**
 - 5.8 CBS architecture Variations**

5. Core Banking System

- Banking Overview
- Computerisation in Indian Banks
 - Drivers And Options
- CBS Overview

- Architecture And Parameterisation
- Understanding CBS Options
- CBS Check Points
- Discussion

5.1 Banking Overview

- Basic Banking
 - Accepting Deposits
 - Lending to Borrowers
- Remittance Services
 - Draft (DD)
 - Mail Transfer (MT)
 - Telegraphic Transfer (TT)

Banking - Overview

- Trade Finance
 - Bills – Collection, Purchase, Discount
 - Letter of Credit
 - Bank Guarantee
- Safe Deposits and Lockers
- Foreign Exchange Services
- Investments and Treasury Operations
- MIS And Statutory Reporting
- Risk Management
- Consultancy Services
- Loan Syndication
- Mergers And Acquisitions
- Portfolio Management

5.2 Conventional Bank Branch - Set up - Traditional

- Multiple Counters – Dedicated Tasks
- Strong Visual Monitoring
 - Each Staff Handles Only His/ Her Books
 - Easy to See Who is Accessing/ Doing What
 - Absent/ Former Staff Can Not Enter Transactions
- Transactions go Through Multiple Hands
 - Easy Error Trapping

5.3 Computerisation & Indian Banks

- Reconciliation Applications
- Rangarajan Committee Reports
 - First Report
 - ALPMs And Back Office Systems
 - Second Report
 - Branch Level LAN
 - Administrative Office – Data Processing Centres
- MIS Applications
 - Consolidation of Branch Returns
 - Business Performance Tracking
 - Payroll Application
 - Personnel Inventory
- Industrial Relations Issues
 - Slow Computerisation
- Changes Started With
 - Railway Reservation – Customer Need
 - Liberalisation in Early 1990s
 - New Private Sector Banks

5.4 Drivers & Options

- Cut Down Errors And Delays
 - Interest Calculations
- Customer Expectations
 - Prompt Attention
 - Transaction Accuracy
 - Any Where Any Time Banking Service
- Improved Compliance
- Common Packages in Use in Banks
 - ALPM – Mostly Homegrown
 - TBA – BankMaster, ISBS, BASIS
 - CBS
 - Off-The Shelf Packages - Flexcube, Finacle, TC4, Bancs24, Prophecy, Profile
 - In-house Packages
- Typical Approaches Adopted by Banks
 - Manual Operations in Rural Areas
 - ALPMs in Semi-Urban/ Urban Areas
 - Total Branch Automation (TBA) Solution in Metros/ Urban Areas
 - Branch Level Computerisation
 - Multiple Vendor Solutions
 - Multiple Platforms
 - Transfer of Data to Central Location
 - Tapes, Floppies

5.5. TBA Architecture Computerisation & Indian Banks - Options

- Typical Approach Adopted by Banks
 - Core Banking Solutions (CBS)
 - Centralised Database
 - Branches Linked to Central Data Centre (CDC)
 - Smaller Banks - Link All Branches
 - Larger Banks - CBS Co-Exists With Other Approaches
 - Single Vendor And Solution
- Mandatory Modules
 - Customer Maintenance
 - Deposit Accounts
 - Loans
 - Trade Finance
 - Clearing, Collections, Remittances
 - General Ledger, Reports, Queries
 - Maintenance of Interest Rates, Charges,
- Additional Modules
 - Safe Deposits, Lockers, Inventory
 - Treasury Operations
- Additional Features – Support For
 - Telebanking, Internet Banking, Mobile Banking
 - ATM Network
 - Customer Relationship Management

Client

- Teller : browser or client/server
- Customer : branch, ATM, POS, telephone, Internet; TV browser, PDA, mobile phone etc.

5.6 Physical Design CBS – Overview

- CBS Revolves Around Parameterisation
 - General Ledger
 - Products
 - Interest Calculation Methods
 - Service Charges
 - User Rights Management
 - Transaction Level Features
 - Interfaces to SWIFT, ATM, RTGS and so on

5.7. General Ledger

- User Defined GL
- Account Linked to GL,
- Product & Customer
- GL Properties Selected
 - Type - Asset, Liability,
 - Direct or Indirect
 - Posting Restrictions
 - Head Office, Branch, Both
 - Balance Sheet Position

General Ledger - Some Examples

Interest

- Interest Accrual at Defined Frequencies
- User Definable Calculation Base(360/365 Days)
- Recalculation Facility For Change in Rate, Change in Limits And Value Dated Transaction
- Flexible Interest Rate Definition
 - Tiered Interest
 - Fixed Rate And Floating Rate
 - Preferential at Customer And Account Levels
 - Normal, Penal And Additional Interest Computation Facility
- Interest On Savings / Checking Account
 - Minimum Balance
 - Monthly Average Balance
 - End of Day Balance

Service Charges

- Facility to Define Event Based Charges
 - Account Opening, Maintenance, Closure
 - Issue of Cheque Book
 - Transaction Charges
 - Remittances
 - Trade Finance Transactions
 - Statement Printing Charges
 - Standing Instructions
- Multiple Charges For Each Event
- Customer Dependent Event-wise Charges

- Flexible Charge Definition
 - Minimum, Maximum, Percentage With Slabs And Rounding Definition, Fixed, Complex (Formula Based)
- Modifications Are Allowed as an Exception
- Charge Recovery at a Later Date
- Single Charge Debit Account For Customer
- General Charges Template For All Miscellaneous Charges

Central Customer Limit & Liability Maintenance

- Customer Limit Set Globally Across Bank
 - Utilisation Can Be Controlled Across Bank
- Maintain Limits And Updates Liabilities at Node (Facility Or Group of Account) Level
- Multi-currency Limits

5.8 CBS Architecture Variations

- Single Central Database Approach
 - Finacle
- Branch Server + Central Database
 - Flexcube, Bancs24
- Cluster Approach
 - Branches Grouped Into Clusters
 - Clusters Connected to Central Server
- TBA at Branch + CBS at CDC
- CBS Variation – Cluster Approach
 - Branches Grouped into Geographic Clusters
 - Cluster Server at Hub Location (Mini-CBS)
 - Cluster Hubs Linked to Central Site

- Central Server Maintains Aggregate Data
- Transactions Across Branches Routed to Appropriate Cluster Server
- Some Options Need Branch Server
- Implications
 - Subset of Branch Data Maintained
 - Limited Off-Line Transactions Possible
 - Part of Batch Operations Can be Done
 - Some Transactions Directly Hit The Host

CBS – Architecture Differences

- Branch Server - Potential Benefits
 - Uninterrupted Customer Service
 - Load Sharing
- Branch Server - Possible Downside
 - Data Synchronisation
 - Withdrawals in ATMs And Other Branches
 - Software Upgrades
 - Parameter Downloads
- Facility For Extending Functionality
 - Add Programs to Existing Menu System
 - Scripting Facility
 - Report Writing Tools
- Linkages Between Modules
 - Separate or Single Log-in
- Network Usage & Hardware Resources

Computerized Environment Typical Sequence of Operations

- Begin of Day Operations
- Transactions Input and Authorization
- Completion of Branch Operations
- Reports Checking
- Backup Operations
- End of Day (EOD) Operations
- BS Package – Extent of Fit For The Bank?
- CBS Will Require
 - Customization
 - Enhancement
- Business Process Reengineering Needed

CBS Environment Check Points

- Change in People's Attitude
 - Resistance to Change
 - New Terms
 - New Screens
 - New Report Formats
 - Change in Work Procedures
 - Change in Power Equations in Offices
- Parameter Changes in CBS
- Authorisation Process Required
- Correctness to be Verified
- Typical Problems
 - Backdated Changes in Parameters
 - Holiday Marking – Minimum Notice Period

- Sanction Terms Not Supported
- Typical Problems in Loans & Advances
 - Sanction Terms Not Supported
 - Mismatch Between System Generated And Sanctioned EMI
 - Security Values/ Stock Statements Not Updated
 - Repayment Options
 - Advance Payment
 - Prepayment
- Management of Users
 - Centralised in CBS
 - Branch Level in TBA
- Ideally Handled by Non-IT Personnel
- Typical Problems
 - Periodic Cleaning Required
 - Leave Handling
 - Single Vs Multiple IDs
 - User Acknowledgement Register

The following figure illustrates the CBS:

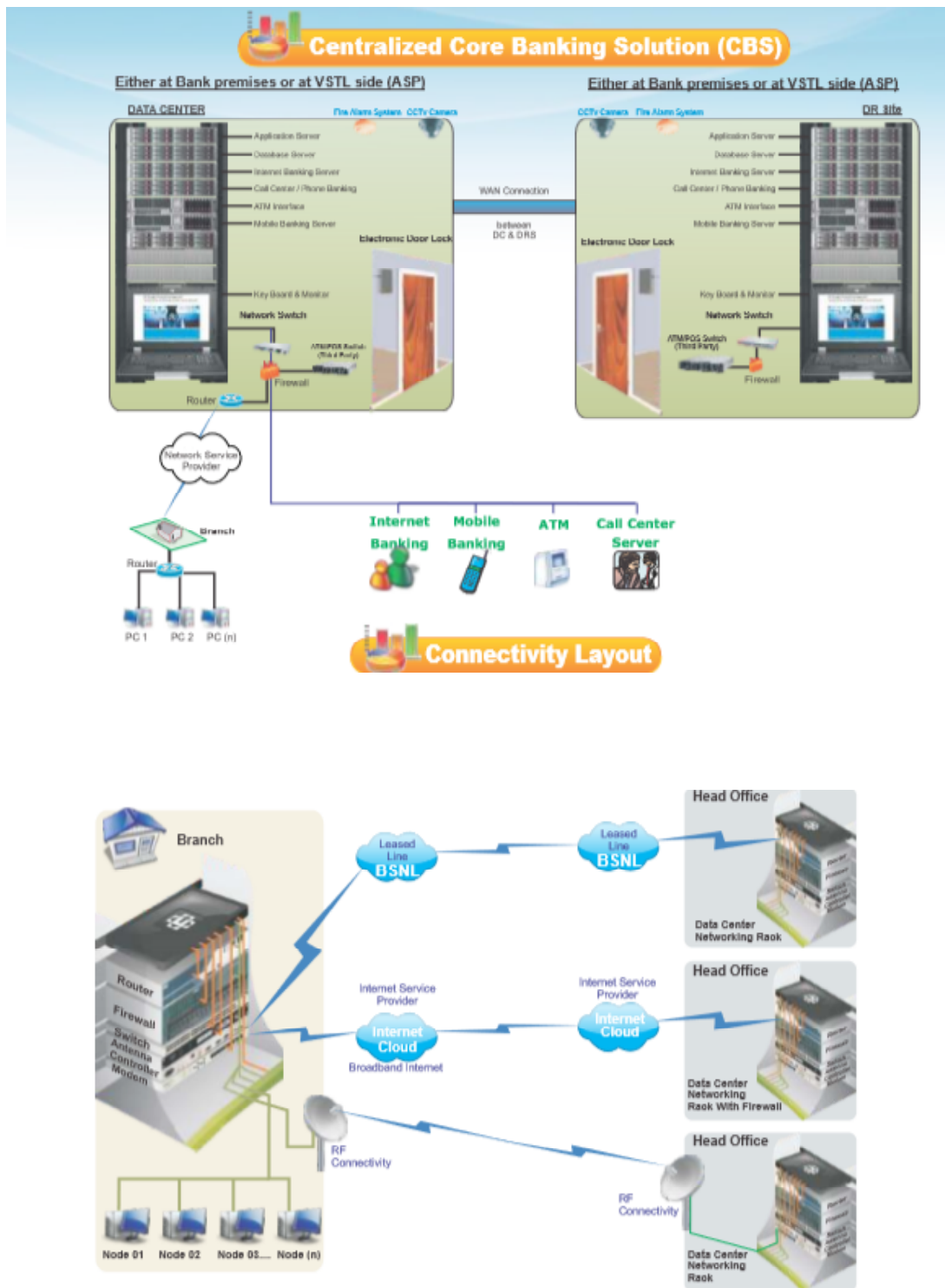


Figure 5.1: Core Banking solutions and Connectivity Layout

(Source: Core Banking systems by VIRMATI)

The data centre Layout of centralized core Banking Solution (CBS) is illustrated in the following figure :

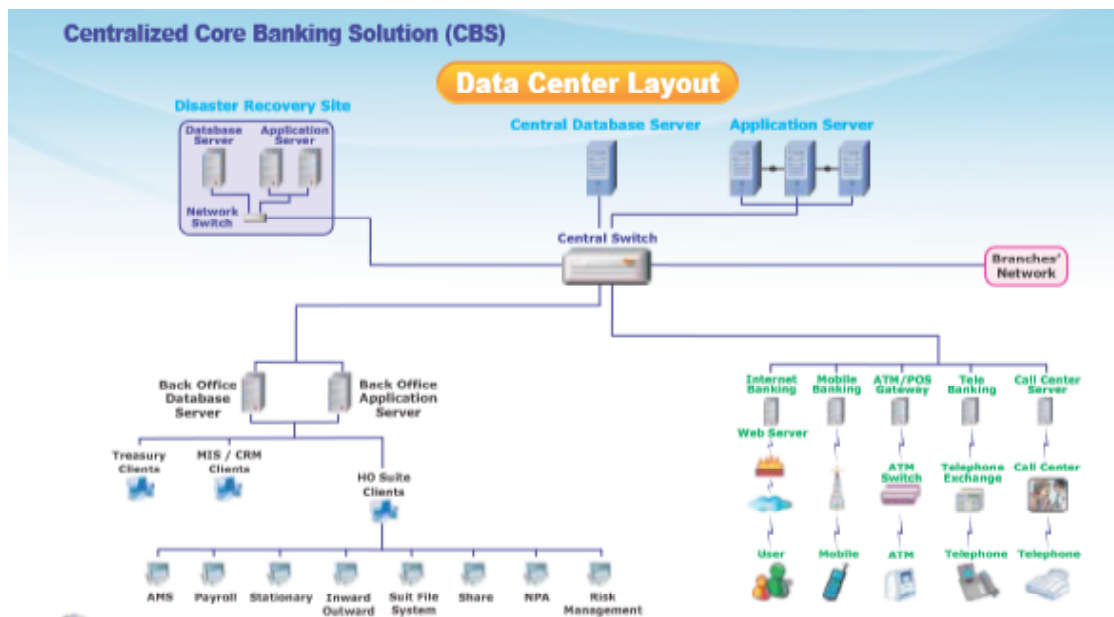
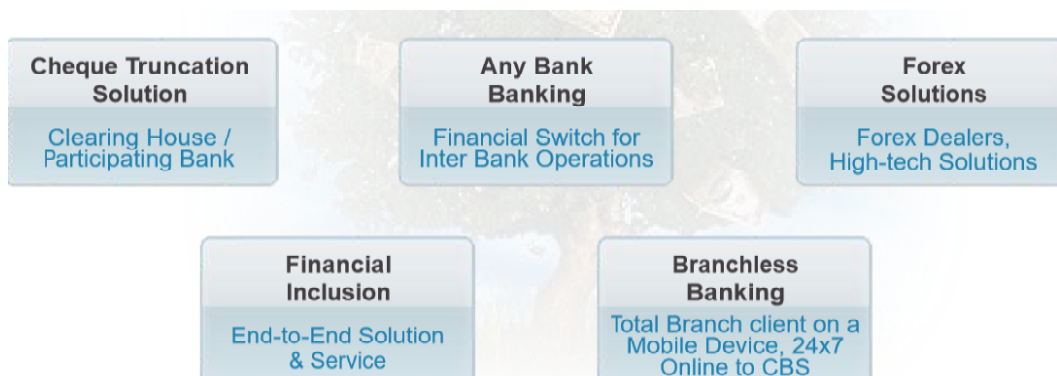


Figure 5.2: Centralized core banking Solutions (CBS)
(Source: Core Banking systems by VIRMATI)



Application related parameters and freedom available to users - Empowerment of users

Multi-lingual Support	Since this solution is browser based & Unicode Compliant the front end can be changed to suite any language, supported on the respective platform
Multi currency Support	The solution supports multi currency for financial statements
Multi Server Environment	Supports multi server environment
Multi-Delivery Channels	24x7 delivery channels can be integrated at Enterprise Level
Disaster Recovery Management	It supports Disaster Recovery Management Norms, Unique Options
Network Control Management	It supports control of Network in terms of Authorized Entities, Authorized Users & Authorized Terminals or Devices etc
Security Management	It supports various application level standards on Security
Customer Types	It supports multiple customer types, customer grouping and gradation. The Service accessibility and charges depend on such gradation
Service Branch	It supports service branch handling of clearing operations
New Banking Products	New launches can be easily configured as the solution is highly parameterized.
Unique Access Control	It has unique concept of hierarchy-based -ID and password for delivery channels
Maker Checker Concept	For 100% operations. Highly parameterized
Bank Account Opening	It supports bank account opening

Special application for user's empowerment is illustrated.

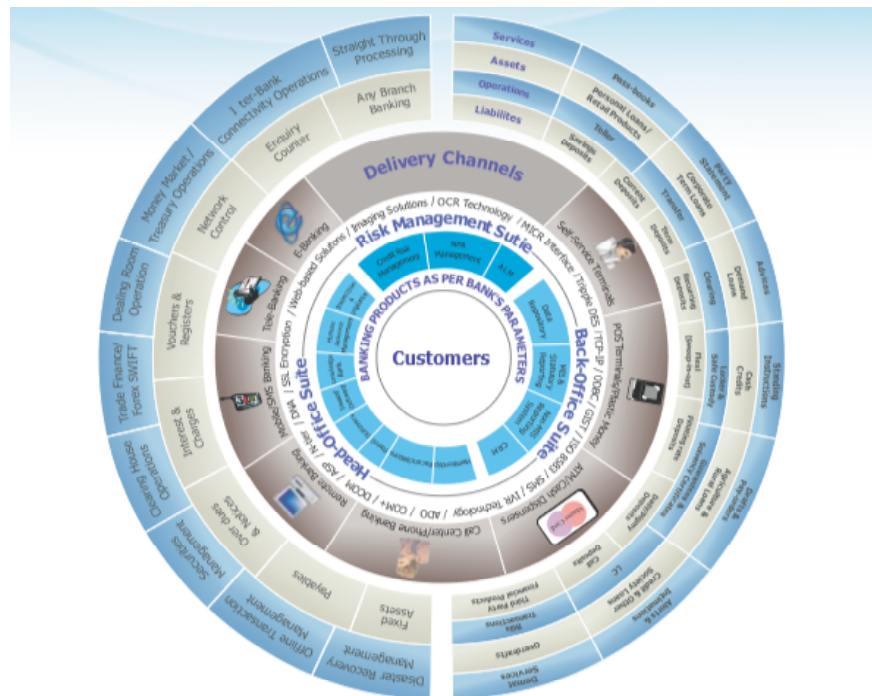


Figure 5.3: Customer for bank – same time aligned to Bank/ agent



Multi-Branch Banking

Enables banks offer complete flexibility to their clients to access and operate their accounts from any branch and enjoy the facilities of Banking Anywhere. Various functionalities available across the branches are:

Cash Anywhere Pay Anywhere Collect Anywhere

Multi-Currency Trade Finance & Forex

Multi-currency Trade Finance operations
Export Bills & L/C, packing credit, Inward Remittances, Import Bills & L/C, Outward Remittances, Forward Sales & Purchase Contracts & Reports, SWIFT Interface

Forex transactions
Currency Sales, Purchase and Exchange, Traveling cheques and instruments Also, Multi-currency deposit accounts

Database Management

- Realtime update at both, primary and secondary database servers, using SQL mirroring/Oracle Guard
- Support Storage area networks (SAN)
- History of database for MIS

Platforms

- CBS solutions is built on 3 tier architecture
- It is browser based and compatible with IE, Netscape Navigator, Mozilla Firefox and so on.
- Development tools are Microsoft ASPNET 2005, JavaScript and Business Object 11.5
- The Application Server is IIS web Server on MS windows 2008
- It is ODBC compliant and it can work with oracle or MS SQL

Back Office Suite

This account based MIS Solution

This account – based MIS solution meant Back Office is largely based on Banks Assets (Loan & Advanced) as well as liabilities (Deposits), Expenses and Incomes. The major modules are:

- Statutory Reporting
- Management Information System (MIS)
- NPA Management

H.O Suite
Reconciliations
Memberships & Dividends
Human Resource Management
Payroll
Stationary /inventory
Fixed asset management
Investment/Treasure Management
Admin
Credit Appraisal & Monitoring System (CAMS)
Suit File System
Local Office/Partner

Summary

Computerisation & Indian Banks

- Reconciliation Applications
- Rangarajan Committee Reports
- First Report
 - ALPMs And Back Office Systems
- Second Report
 - Branch Level LAN
 - Administrative Office – Data Processing Centres
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 - Business Performance Tracking
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Physical Design CBS – Overview

- CBS Revolves Around Parameterisation
 - General Ledger
 - Products
 - Interest Calculation Methods
 - Service Charges
 - User Rights Management
 - Transaction Level Features
 - Interfaces to SWIFT, ATM, RTGS and so on

Check your Answers

Explain the following:

- a) CBS architecture differences
- b) Implication of Central Customer limit & Maintenance
- c) Examples of General Ledger

Lesson - 6

Money Laundering and Anti Money Laundering

Objectives

After reading this lesson you will be able to understand the following:

- Money laundering Definition
- Common factors in money laundering
- Stages of money laundering
 - o Placement
 - o Layering
 - o Structuring
 - o Integration
- Money Laundering Reporting Officer (MLRO)
 - o The duties and responsibilities of the - The role of the MLRO
 - o Generating management information
 - o Common MLRO problems - **Recognition, handling and reporting transactions**
 - o The legal obligation to report
 - o Designing an effective internal reporting system
 - o The MLRO's evaluation process
 - o **Post-reporting considerations**
 - § Handling the risk of committing the tipping off offence
 - § Constructive trusteeship - Responding to discovery and enforcement orders
 - § Terminating relationships
 - § Subsequent client review techniques
 - § Secure record retention

- Dealing with the authorities
- Customer Due Diligence and Risk Profiling.
- Understanding anti-money laundering tools and strategies in Banks
- **Security Controls typically available and Case studies**
 - o Log of User activities in the application - Logging exceptional events
 - o Audit tools to analyze data for exceptions
 - o Change management procedures
 - o Internal data consistency checks
- **Anti-money laundering and legal frameworks**
 - o International and regional bodies - International initiatives - Terrorist financing
 - o The US Patriot Act
 - o The UK Model – The Jersey Strategy on Terrorist funding and money laundering
 - o National anti money laundering frameworks
 - o PMLA 2002- Objectives and salient features
 - o Financial Intelligence Unit India (FIU-IND)
 - o Financial Action task Force (FATF)
 - o Enforcement Directorate
 - o Serious Fraud Investigation Office (SFIO)
 - o KYC

VARIOUS HAPPENINGS OF CONCERN GLOBALLY

- BLASTS in Pune, Hyderabad, Bangalore, Mumbai, Varanasi, Peshawar, etc
- Arms dealing, drug trafficking, betting by known and unknown person
- Un-bridled access to such bad elements and flow of unaccounted money of huge volume throughout the year.
- Lack of strong effective counter mechanism, loss, enforcements.

Structure

- 6.1 Definition
- 6.2 Stages of Money Laundering-Brief
- 6.3 Legal Acts Enacted
- 6.4 Cash Transactions in Money Laundering
- 6.5 Money Laundering Reporting Officers [MLRO]
- 6.6 Laundering Process
- 6.7 Money Laundering Process – Detailed
- 6.8 Banks Role in Laundering
- 6.9 Anti-Money Laundering Legal frameworks

6.0 What is Money Laundering ?

- Money laundering is the process of transforming the profits of crime and corruption into ostensibly “legitimate” assets
- Money laundering is the process by which large amounts of illegally received money is given the appearance of having originated from a legitimate source.
- money laundering has criminal dimensions related to black
- It is the way to convert the black money into white money.
- Money laundering must involve a predicate crime such as the violation of Indian Penal Code, IPC, Narcotics, Prevention of Corruption Act and Human Trafficking.
- In India, stashing Black money a civil crime, while money laundering has criminal
- dimensions related to black money.

Some techniques of money laundering

Hawala: Hawala is an alternative or parallel remittance system. It exists and operates outside of, or parallel to ‘traditional’ banking or financial channels.

Structuring deposits: Also known as smurfing, this method entails breaking up large amounts of money into smaller, less-suspicious amounts.

Third party cheques: Utilizing counter cheques or banker's drafts drawn on different institutions and clearing them via various third-party accounts. Third party cheques and traveler's cheques are often purchased using proceeds of crime.

Credit Cards – Clearing credit and charge card balances at the counters of different banks. Such cards have a number of uses and can be used across international borders.

Causes of money laundering:

- No deals for sharing tax information with other countries
- Availability of instant corporations
- Corporate Secrecy Laws – as the corporate law of certain countries enables launderers to hide behind shell companies.
- Excellent Electronic Communication
- Tight Bank Secrecy Laws
- A Government that is Relatively Invulnerable to Outside Pressures
- A high degree of Economic Dependence on the Financial Services Sector
- A Geographical Location that Facilitates Business Travel to and from rich neighbors.

Harmful effects of money laundering:

Terrorism:

Money Laundering serves as an important mode of terrorism financing.

Terrorists have shown adaptability and opportunism in meeting their funding requirements.

Threat to banking system:

Across the globe, banks have become a major target of Money Laundering operations and financial crime because they provide a variety of services and instruments that can be used to conceal the source of money.

Threat to economic and political stability:

- The infiltration and sometimes saturation of dirty money into legitimate financial sectors and national accounts can threaten economic and political stability

- Thus money laundering is an activity which is capable of corrupting a chain of financial institution
- money laundering is an activity which is capable of corrupting a chain of financial institution

Regulation of money laundering in India:

- With its growing financial strength, India is vulnerable to money laundering activities even though the country's strict foreign exchange laws make it difficult for criminals to launder money.
- International Narcotics Control Strategy Report by Bureau for International Narcotics and Law Enforcement Affairs emphasizes India's vulnerability to money-laundering activities.
- The report highlighted India's emerging status as a regional financial center, its large system of informal cross-border money flows, and its widely perceived tax avoidance problems all contribute to the country's vulnerability to money laundering activities.
- Some common sources of illegal proceeds in India are narcotics trafficking, illegal trade in endangered wildlife, trade in illegal gems (particularly diamonds), smuggling, trafficking in persons, corruption, and income tax evasion. Because of India's geographical location between the heroin-producing countries of the Golden Triangle and Golden Crescent, India continues to be a drug-transit country.

Solutions:

- More strict laws related Anti-money laundering is necessary because money laundering tends to corrupt even the most professional players in the market.
- There is a need to sensitize the Private Sector about their role in anti-money laundering activities
- Continuous up-gradation and dissemination of information is necessary
- There is a need to build a balance between financial confidentiality and this confidentiality turning to a money-laundering haven.

Global steps to counter money laundering problem:

International efforts to combat money laundering: International regimes attempt to monitor or regulate international relations and activities. By establishing norms, rules, and procedures agreed to in order to regulate money laundering, international regime reduce the disparities between divergent legal systems.

United Nations Convention against illicit traffic in Narcotic Drugs and psychotropic substances: India's urgent need to control drug trafficking reflects growing awareness within the international community that drug trafficking and money laundering must be suppressed. The United has played an integral role in the global community's fight against illicit drug trafficking and money laundering.

Financial Action Task force on money laundering: Intergovernmental groups have also taken action against the rising levels of global money laundering. The task force was created by G-7 countries to curb the occurrence of money laundering.

Basel Committee Minimum standards: The Basle committee on Banking regulations and supervisory practices consists of officials from the central banks and supervisory authorities of eleven major industrialized nations and Luxembourg. In 1998, the committee adopted a statement of principles that targeted money laundering.

Laws related to money laundering:

1. The Prevention of money Laundering Act 2002(PMLA):

- It comes into force in 2005.
- It defines money laundering offence and provides for the freezing, seizures and confiscation of the proceeds of crime.
- The provisions of this act are applicable to all financial institutions like banks, mutual funds, insurance companies, and their financial intermediaries.
- The agency monitoring the anti-Money laundering activities in India is the Financial Intelligence Unit (FIU-IND). The agency is responsible for coordinating and strengthening efforts of national and international intelligence, investigation and enforcement agencies in pursuing the global efforts against money laundering and related crimes.

- According to the act, there can be punishment of imprisonment upto 3-7 years with fine upto 5 lakh rupees. But in case of offences done under Narcotic Drugs and Psychotropic Substance Act 1985, maximum punishment is extent to 10 years rather than 7 years.

2. The Prevention of Money Laundering (Amendment) Act 2012:

- The act was enacted in 2002, but was amended thrice, first in 2005, then in 2009 and then 2012.
- PMLA (Amendment) Act, 2012 has enlarged the definition of money laundering by including activities such as concealment, acquisition, possession and use of proceeds of crime as criminal activities.
- The amendment has introduced the concept of Corresponding law to link the provisions of Indian law with the laws of foreign countries and to provide for transfer of the proceeds of foreign predicate offence committed in any manner in India.
- It also introduced the concept of 'reporting entity' which would include a banking company, financial institution, intermediary or a person carrying on a designated business or profession.
- The amendment act has removed this upper limit of fine which was levied by 2002 act
- The act has provided for provisional attachment and confiscation of property of any person (for a period not exceeding 180 days).
- The act has delegated the powers upon the Director to call for records of transactions or any additional information that may be required for the purposes of investigation.

The government needs to step in to expedite a probe into what could be a massive money laundering exercise. This will help in better results than waiting for millions of missing investors to turn up.

6.1 Definitions of Money Laundering

Money laundering is defined as:

- The act of making money that comes from source 'x' look like it comes from source 'y'.

- Washing dirt money in a financial system and makes it clean throughout the transactions
- Disguising the origins of money obtained through illegal activities so it looks like it was obtained from legal source

Let us now see how it happens knowingly or unknowingly

6.1.1 Prevention of Money Laundering Act 2002

- Section 3 of the Prevention of Money Laundering Act, 2002 (PMLA) has defined the “offence of money laundering” as under:
- “Whosoever directly or indirectly attempts to indulge or knowingly
- Assists or knowingly is party or is actually involved in any process
- Or activity connected with the proceeds of crime (including its
- Concealment, possession, acquisition or use) and projecting or
- Claiming it as untainted property shall be guilty of offence of
- Money laundering”.
- Money launderers use the banking system for cleansing ‘dirty money’ obtained from criminal activities with the objective of hiding/disguising its source. The process of money laundering usually creates a web of financial transactions so as to hide the origin and true nature of these funds.

6.2 Stages of Money Laundering in Brief

PLACEMENT: This is the first stage of money laundering. In this stage the cash, generated in stages, is placed in small unrecognizable quantities, through illegal or money laundering activities, into the financial system.

LAYERING: While transferring funds, instead of ‘straight’ transfers from one point to another, a complex series of transfers are made. Layering ultimately conceals or obliterates the source of funds, the nature of funds and/or their ownership.

STRUCTURING: Keeping the amounts lower than that fixed for reporting and building up similar transactions till the amount placed to be laundered. For example, transaction in cash below Rs.10.00 lakh (to avoid CTR), making Demand Draft below Rs.50,000/- (to avoid giving

PAN no), planning maturity value of TDR below Rs.20,000 /- to have payment only in cash;

INTEGRATING: This is the stage where tainted money eventually merges, quietly, into the legitimate economic and financial systems as 'clean' money. The tainted money assumes a legitimate look at this stage.

6.3 Legal Acts Enacted

- Government of India along with Reserve Bank of India are the central agencies monitoring and issuing guidelines, instructions to counter money laundering activities.

6.3.1 PMLA

- The prevention of Money laundering Act, 2002 (PMLA) enacted to combat money laundering and financing of terrorism.
 - o Enacted on 17th Jan, 2003
 - o Brought into force from 1st July, 2005
- Administered by:
 - o Financial intelligence unit (FIU) for verification of identity of client, maintenance of records and reporting
 - o Enforcement Directorate (ED) for investigation of and prosecution for money laundering offenses.
 - o Finance Action Task Force Inter-governmental body having 36 Member countries.

6.3.2 Financial action Task Force (FATF)

- FATF has classified countries into High, Medium & Low Risk based on deficiencies in AML/CFT guidelines. A few examples of High Risk countries: - Iran, North Korea, Indonesia, Myanmar, Algeria, Pakistan, Iraq, Afghanistan, Argentina etc.
- Medium Risk countries – India, United Kingdom, Japan, Germany, France.
- Low Risk countries – Norway, Finland, Sweden, New Zealand, Denmark.

6.4 Cash/Transactions in Money Laundering

- Nobody knows the ownership of cash. Source of cash not known – Anonymity; cash changed hands is not known; changing form of cash (gold, etc.). Counterfeit currency notes are in circulation.

- Transactions in accounts of customer and non-customers are another case for money laundering.
- Large and multiple transactions and frequency of such transactions between accounts and multiple accounts are other forms of money laundering.
- To counter such nefarious actions, Banks have to maintain records, furnish information, and verify identity of customers and report of financial intelligence unit – India (FIU-IND) besides their own Head Office through a specially appointed top management official designated as “Principal Officer”.

6.5 Money laundering reporting officers (MLRO)

This term (usually abbreviated to MLRO) is used to describe the nominated officer appointed under Regulation 20(2)(d) of the 2007 Regulations and as referred to in Section 331 Proceeds of Crime Act 2002(POCA).

6.5.1 Why have an MLRO?

Regulation 20(2)(d)(i) requires that all businesses within the regulated sector must have a nominated officer to receive Disclosures under Part 7 of POCA and the Terrorism Act, and to make disclosures to the Serious Organised Crime Agency (SOCA).

Regulation 20(3) provides that there is no requirement to have a nominated officer in the regulated sector if you are an individual who provides regulated services but do not employ any people or act in association with anyone else.

6.5.2 Who should be an MLRO?

The role of the MLRO carries significant responsibility and should be undertaken by an appropriately experienced individual.

An MLRO should be of sufficient seniority to make decisions on reporting which can impact a business's relations with its clients and its exposure to criminal, civil, regulatory and disciplinary sanctions. They should also be in a position of sufficient responsibility to enable them to have access to all of the business's client files and business information to enable them to make the required decisions on the basis of all information held by the business.

Businesses authorised by the Financial Conduct Authority (FCA) (one of the successors to the Financial Services Authority) will need to obtain the FCA's approval to the appointment of the MLRO as this is a controlled function under section 59 of the Financial Services and Markets Act 2000.

6.5.3 Role of the MLRO

An MLRO is responsible for ensuring that, when appropriate, the information or other matter leading to knowledge or suspicion, or reasonable grounds for knowledge or suspicion of money laundering is properly disclosed to the relevant authority. The decision to report, or not to report, must not be subject to the consent of anyone else. The MLRO will also liaise with SOCA or law enforcement on the issue of whether to proceed with a transaction or what information may be disclosed to clients or third parties.

The size and nature of some businesses may lead to the MLRO delegating certain duties regarding the business's anti-money laundering/counter terrorist financing obligations. Delegation does not relieve the MLRO of responsibility. In some large businesses, one or more permanent deputies of suitable seniority may be appointed. The reporting channels should be well known to all relevant employees. All businesses will need to consider arrangements for temporary cover when the nominated officer is absent.

MLRO's may also take responsibility for:

- Training within the business
- Advising on how to proceed with work once an internal report and/or SAR has been made in order to guard against risks of tipping off or prejudicing an investigation
- The design and implementation of internal anti-money laundering systems and procedures

If this role is not undertaken by the MLRO, these responsibilities should be taken on by another sufficiently senior and skilled person within the business. This person should work closely with the MLRO.

Money Laundering Reporting Officer MLRO Every practice should have a 'nominated officer' officer or a Money Laundering Reporting Officer (MLRO) as provided for by Part 7 of the Proceeds of Crime Act 2002 or Part 3 of the Terrorism Act 2000. Regulation 20 of the 2007 Regulations require this person to be appointed, to have procedures where staff make internal

suspicious activity disclosures to the MLRO and a defined policy for the MLRO to consider any such internal reports to determine if a SAR should be made. A breach of regulation 20 can result in an offence (regulation 45).

The presence of an appointed MLRO ensures that the direction of AML/CTF policy is centralised and that submissions to NCA or the relevant Financial Intelligence Unit (FIU) are consistent.

Internal systems and policies

1. The MLRO should have a degree of seniority and authority within the business to drive policy decisions and actions. He/she should have undergone sufficient level of training and hold suitable qualifications and experience. Arrangements should be in place for when the MLRO is absent.

2. In conjunction with the senior management if applicable, the MLRO should oversee the implementation and adherence to risk based policies and processes on: a) customer due diligence and ongoing monitoring, b) reporting processes, c) keeping of records, d) internal controls, e) practice AML/CTF risk assessment, f) training of staff, g) communication of ML/TF policy and issues within the business

3. The MLRO's duties include establishing an internal AML/CTF policy which will set out on record how all risks considered in the practice AML/CTF risk assessment, surrounding clients, services, locations served and delivery channels, are managed. (See CCAB Guidance, para. 3.7-3.8)

4. The MLRO should ensure that the practice AML policies are recorded and monitored for effectiveness. The risk assessment of the business should be conducted by the MLRO and they have responsibility for monitoring internal compliance and effectiveness of controls.

5. The MLRO should work with management to ensure that the training provided to staff is suitable for the practice on a risksensitive basis.

Evaluation of internal reports

6. Sections 327-329 of POCA 2002 outline the MLRO's responsibilities surrounding disclosures where there are suspicions or knowledge of specific offences around ML/TF.

The MLRO should also be fully aware of their responsibilities around requesting consent and the types of disclosures as laid down in sections 335339 POCA 2002.

7. A failure to make a disclosure or a SAR where one should be made can have serious repercussions for the practice and therefore the MLRO position carries great responsibility. Once a junior staff member makes a report to the MLRO, the MLRO then assumes responsibility for the suspicion reported. The MLRO should remind staff of their obligations regarding the offence of tipping-off.

8. When evaluating an internal report, the MLRO should consider the complete client profile and history of the business relationship, including the initial client onboarding CDD information and ongoing monitoring information. The MLRO must review the patterns of activity or services provided to the client up to the present time including the volume, frequency, size of business done. Consider how long the client has been with the practice? is it an entirely new client or a long-standing well known one? The importance of good CDD is seen in this instance as it gives the MLRO a detailed understanding of who the client is and whether suspicions are well founded.

9. The MLRO may also require further information to be obtained, from the client if necessary. The Joint Money Laundering Steering Group state that any approach to the client should be made sensitively, and probably by someone other than the MLRO, in order to minimise the risk of alerting the client that a disclosure to the FIU may be being considered.

10. The MLRO should consider if professional privilege will exempt the practice from making a report where the knowledge or suspicion comes to the practice from privileged circumstances, provided the information received is not with the intention of furthering a crime (section 330 POCA). Examples of Privileged circumstances are outlined in the CCAB Guidance para. 7.3.

MLRO responsibility to provide sufficient content in the SAR 11. The MLRO has a duty to include as much relevant information as possible about the client and the activity which is suspicious. The SAR should be in the format outlined in s. 339 of POCA.

12. The MLRO must ensure that the content of the SAR includes as much useful information as possible: on the identity of the suspect (client's occupation/company's business and National Insurance etc); as much detail as possible to explain the basis on which the suspicion or knowledge of the money laundering is founded; the whereabouts of the illicit property if known.

Ensure that your reasons for suspicion are specific and make it clear what services you are actually providing/proposing to provide when the suspicion arose.

13. Additional information to be included in a SAR: Name of who the reporter, the date, who is suspected or information assisting identification who is suspected or information assisting identification process, name of victim(s) if applicable, other parties involved and how, facts of the matter giving rise to suspicion, whereabouts of any criminal property, what the involvement with the business is e.g. providing book-keeping service. Without good information it is difficult for law enforcement to provide consent to act if the MLRO has requested consent. The MLRO must make a report as soon as possible once a suspicion arises.

14. A decision by the MLRO not to report externally must be documented and have sufficient reason and detail outlined describing how the decision was made. The details of this should be filed separately from the actual client file to ensure the client is not tipped-off accidentally.

15. If the practice has already carried out an act for a client without realising that the act could amount to a money laundering offence, the MLRO should immediately make an authorised disclosure under POCA.

16. The MLRO should ensure that once a SAR is submitted with a request for consent that the practice does not do the act in question until the notice period has elapsed (i.e. 7 working days from day after SAR is made).

17. If consent to act is refused by the NCA or relevant FIU, the MLRO must wait until the 'moratorium period' has elapsed which is a period of 31 days starting with the day on which the person receives notice that consent to the doing of the act is refused.

18. Where consent is received or refused the NCA or relevant FIU will communicate this to the MLRO and it is the duty of the MLRO to ensure that the staff or officers act accordingly.

Case Scenario

The Sahara Group's case calls for a thorough probe to reveal all its money laundering dimensions.

- It has been about five years since the Supreme Court ordered the Sahara Group, led by Subrata Roy, to refund money that it borrowed investors without sufficient regulatory clearance.

- The Securities and Exchange Board of India, was tasked by the apex court to oversee the actual transfer of money from the Sahara Group to investors.
- The total amount, including interest on the initial principal, that need to be refunded to investor has bulged to about Rs 40,000 crore.
- Of this, SEBI has received an aggregate amount, including interest earned on deposits, of about Rs 14,487 crore from the Sahara Group.
- As per the SEBI's latest annual report, on March 31, 2017 only about Rs 85.02 crore,
- including interest of about Rs 38.05 crore, of this amount has actually been returned to investors.
- The Group failed to satisfy the apex court's request to provide evidence of the source of funds used to make the claimed return payments.
- The Enforcement Director began proceeding in 2014 against the Sahara Group under the prevention of money laundering act.

6.5.4 Know your Customers (KYC)

Reserve Bank of India has specified Know Your Customers (KYC) standards to be followed by banks and measures to be taken in regard to Anti Money Laundering (AML) and Combating of Financing of Terrorism (CFT). The guidelines incorporate:

- Obligations cast on banks under the Prevention of Money Laundering Act (PMLA), 2002
- Recommendations made by the Financial Action Task Force (FATF) on AML standards and CFT
- Customer Due Diligence (CDD) for banks as stipulated by the Basel Committee on Banking Supervision

The primary objective of the KYC, AML & CFT Policy of the Bank, hereinafter called the Policy, is to prevent the bank from being used, intentionally or unintentionally, by criminal elements for money laundering or financing of terror activities. Purpose proposed to be served by the Policy are:

- 1) To prevent criminal elements from using the Bank for money laundering activities
 - 2) To enable the bank to know/understand the customers and their financial dealing better which, in turn, would help the bank to manage risks prudently
 - 3) To put in place appropriate controls for detection and reporting of suspicious activities in accordance with applicable laws/laid down procedures
 - 4) To monitor cash transactions for filing of Cash Transaction Reports
 - 5) To file Counterfeit Currency Reports in case of detection of counterfeit currency
 - 6) To comply with applicable laws and regulatory guidelines.
 - 7) To ensure that the concerned staff are adequately trained in KYC/AML/CFT procedures.
 - 8) To facilitate monitoring of transactions in Non-profit Organizations (NPO) and file NPO transactions Reports.
- The policy/procedural guidelines are applicable to all domestic branches/offices of the bank and are current.
 - The instructions shall also apply to the branches and majority owned subsidiaries of the Bank located in countries which do not or insufficiently apply the FATF recommendations, to the extent local laws permit. When local applicable laws and regulations prohibit implementation of these guidelines, the same should be brought to the notice of RBI and the host is a variance in KYC/AML/CFT standards prescribed by RBI and the host country regulators, branches abroad/overseas subsidiaries will be required to adopt the more stringent regulation of the two.

IMPORTANT ELEMENTS OF KYC GUIDELINES ISSUED BY RBI, SEBI & IRDA

The KYC/AML/CFT policy of the bank has the following key elements:

- **Customer Acceptance Policy** : lays down the criteria for acceptance of customers. A customer is a person or entity that maintains an account. A person or entity that has a business relationship with the bank; one on whose behalf the account is opened (Beneficial Owner); any person or entity connected with Financial transaction.
- **Customer Identification Procedures (CIP)** : Customer identification requires identifying the customer and verifying his/her identity by using reliable, independent source documents, data or information. Thus, the prospective is actually who he/

she claims to be. The second requirement of CIP is to ensure that sufficient information is obtained on the identity and the purpose of the intended nature of the banking relationship. This would enable risk profiling of the customer and also to determine the expected or predictable pattern of transactions. Mandatory information required for KYC purpose, which the customer is obliged to give, shall be obtained at the time of opening an account or during periodic updation. Other optional customer details/additional information, if required, may be obtained separately after the account is opened with the explicit consent of the customer.

- **Risk Profiling of Customers:** Each and every account of the Bank is classified as “Low”, “Medium” or “High” Risk accounts for the purpose of monitoring and reporting of suspect transactions.

A few examples of High Risk Customers/Accounts are as under:

- 1) Politically Exposed person of foreign origin
- 2) Bullion Dealers/Jewellers
- 3) Non Resident Customers (NRIs)
- 4) Trust Charities, NGOs & Organisations receiving donations from India & abroad.
- 5) Non face-to-face Customers
- 6) Customers domiciled in/having transactions with High Risk Countries

A few examples of High Risk Customers/Accounts are as under:

- 1) Non-Bank Financial Institution
- 2) Stock Brokers
- 3) Import/Export customers
- 4) Telemarketers
- 5) Pawn shops
- 6) Auctioneers
- 7) Venture Capital Companies
- 8) All in-operative accounts

- 9) Individual account holder with Credit/Debit summations of Rs.50.00 lacs to below Rs.2.00 crores per annum
- 10) Non individual account holders with credit debit summations of Rs.2.00 crores to below Rs.10.00 crores per annum.

A few examples of Low Risk Customers/Accounts are as under:

- 1) Salaried Employees (whose salaries are well defined)
- 2) Customer belonging to lower economic strata, accounts opened under financial inclusion
- 3) NGOs/NPOs promoted by UN or its agencies.
- 4) Government owned Companies / Departments and (State/Central), PSUs, JVs with Govt., Regulators, FIs, Statutory bodies etc.
- 5) All customers not classified either as High/Medium Risk Categories
- 6) Individual Account holder with Credit/debit summations below Rs.50 lacs per annum.
- 7) Non individual account holders with credit debit summations below Rs.2 crores p. a.

Monitoring of Transactions- Each Bank has a dedicated Anti Money Laundering Monitoring officer for the Bank who is also designated as the Principal Officer (AML/CFT) of the Bank. With the aid of a suitable software programme, Alerts are generated for abnormal transactions. Each Bank have devised their own programmes for generation, analysis and reporting of Alerts of the previously discussed reports.

Alerts are of 2 types: On-line Alerts (Generated on daily basis with regards to certain defined parameters) and Off-line Alerts (Generated on a weekly basis); The AML Cell of each Bank Examines all the alerts to find out: Where no further probing is required, alerts are closed by AML Cell if further probing is required or not. Alerts requiring further information are sent to respective branches/units for further information.

On receipt of responses AML Cell evaluates the nature of alerts & responses received & decides to file or not to file STR

In terms of rules of PMLA 2002, banks are obliged to file following reports to the Financial Intelligence Unit-India (FIUIND) which has been set up as a national unit under Ministry of

Finance, for inter-alia, collecting, analysing, disseminating information in respect of financial transactions: -

- i. Cash Transactions Reports (CTRs)-(for Cash transactions of Rs.10 lacs or its equivalent and above and over)
- ii. Counterfeit Currency Reports (CCRs)
- iii. Suspicious Transactions Reports (STRs)- irrespective of any amount but abnormal to the profile of the Customer or enquiry
- iv. Non Profit Organisations Transactions Report (NTRs)-(of receipts of Rs. 10 lacs or its equivalent and above)
- v. Cross Border Wire Transaction Reports

Risk Management: Each Bank has identified a top management official to oversee the compliance of the AML/KYC guidelines/policies of the Bank and report to appropriate authorities (FATF, FIU-IND, RBI), in time.

KYC be carried out while

- (a) Opening a new customer account;
- (b) Existing accounts for which KYC was not done
- (c) While granting other banking facilities, such as locker access
- (d) When there are significant changes in signatories, mandate holders and beneficial owners
- (e) When non-account holders approach the Bank for one of transactions

Key Point for Remittance

Remittance of funds by way of Demand Drafts, Mail, Telegraphic transfer or any other mode and issue of Traveller's Cheque (TC's) for value of Rs. 50,000/- & above is effected by DEBIT to Customer's Account or Against Cheque, BUT NOT AGAINST CASH

DOCUMENTS REQUIRED FOR OPENING ACCOUNTS

Individuals (Documents acceptable as proof of identity/address)	• Passport • Voter's Identity Card • Driving Licence • Aadhaar Card • NREGA Card • PAN Card Any one document towards proof of identity and proof of address (either permanent or current)
Minors	*If minor is less than 10 years of age, ID proof of the person who will operate the account be obtained. *In case where minor can operate the account independently, KYC procedure for identification/address verification as in the case of any other individuals would apply.
NRIs	Passport and Residence Visa Copies, duly attested by • Foreign offices • Notary Public • Indian Embassy • Officers of correspondent banks whose signatures are verifiable through an authorized (A/B category Forex handling branch) branch of the Bank
For Proprietorship, Partnership, Ltd. Cos Trusts, Societies etc.,	In addition to the items mentioned in para 1 above for the individuals representing the units, appropriate deeds, permits, licences, agreements, certificate from applicable govt. authorities are to be obtained.

Laundering Process

The word "launder" means to wash iron clothes. Now in addition it also means informal transfer "of illegally obtained money "to conceal its origin".

Money laundering is variously defined. Indian law has defined (ref2) “money laundering in the prevention of money laundering act 2002”.

“Whoever directly or indirectly attempts to indulge or knowingly assists or knowingly is a party or is actually involved in any process or activity connected with the proceeds of crime and projecting it as untimed property shall be guilty of money laundering”

More comprehensive definitions of money laundering are:

1. “The concealment, conversion, or integration of tainted money or transfer of property knowing that such property is derived from serious crime, is money laundering. In other words, any action for the purpose of concealing or disguising illicit origin of the money or property or of assisting any person(s) who is involved in committing such an offence or offences to evade or save the person from legal consequences of his action and the concealment or disguise of the true nature, source, location, disposition, movement, rights with the respect to, or ownership of property, knowing that such property is derived from serious crime, is money laundering”.
2. “Money laundering is the process by which large amounts of money obtained from serious crimes (like drug trafficking, terrorists activities, etc) is given the appearance of having originated from a legitimate source”.
3. “Money laundering is the process by which black moneys are introduced into an economy and used for legitimate purposes”.

In common parlance money laundering is conversion of the illegitimate, black, tainted, dirty money, garnered illegally, into “white” legal money which is then used for legitimate investments or other legal, economic or social activities.

Examples:

1. A renowned company some years ago, floated a money deposit scheme in which they collected about 500 crores of rupees in about 2 years (1993-95). Some of it was used for purchase of new machinery. But most of it, more than 400 crores of rupees was laundered away by the management. (CH)¹
2. The enforcement directorate some years ago investigated a case relating to a famous tobacco company. The company has setup a number of front companies with the assistance of some Indian working abroad. The company propped up there tottering

image by showing huge profits through export, using this front companies to show transfer huge amounts buy these companies to the tobacco company. The money was generated through over invoicing of the exports. (CH2).

3. One drug smuggler, in USA, had laundered six hundred million dollars in a period of about six years before he was apprehended. He registered a firm in Nassau, Bahamas and deposited the proceeds of the smuggling in the name of that company in a bank in Bahamas which had branches in USA. After gestation period he would transfer the money from Bahamas to USA, through the local branch of the bank in USA.(CH3)

Money laundering is a criminal activity, carried out in secrecy known to only those who are concerned with the transaction(s). Money laundering is usually done through organized crime syndicates. The bank or the usual vehicle for all the process: to hide (or to smurf) to transfer the moneys to laundering havens, to convert black money into white money where and to transfer the money back to the launderer. The bank and crime syndicate are its indispensable arms. The following are its salient features:

- Money laundering involves large amounts of illegitimate money (or its equivalent)
- Money is obtained from serious crimes
- Money laundering is carried out in secrecy. only involved persons know about it
- The process is carried out by professionals usually international crime syndicates
 - o They have vast resources in men, materials and facilities.
 - o They use latest technologies to move the money from one place (or one country) to another. Electronic transfer is the usual mode.
 - o They often mix legitimate money with black money to transform the whole into white money.
 - o They hire expertise whenever needed.
- National, multinational (trans-national) banks are necessarily involved in the process, most of the times
- Money laundering is multi trillion dollar 'business' annually, worldwide.
- It is like an economic explosive bomb

- It can seriously damage the economy of any country
- In India the main money launderers are:
 - o Some politicians: kickback money!
 - o Kickbacks are common in practically all major national or international deals. Bofors gun purchase case of the 80s (still simmering!) and the 'oil for food' (2006) are glaring infamous examples.
 - o Bribe money when 'earned' in substantial amounts (the chairman of a public service commission made crores of rupees. He allegedly took about half a crores of rupees from everyone he selected for gazetted posts. Such 'earnings' need laundering.
 - o The business corporate and magnets hide tax evaded money or money obtained from black market. They launder such ill-gotten money.
 - o Tax evaded money and the tax saved there on is black money. Here the money earning may be illegal, but the secreted money and the tax thereon becomes black money. Money earned 'earned' through black marketing is also black money.
 - o Expert criminal importers and exporters rake in huge amounts of money by under or over invoicing as the situation demands. They may also substitute import or export licenses for more lucrative goods than the ones they are permitted to import or export.
 - o Crime syndicates: extortionists, drug smugglers, human trafficker, bootleggers, brothel owners, mercenary killers, terrorists, etc often have illegal money to launder:
 - § Narcotic drug trade in its various forms is flourishing everywhere.
 - § It generates pots of tainted money. It is laundered frequently. The business is expanding exponentially.
 - § Kidnapping, extortions, contract killings, robberies, murderers bring pots of illegal money.
 - § It is made legal through laundering.
 - § Smuggling of gold, weapons, narcotics and other goods brings in huge illegal profits. The money is laundered.

- § Bootlegging is a lucrative business to produce money. The money is often laundered to make it legal.
- § Prostitution and gambling syndicates collect large amounts of money. The tainted money is laundered to make it respectable.
- § Frauds: Bank frauds, share and stocks scams, chit fund frauds, stamp scams, spurious lands, even multiplications of SIM cards of cell phones make it possible to rake in millions. The money needs laundering.
- § Terrorists make money in many ways, they get money from their mentors, they loot their victims, they trade narcotics, they push fake currencies in the market and they collect ransoms. If they survive the ordeal they need money laundering.
- o Money laundering does not involve the common man nor does he witnesses the process; except when he is directly affected in the collection of illegal money in situations such in drug trafficking, prostitution, extortions, robberies, and the like. Most of the Indians do not know what money laundering!
- o Money laundering as global ramifications:
 - § Money laundering is really a big business.
 - § The exact figures are not known but it is estimated that, worldwide about 50 trillion rupees are laundered annually! And 'businesses' is expanding in value, volume and space like the world after big bang!

6.7 Laundering process in detail

It is usually a multistage process:

1. Collection of illegal money.
2. Hiding the money by 'smurfing' depositing the money in small amount in a number of banks) or transferring the money to foreign banks. The launderers call it 'Placement', 'covering', or 'layering'.
3. Conversion (changing) the black money to white money, is done mostly through fake business and fake companies both exist in papers only. The process is called 'conversion' or 'cleaning'.

4. Retransfer of money to its owner to the place of origin (or elsewhere), as 'white' money to be used as such. The process is called 'integration' or 'legitimization'. The launderer may use the money, now, as white money, in legitimate pursuits.

6.7.1 Placement

Placement is shifting the tainted money from the site of its origin to another site(s) where the people do not know the tainted nature of the money or the criminal launderer. It is the first step in the laundering process. It is really 'displacement' of the money to be laundered from the site of origin.

In placement of the black money, the launderer uses various modes.

- He may smurf the money
- He may send the money to foreign land through Hawala or laundering agents. Once it is out of the country safely, it is safe there. Most of the headache of the launderer is over.
- He may use other methods.

a) Smurfing

Placement of money is called 'smurfing' when the money is deposited, in convenient amounts, in a number of banks and their branches. The deposits may be in the banks of the country of origin, or in some foreign banks.

In smurfing the tainted money is deposited in small amounts. The size of the amounts is determined by the government rules. Usually, it is the highest amount for which the bank is not required to notify the government agencies. The deposits are repeated a number of time, in a number of branches of a number of banks, in a number of towns or cities. The actual number of deposits, in smurfing, depends mainly upon the amount of ill gotten money involved, the notoriety of the collection of the illicit money and the government rules prevailing at the time.

In a case of small time operation, a person wanted to convert a crore of rupees, he had collected as bribe money into white money. He contacted a money laundering syndicate on the recommendations of a close friend; he handed over the money to the representative of the syndicate. They charged 10% commission within 24 hours he got confirmation from his relative in USA that he had received the right amount in his account. The 'owner' received the money as gift cheques over a period from his relative. (CH5).

b) Transfer to foreign lands

Crime syndicates usually handle the foreign transfers. They have accounts with the banks carrying out the nefarious business and have proper contacts and rapport with their counterparts there. The country chosen carefully:

- It should have low taxation rates
- It protects the privacy accounts
- Its government does not ask too many questions about the origin of the money
- Its banks are technology savvy with electronic funds transfer facilities for transferring money to and fro. Thus they are able to receive and transfer huge amounts of black (and white) money instantly to intended safer places, without leaving any substantial trails.

c) Other methods

There are other methods of placement of money:

- The criminal simply hides the money at a distance place, a farm house or a building owned and controlled by him.
- Places the money in bank lockers, risky!
- Places the money with trust worthy relatives and to friends. Somewhat risky.
- Open letter(s) of credit(s) in bank for bogus foreign companies.
- Purchases fully paid insurance policies
- Purchases bank drafts. Traveller's cheques, foreign currency drafts.

6.7.2 Layering

The black money is kept anonymous till it has been cleaned and made white. The 'layering' or 'covering' of the black money has been facilitated by:

- **Secrecy laws in banking:** In most of the countries, the secrecy laws in banking used to help the launderer to keep the black money anonymous. The countries are looking into these laws. New laws do not permit secrecy of accounts and bank transactions. Government of India has almost completely done away with the parochial laws. Such as Switzerland, this was extensively used by the Indian

Launderer. The secret and numbered account there provided them with needed safety and secrecy.

- **The money is mixed with legitimate money** in transactions where substantial amounts are paid over and above the value mentioned in the papers. The excess money paid is tainted money.
- **Layering of black money has not been a problem in India.** The launderer could keep the loot in his home. (It has been found in the back yard in some cases!), with bank lockers, or in his farm house with impurity, unless he goes for an extravagant spending spree. In short if he is mum the black money is anonymous.
- There may be murmurs but never din and he gets away with the money most of the time. The laundering business is becoming a big business. Huge amounts are involved and number of persons who want to launder money and the cases of laundering are also increasing exponentially. The criminal has now to do his home work thoroughly to hide the money properly though he has ample choices of hide modes. He chooses one most suited to the given situation. The amount and extent of the noise (the murmur, the whisper, the talk of the town or the open media war!) about the tainted money or the main determiners. However bank lockers (is required number, in different banks or cities) are the hot favourites with the launderer of the small or medium stature.

a) Electronic Fund Transfers (EFT) shifts the money across the borders in no time and makes the black money anonymous. Any amount can be transferred to any place in the world, where EFT facility exists. The transfer does not take practically any time. It leaves no traces of the transfer. This money transfer is intermingled with thousands of legitimate transfers of which it forms only a fraction. It is difficult to identify the shady transfer. The tainted money becomes anonymous instantly. The wire transfer black money to foreign lands, in the name of a company or agency or even in the name of an individual. The mode layers any amount.

b) Instant Bank

The crime syndicate helps to create (purchase) and instant bank for the client in the foreign lands. The bank usually exists only in papers. The account of the money of the banks which the client sends to the bank goes to another bank which keeps the accounts of instant bank. If investigation starts the “instant banks” is destroyed along with the record and does, no trail of the tainted money is left for the investigator.

c) Serial accounts

The launderer opens a number of accounts in different branches of banks at different places the first bank is usually in the country of origin of the tainted money itself. After opening the account or after sometime, the customer instructs the bank to transfer all new deposits to a second bank, usually in a foreign land. He also instructs the bank to inform him if any law enforcement agency enquires about the same. In the eventuality he asks the second bank to transfer the amount to still another bank or to another branch of the same bank. The law enforcement agencies lose the trail quiet frequently.

There are quite a few financial institution, 'chartered accountants' firms, law firms, and shady trusts which transfer the black money to offshore banks. Modus operandi matches with that of the crime syndicate crooks.

d) International organized crime syndicates transfer the black money to foreign countries, by various modes. They are frequently used to transfer the money to make the money anonymous.

If money laundering is to be tackled successfully, it has to be tackled in the banking process involved in the work, when the money smurfed, collated and transferred to offshore banks and transferred to the launderer. Banks carrying out the process have to be watched.

6.7.3 Conversion

The conversion of the black money into white money involves corporation, agencies or business houses, real or fake, banks and other institutions. The process of conversion requires business activities flow of money and profits. They are 'created' and 'manipulated'. In fact the tainted black money is circulated. It is deposited in such organization which, after subjecting it to conversion processes (investment, profit, etc). Retransfer the money through banks to the sources of its origin or to the place where the launderer wants the same to be transferred. The cleaning process is also called 'legitimatization'.

The conversion of the illicit black money starts with its wire transfer to offshore banks, if the money is shown as profits 'earned' through some fake business activity, it stands converted into white money. The processes are varied and complex when large amounts are involved. The ingenuity and the resources of launderer's agents pay the pivotal role. The following methods of conversion are known:

- Bogus companies are created abroad, having financial resources, mostly in papers. They invest the laundered money as white moneys in the companies of the owner of the black money launderer in the country of origin (or elsewhere). The invested money becomes white money.
- Loans from local or foreign bogus banks or financial institutes are shown, of course, with all the legal strapping (receiving money through banks, receipts issued, and creation of legally valid looking agreement papers etc).
- Overvalued invoices are sent and their payment is received from the tainted money already sent. Invoicing process launders the money and makes it clean.
- Under invoicing of the goods received from abroad and sold at a much higher price (at the market price of the goods). Rakes in the tainted money as non tainted money.
- Benami properties are purchased with the black money in various countries and sold to get liquid money when needed, of course, after intervals.
- Stocks and shares are purchased in real or fake names and liquidated when liquidity is required.
- Purchase of lands and houses in big cities and at other distant places is another convenient conversion mode, frequently employed in India. Often the seller gives receipts of less than 50% of the sale price. The deals may be in real or fake names.

In an interesting case, a licensed foreign exchange dealer adapted an interesting practice. He created a rapport with a number of travel agencies. He could collect the passport numbers and flight ticket numbers of the persons who had gone abroad. He would show that he had sold the permitted amount of dollars to those travellers, without any such transactions. The 'proceeds' would be used to turn the black money into white money.(CH).

Rafking was computer expert. He wanted to get rich quick. He could not do so through the legal means. Frustrated he walked in Los Angeles bank. He knew the crew working in EFT section of the bank, as he had programmed the system. While gossiping he noticed how the money was transferred and learnt its code number.

Later, from a call booth he contacted the bank official with a little effort he got transferred 10.2 million dollars to diamond dealer in Switzerland with whom he had already had a deal to purchase diamonds.(CH)

Launderer can convert the tainted money into white money, himself. But it is risky. He might blunder and fall into the hands of the law enforcement agencies. The clever person, therefore, does not take the risk. He relies upon professional laundering syndicates but ensures that:

- The money and its ownership remains secret till it is converted into white money
- The white money is not linked to the involved black money, place or the activity.
- The money is not lost to a cheater agent (crime syndicate if one is used for conversion) or some other person. He tries to keep the control over the money as far as possible. But the judicious choice of crime syndicate and the bank handling the business is a better bet.

Numerous shady but well established crime syndicates and financial institutions exist whose main business is to launder money. The business is flourishing. They usually charge 10% commission. They use EFT to transfer large sums to foreign countries, show business activities and profit and then send the money in convenient desired form to the client. They carry out the work efficiently, having tentacles in most of the laundering centres of the world, in countries where lax tax laws prevail, where the privacy of bank accounts is honoured and where the law enforcement agencies do not enforce harsh and stringent countermeasures and often turns Nelson's blind eye towards shady and shoddy accounts of foreign origin.

There are dozens of such places in the world where the hiding and conversion is carried out with 'efficiency', the Cayman Island being one of the most important such laundering centre. The crime syndicates usually have legitimate business also to cover the conversion process in these shady dealings.

6.7.4 Integration

The final act in the laundering process is integration. It is to mix the converted money with the legitimate money of legitimate business (hotels, restaurants, real estate business or construction works) in the country of origin.

The 'black money' (all types of tainted money) after the cleaning is returned to the launderer. The owner is a respectable man now and a man of substance as he possesses substantial amount of money and can purchase men in power or power through men.

How does he receive the converted money?

The process is sometimes called integration of money. The following are the usual methods to collect the converted money:

- Use of credit cards: – when the money is deposited in foreign banks, international credit cards are convenient to draw or expend the money. The bank secrecy laws are also applicable to the accounts of the credit cards, in most of jurisdiction. Visa cards and master cards have been used for the purpose extensively.
- Formation of fake partnership: - a partnership deed is executed with a fake foreign firm which invests the black money with a firm of the owner of the black money in the latter's country. Over the years the partner disappears (often legally), leaving the money and the assets to the illicit money owner, the launderer.
- Loan subterfuge:- bogus firms or business houses are opened, often in papers 'Loans' from the foreign banks where the black money was processed or obtained. These 'loans' are in fact the original black money now termed 'Loan'. The launderer never returns the money. He may dissolve the company after suitable period declaring it insolvent. This is the frequent method employed when large amounts of the cleaned money are to be collected.
- Land sale:- the launderer arranges to sell land to a foreign fake company to establish business in the country. Highly inflated price are charged. The 'fake purchaser' hiked priced from the black money, received by him earlier.

Other modes:-

- o The owner of the illicit money becomes a fake consultant or an official of a foreign fake or real firm and remunerations (which comes from the black which he had already deposited in the bank) for the same on regular intervals.
- o He gets 'gift' cheques from his 'kith and kin'. The gift money comes from illicit money the recipient deposited with his 'kith and kin'.
- o He 'bins' a lottery of high value in fact either he purchases from the real winner: or, he fakes a win joining hands with some fraudsters. Launderer thus may hoodwink the law enforcement agencies of the country.

- o He uses under and over invoicing for imports and exports he covers the difference from the illicit money. He continues to do till he collects the total amount of the illicit money he has secreted.

6.8 Banks role in laundering

Banks are the life line of laundering 'businesses', as of other businesses. In fact money laundering of substantial sums of money starts with the bank (smurfing) and ends with the bank, when a bank sends back the converted black money to the launderer as white money.

A launderer misuses the bank in many ways for smurfing:

- o Savings accounts
- o Current accounts
- o Fixed deposits
- o Credit card accounts
- o Debit card accounts
- o ATM accounts

Shifts the illicit money from site(s) of origin to the site(s) of safety may be foreign banks through electronic transfers or traditional modes.

- o Helps smurf large amounts of money in a number of banks and the branches in small lots, though government of India has seen through the game and have come up with laws to check the repeated deposits.
- o Provide lockers to hide the money safely. The launderer frequently uses the lockers to keep cash or its equivalent: diamonds, gold etc. He also hides the shady property deals documents in the locker.
- o Change the money form: cash to bank drafts or travellers cheques, prepaid life insurance.
- o Provide mobile and serial banking facilities. They assist the launderer to hide the money at pinch.
- o Off-load the tainted money from the banks at the site of origin to foreign instant banks and save the launderer from immediate fear of being caught with the loot.

- o Allow opening of numbered accounts. Here the banks and the others do not know the persons who opened and maintain the accounts. The number alone is the link between the bank and the account holder.

Banks are the main instruments for the conversion of tainted money:

- o Provide 'loans' to propped up companies and business houses created by the launderer. They use the tainted money as the white money for this 'loans'. When these companies become insolvent the banks (fake) write off the loans as 'bad debts'.
- o Appoint a launderer as consultant or director and pay regular remunerations to the appointee from the tainted money deposited with them till it is finished.
- o Transmit money to the launderer as profits on behalf of business houses fake or real.
- o Manipulates stocks and shares with the launderer money and pay him the 'profits' or 'sales proceeds'.
- o Help the launderer bogus firms and companies and clean the money while handling their business.
- o Handle under invoicing and over invoicing processes for the launderer.

The banks provide the launderer white money in various forms:

- o As profits of its real or fake enterprises abroad.
- o As sale proceeds of his shares (fake or valid) abroad.
- o As remunerations for 'services' (only in papers) rendered as 'consultant' or director of enterprise.

6.8.1 Government of India instructions

Govt of India has seen through the immense role of the banker in money laundering. They have enacted stringent laws against the Bunco banker. Section 3 of the act is the pointer towards the same:

“Whoever directly or indirectly attempts to indulge or knowingly assists or knowingly is the party or is actually involved in any process or activity connected with the proceeds of crime and projecting it as untainted money shall be guilty of money laundering”.

In other words, a person, a bunco banker, a company (a banking company or any other financial institution) commits the offence of money laundering if he/ it is involved in any way, in a instruction connected with proceeds of crime. In short, is the culprit if he, knowingly does so directly or indirectly:

- o Acquires, owns, possesses or transfers any proceeds of crime; or
- o Enters into any transaction connected with the proceeds of crime; or
- o Conceals or aids in the concealment of the proceeds of crime.

The interpretation of the meaning of money laundering is significant. The bank could plead ignorance about the transactions previously. However, the 2005 rules have made it mandatory to the banker to know both the transactions (of the individual or of a company or intermediary) and also the clients who are carrying out business. In fact a very onerous duty (or, is it a burden?) has been imposed on the banks to assists the authorities in the fight against laundering.

According to this legal provisions whenever money is laundered through banks with the knowledge they are committing, willy-nilly, criminal act of laundering and today most of the money laundering transactions are through banks. They, does, are involved in laundering. The bankers have to know the related law, the mechanics of laundering collusion knowingly or unknowingly. In some countries the secrecy of accounts is still sacrosanct and they continue with these laws. Others have modified the laws for transparency of the transactions. Legislation in India has not only done away with the secrecy of bank accounts but the bank also required to assists the authorities to curb the menace of money laundering, in a big way. If they do not inform they can be fined: “he may, by an order, levy a fine on such banking company or financial institution or intermediary which shall not be less than 10,000 Rupees and may extend to 1 lakh rupees for each offence”.

The amount of fine is too low as to act deterrent to the banker. In fact, it is mockery vis-a-vis equation of crime and punishment. The amounts involved in laundering are in crores of rupees. The launderer can pay the fine (to the banker) as business expenses. It is not even a fraction of an amount the launderer pays as an amount as commission to the crime syndicate

which turns its black money into white money. A better deterrent has to be devised if it is to be made a real deterrent.

Govt of India has specified the role of the banks in the fight against money laundering in the prevention of money laundering act and the rules (2). Clear instructions issued to the banks and other financial institutions including insurance companies and intermediaries or in GSR 444 (e), dated 1st July 2005. The salient features of these rules are:

- Know the client – the banker must know his client properly. He ascertains the identity of the customer at the time of the opening account (savings, current, fixed, ATM, Credit card, mutual fund, etc) or soon a after. In case of individual the identity is established from ‘officially valid documents’:
 - o Voters identity cards issued by the election commission
 - o Passport
 - o PAN number (issued by the income tax department)
 - o Driving license
- Know the clients financial status – the banker must know the clients financial status, his business or job, all of his addresses including his permanent address.
- The banker must keep a copy of his recent photograph.
- If the client happens to be a company or firm; the companies incorporation documents, partnership documents, resolutions of the directors of the company, conferring powers on various individuals and power of attorney holding offices who are to transact business on behalf of the company must be scrutinized and relevant copies kept.
- The account holders have to be real persons not pseudonymous.
- The banker must maintain records of all cash transactions involving more than 10 lakhs of rupees.
- If there are a series of similar cash transactions in a month, involving amounts even less than 10 lakhs, the banker has to keep the record. He must produce the record to the central government specified authorities on demand.

The prevention of money laundering act 2002 finishes secrecy laws. However the defaulter banker has been given preferential treatment. If he is a party to the laundering process he may be fined 10,000 to 1 lakh rupees. The amount is chicken feed when compared with the amounts that is laundered (in crores of rupees).

6.8.2 Maintain records

Banking companies, financial institutions and intermediaries to maintain records:

- Every banking company, financial institutions and intermediary shall:
 - o Maintain a record of all transaction, nature and value of which may be prescribed whether such transactions comprise of a single transaction or a series of transactions integrally connected to each other and where such a series of transactions take place within a month.
 - o Furnish information of transactions to in clause (a) to the director within such time may be prescribed;
 - o Verify and maintain the records of the identity of all the clients, in such manner as may be prescribed.

Provided that where the principle of a banking company or financial institution or intermediary as the case may be has the reason to believe that a single transactions or a series of transactions integrally to each other have been valued below the prescribed value so as to defeat the provision of the section, such officer shall furnish information in respect of such transaction to the director within the prescribed time.

The above record has to be maintained for 10 years after the cessation of the transaction.

The enactment is elaborated in the rules relating to the maintaining of records of transaction (nature and value) GSR 444 (e). Dated 1st July 2005 for handling the bank records information etc.

6.9 Anti-money laundering and legal frameworks

- o International and regional bodies - International initiatives - Terrorist financing
- o The US Patriot Act
- o The UK Model – The Jersey Strategy on Terrorist funding and money laundering

- o National anti money laundering frameworks
- o PMLA 2002- Objectives and salient features
- o Financial Intelligence Unit India (FIU-IND)
- o Financial Action task Force (FATF)
- o Enforcement Directorate
- o Serious Fraud Investigation Office (SFIO)
- o KYC

6.9.1 Anti Money Laundering and legal framework

International and regional bodies that combat money laundering

Money laundering is a global phenomenon that thrives parasitically on the world's financial markets. It ignores national boundaries in much the same way that the financial market place does. It has been accepted that this problem requires both national and international countermeasures.

Economies built on the back of criminal activity are unstable and pose significant risks to neighbouring countries and those with close economic ties. The impetus for international governmental action against money laundering also stems from a fear of national or regional economic destabilisation resulting from over exposure to illegally derived funds.

The global financial market provides criminals with unlimited opportunities to launder money. Global efforts to prevent money laundering are only as strong as the weakest national AML framework. International initiatives have been designed to encourage national governments into taking or strengthening measures to prevent money from being laundered within their borders. Such initiatives come from international bodies like for example the Financial Action Task Force (FATF) and prompt regional action such as EU Directives, which in turn result in national legislation.

The criminalization of money laundering is a relatively recent legal development and all the international efforts to prevent money laundering are relatively young:

- The first international initiative during the early 1980s focused on the prevention of the laundering of funds derived solely from drug trafficking.

- Then the focus was expanded to include the prevention of the laundering of the proceeds of a wider range of criminal activity.

After the terrorist attacks in the US on 11 September 2001, the international community recognized the importance of preventing the financing of terrorism and international and regional bodies previously responsible for money laundering prevention expanded their remits to include the prevention of terrorist financing.

Some of the most influential international and regional bodies in the anti money laundering

A. The Bank for International Settlements (BIS)

The Bank for International Settlements was established on 17 May 1930. It is the world's oldest international financial institution.

The head office is in Basel, Switzerland and there are two representative offices: in the Hong Kong Special Administrative Region of the People's Republic of China and in Mexico City.

The BIS currently has 60 member central banks, all of which are entitled to be represented and vote in the General Meetings. Voting power is proportionate to the number of BIS shares issued in the country of each member represented at the meeting.

Board of the BIS

The Board of Directors of BIS may have up to 21 members, including six ex officio directors, comprising the Governors of the founding members (central banks of Belgium, France, Germany, Italy and the United Kingdom and the Chairman of the Board of Governors of the US Federal Reserve System). Each ex officio member may appoint another member of the same nationality. The Statutes also provide for the election to the Board of not more than nine Governors of other member central banks.

The Board of Directors elects a Chairman from among its members for a three-year term. The Board also elects a Vice-Chairman.

Mission

The mission of the Bank for International Settlements (BIS) is: (a) serve central banks in their pursuit of monetary and financial stability, (b) to foster international cooperation in those areas and (c) to act as a bank for central banks.

In broad outline, the BIS pursues its mission by:

- promoting discussion and facilitating collaboration among central banks;
- supporting dialogue with other authorities that are responsible for promoting financial stability;
- conducting research on policy issues confronting central banks and financial supervisory authorities;
- acting as a prime counterparty for central banks in their financial transactions;
- serving as an agent or trustee in connection with international financial operations;

The BIS has no authority to make legally binding pronouncements but formulates broad supervisory standards and guidelines and recommends best practices. The most important of these for Money Laundering is called Basel Principles – Prevention of criminal use of the banking system for the purpose of money-laundering and has been issued in 1988.

The above Statement is a general statement of ethical principles which encourages banks' management to put in place effective procedures to ensure that all persons conducting business with their institutions are properly identified; that transactions that do not appear legitimate are discouraged; and that cooperation with law enforcement agencies is achieved. The Statement is not a legal document and its implementation will depend on national practice and law.

Basel Committee

The Basel Committee on Banking Supervision provides a forum for regular cooperation on banking supervisory matters. Its objective is to enhance understanding of key supervisory issues and improve the quality of banking supervision worldwide. It seeks to do so by exchanging information on national supervisory issues, approaches and techniques, with a view to promoting common understanding. At times, the Committee uses this common understanding to develop guidelines and supervisory standards in areas where they are considered desirable. In this regard, the Committee is best known for its international standards on capital adequacy; the Core Principles for Effective Banking Supervision; and the Concordat on cross-border banking supervision.

The Committee encourages contacts and cooperation among its members and other banking supervisory authorities. It circulates to supervisors throughout the world both published and unpublished papers providing guidance on banking supervisory matters. Contacts have

been further strengthened by an International Conference of Banking Supervisors (ICBS) which takes place every two years

B. The International Monetary Fund (IMF)

The IMF was established in 1945 as a specialized agency of the United Nations system within the framework of the Bretton Woods Accords.

- It has now 188 member countries
- The founders aimed to build a framework for economic cooperation that would avoid a repetition of the disastrous economic policies that had contributed to the Great Depression of the 1930s and the global conflict that followed.
- It was to do this by (a) providing a forum for cooperation on international monetary problems; (b) facilitating the growth of international trade, thus promoting job creation, economic growth, and poverty reduction; (c) promoting exchange rate stability and an open system of international payments; and (d) lending countries foreign exchange when needed, on a temporary basis and under adequate safeguards, to help them address balance of payments problems.
- Over time, the IMF has concentrated more on surveillance of individual countries and providing technical assistance to them. It (a) monitors economic and financial developments in member countries; (b) gives policy advice on macro-economic policy, balance of payments, regulation and supervision of banks and other financial institutions and structural policies (such as labour market policies); and (c) provides training.
- The IMF supports its membership by providing (a) policy advice to governments and central banks based on analysis of economic trends and cross-country experiences; (b) research, statistics, forecasts, and analysis based on tracking of global, regional, and individual economies and markets; (c) loans to help countries overcome economic difficulties; (d) concessional loans to help fight poverty in developing countries; and (e) technical assistance and training to help countries improve the management of their economies.
- In reviewing the prudential regulation of particular jurisdictions, the IMF analyses the role of regulatory and law enforcement bodies in the prevention of money laundering. Unlike FATF, the IMF does not seek to set international AML standards;

instead it plays an important role in evaluating jurisdictional compliance with them.

C. United Nations Office on drugs and Crime (UNODC)

Set up by the United Nations in 1997 to implement the United Nations International Drug Control Programme (UNDCP) and the Crime Prevention and Criminal Justice Programme (CPCJP), both of which were established by resolutions of the UN General Assembly in 1991.

Its overall aims are to assist UN member states in the struggle against illicit drugs, crime and terrorism. It does this in three main ways: (a) research and analytical work; (b) helping countries and implementing treaties and drafting domestic legislation; and (c) field-based or on the ground assistance, which includes training judicial officials.

UNODC's Global Programme Against Money Laundering (GPML) helps member states to:

- Introduce legislation against money laundering and to develop and maintain strategies to combat money laundering
- Encourage AML policy development
- Raise public awareness about money laundering, and
- Coordinate joint AML initiatives by the United Nations with other international organizations

It has developed, in collaboration with the UNDCP Legal Advisory Section, model laws for both common law and civil law legal systems to assist countries in setting up their AML legislation in full compliance with international legal instruments against money laundering.

D. The Commonwealth Secretariat

The Commonwealth Secretariat provides guidance on policy making, technical assistance and advisory services to Commonwealth member countries. It supports governments to help achieve sustainable, inclusive and equitable development.

Its work promotes democracy, rule of law, human rights, good governance and social and economic development. It is a voice for small states and a champion for youth empowerment.

Priority areas of work are agreed at Commonwealth Heads of Government Meetings, which occur every two years. The next summit is in Malta in 2015.

- Established by Commonwealth heads of governments in 1965
- The principal inter-governmental organization of the Commonwealth
- In 1993 the Commonwealth Secretariat released an anti money laundering common law model, followed in 1996 by a set of Commonwealth Guidance Notes for the Financial Sector
- In July 2000 it published an updated version of its guidance notes entitled A Model of Best Practice for Combating Money Laundering in the Financial Sector
- In response of the UN resolution 1373, which followed 11 September 2001, the Commonwealth set up the Commonwealth Committee on Terrorism (CCT). That reported in 2002: the report provided drafting instructions for model counter terrorism legislation.
- Both the Model Law on Money Laundering and the Model of Best Practice were reviewed in 2004.

E. The European Commission and Council

The role of the EU is important in the AML arena and provides an excellent example of the manner in which national governments have been required to implement legislation to forestall AML legislation in line with the expectations of a transnational organization. Both the Commission and the Council have introduced measures aimed at forestalling money laundering. The most recent of these is the third EU Directive and recently a revised draft Convention on Laundering, Search, Seizure and Confiscation of the Proceeds of Crime.

F. The organisation for Economic Development (OECD)

- The Organisation for Economic Co-operation and Development (OECD) was officially born on 30 September 1961.
- Other countries joined in, starting with Japan in 1964. Today, 34 OECD member countries worldwide regularly turn to one another to identify problems, discuss and analyse them, and promote policies to solve them. The track record is striking.
- The mission of the Organisation for Economic Co-operation and Development (OECD) is to promote policies that will improve the economic and social well-being of people around the world.

- OECD uses its wealth of information on a broad range of topics to help governments foster prosperity and fight poverty through economic growth and financial stability. It helps ensure the environmental implications of economic and social development are taken into account.
- OECD's work is based on continued monitoring of events in member countries as well as outside OECD area, and includes regular projections of short and medium-term economic developments. The OECD Secretariat collects and analyses data, after which committees discuss policy regarding this information, the Council makes decisions, and then governments implement recommendations.
- Money laundering and harmful tax practices are two issues in which the OECD has taken a significant interest in recent years.
- The OECD made a significant contribution to the global effort against money laundering by spawning the Financial action Task Force (FATF) at its economic summit in 1989.

G. The Financial action Task Force (FATF)

- In response to mounting concern over money laundering, the Financial Action Task Force on Money Laundering (FATF) was established by the G-7 Summit that was held in Paris in 1989.
- Recognising the threat posed to the banking system and to financial institutions, the G-7 Heads of State or Government and President of the European Commission convened the Task Force from the G-7 member States, the European Commission and eight other countries.
- The Task Force was given the responsibility of examining money laundering techniques and trends, reviewing the action which had already been taken at a national or international level, and setting out the measures that still needed to be taken to combat money laundering. In April 1990, less than one year after its creation, the FATF issued a report containing a set of Forty Recommendations, which provide a comprehensive plan of action needed to fight against money laundering.
- In 2001, the development of standards in the fight against terrorist financing was added to the mission of the FATF. In October 2001 the FATF issued the Eight Special Recommendations to deal with the issue of terrorist financing. The continued evolution of money laundering techniques led the FATF to revise the FATF standards

comprehensively in June 2003. In October 2004 the FATF published a Ninth Special Recommendations, further strengthening the agreed international standards for combating money laundering and terrorist financing – the 40+9 Recommendations.

- One of FATF's functions is to review and report on money laundering trends, techniques and methods (also referred to as typologies). To accomplish this aspect of its mission, FATF issues annual reports on developments in money laundering through its Typologies Report. These reports are very useful for all countries, not just FATF members, to keep current with new techniques or trends to launder money and for other developments in this area.
- During 1991 and 1992, the FATF expanded its membership from the original 16 to 28 members. In 2000 the FATF expanded to 31 members, and has since expanded to its current 36 members; 34 jurisdictions and 2 regional organisations (the Gulf Cooperation Council and the European Commission).
- The Financial Action Task Force (FATF) is an inter-governmental body whose purpose is the development and promotion of policies, both at national and international levels, to combat money laundering and terrorist financing.
- The FATF is a “policy-making body” which works to generate the necessary political will to bring about national legislative and regulatory reforms to combat money laundering and terrorist financing.
- The FATF monitors members' progress in implementing necessary measures, reviews money laundering and terrorist financing techniques and counter-measures, and promotes the adoption and implementation of appropriate measures globally.

The Financial action Task Force (FATF) – Self assessment and mutual evaluations

Monitoring the progress of members to comply with the requirements of the Forty Recommendations is facilitated by a two-stage process: self assessments and mutual evaluations. In the self assessment stage, each member responds to a standard questionnaire, on an annual basis, regarding its implementation of the Forty Recommendations. In the mutual evaluation stage, each member is examined and assessed by experts from other member countries.

Members of the FATF are strongly committed to the discipline of multilateral peer review. The mutual evaluation programme is considered as the primary instrument by which

the FATF monitor progress made by member governments in implementing the FATF Recommendations.

The mutual evaluation and assessment processes are fundamental to the work of the FATF and all other AML/CFT assessment bodies. In June 2010, the FATF Plenary approved a set of Key Principles for Mutual Evaluations and Assessments. These Key Principles have been prepared by the FATF in collaboration with the FATF-style regional bodies (FSRBs), the IMF and World Bank. They set out the fundamental objectives, principles and essential underpinnings for the assessment processes, and their observance will enhance the quality and consistency of the mutual evaluation and detailed assessment reports and of the applicable procedures.

In the event that a country is unwilling to take appropriate steps to achieve compliance with the Forty Recommendations, FATF recommends that all financial institutions give special attention to business relations and transactions with persons, including companies and financial institutions, from such non-compliant countries and, where appropriate, report questionable transactions, i.e. those that have no apparent economic or visible lawful purpose, to competent authorities. Ultimately, if a member country does not take steps to achieve compliance, membership in the organisation can be suspended. There is, however, the process of peer pressure before these sanctions are enforced.

The Financial action Task Force (FATF) –NCCT

The FATF in order to encourage all countries to adopt measures to prevent, detect and prosecute money launderers, i.e. to implement The Forty Recommendations, has adopted a process of identifying those jurisdictions that serve as obstacles to international cooperation in this area. The process uses 25 criteria, which are consistent with the Forty Recommendations, to identify such non-cooperative countries and territories (NCCT's) and place them on a publicly available list. A NCCT country is encouraged to make rapid progress in remedying its deficiencies. In the event a NCCT does not make sufficient progress, counter-measures may be imposed. Counter measures consist of specific actions by FATF member countries taken against NCCT-listed countries.

In addition to the obligation of applying special attention to business relationships and transactions from such countries, the FATF can also impose further counter-measures, which are to be applied in a gradual, proportionate and flexible manner. These include:

- Stringent requirements for identifying clients and enhancement of advisories, including jurisdictions-specific financial advisories, to financial institutions for identification of the beneficial owners before business relationships are established with individuals or companies from these countries.
- Enhanced relevant reporting mechanisms or systematic reporting of financial transactions on the basis that financial transactions with such countries are more likely to be suspicious.
- In considering requests for approving the establishment in FATF member countries of subsidiaries or branches or representative offices of banks, taking into account the fact that the relevant bank is from a NCCT.
- Warning non-financial sector businesses that transactions with entities within the NCCTs might run the risk of money laundering.

Finally, these counter measures may include FATF-member countries terminating transactions with financial institutions from such a country. Most countries make a concerted effort to be taken off the NCCT list because it causes significant problems for their financial institutions and businesses with respect to international transactions, as well as their reputation internationally.

Measures to be taken with respect to countries that do not or insufficiently comply with the FATF Recommendations are also recommendations 18 and 19.

Recommendation 18

Financial institutions should be required to implement programmes against money laundering and terrorist financing. Financial groups should be required to implement group-wide programmes against money laundering and terrorist financing, including policies and procedures for sharing information within the group for AML/CFT purposes.

Financial institutions should be required to ensure that their foreign branches and majority owned subsidiaries apply AML/CFT measures consistent with the home country requirements implementing the FATF Recommendations through the financial groups' programmes against money laundering and terrorist financing.

Recommendation 19

Financial institutions should be required to apply enhanced due diligence measures to business relationships and transactions with natural and legal persons, and financial institutions, from countries for which this is called for by the FATF. The type of enhanced due diligence measures applied should be effective and proportionate to the risks.

Countries should be able to apply appropriate countermeasures when called upon to do so by the FATF. Countries should also be able to apply countermeasures independently of any call by the FATF to do so. Such countermeasures should be effective and proportionate to the risks. The following table lists down the members of FATF

Table 6.1: Members of the FATF

	The 36 Members of the FATF		
Argentina	Argentina	Hong Kong	Norway
Australia	Australia	China	Portugal
Austria	Austria	Iceland	Russian Federation
Belgium	Belgium	India	Singapore
Brazil	Brazil	Ireland	South Africa
Canada	Canada	Italy	Spain
China	China	Japan	Sweden
Denmark	Denmark	Republic of Korea	Switzerland
European Commission	European Commission	Luxembourg	Turkey
Finland	Finland	Mexico	United Kingdom
France	France	Netherlands,	United States
Germany	Germany	Kingdom of New Zealand	
Greece	Greece		
Gulf Co-operation Council	Gulf Co-operation Council		

Ref: <http://calert.info/details.php?id=1247>

The US Patriot Act

The official title of the USA PATRIOT Act is “Uniting and Strengthening America by Providing Appropriate Tools Required to Intercept and Obstruct Terrorism (USA PATRIOT) Act of 2001.” To view this law in its entirety, click on the USA PATRIOT Act link below.

USA PATRIOT Act

The purpose of the USA PATRIOT Act is to deter and punish terrorist acts in the United States and around the world, to enhance law enforcement investigatory tools, and other purposes, some of which include:

- To strengthen U.S. measures to prevent, detect and prosecute international money laundering and financing of terrorism;
- To subject to special scrutiny foreign jurisdictions, foreign financial institutions, and classes of international transactions or types of accounts that are susceptible to criminal abuse;
- To require all appropriate elements of the financial services industry to report potential money laundering;
- To strengthen measures to prevent use of the U.S. financial system for personal gain by corrupt foreign officials and facilitate repatriation of stolen assets to the citizens of countries to whom such assets belong.

Below is a brief, non-comprehensive overview of the sections of the USA PATRIOT Act that may affect financial institutions:

Section 311: Special Measures for Jurisdictions, Financial Institutions, or International Transactions of Primary Money Laundering Concern

This Section allows for identifying customers using correspondent accounts, including obtaining information comparable to information obtained on domestic customers and prohibiting or imposing conditions on the opening or maintaining in the U.S. of correspondent or payable-through accounts for a foreign banking institution.

Section 312: Special Due Diligence for Correspondent Accounts and Private Banking Accounts

This Section amends the Bank Secrecy Act by imposing due diligence & enhanced due diligence requirements on U.S. financial institutions that maintain correspondent accounts for foreign financial institutions or private banking accounts for non-U.S. persons.

Section 313: Prohibition on U.S. Correspondent Accounts with Foreign Shell Banks

To prevent foreign shell banks, which are generally not subject to regulation and considered to present an unreasonable risk of involvement in money laundering or terrorist financing, from having access to the U.S. financial system. Banks and broker-dealers are prohibited from having correspondent accounts for any foreign bank that does not have a physical presence in any country. Additionally, they are required to take reasonable steps to ensure their correspondent accounts are not used to indirectly provide correspondent services to such banks.

Section 314 helps law enforcement identify, disrupt, and prevent terrorist acts and money laundering activities by encouraging further cooperation among law enforcement, regulators, and financial institutions to share information regarding those suspected of being involved in terrorism or money laundering.

- Section 314(a)
- Section 314(b)
- Section 319(b): Bank Records Related to Anti-Money Laundering Programs
- To facilitate the government's ability to seize illicit funds of individuals and entities located in foreign countries by authorizing the Attorney General or the Secretary of the Treasury to issue a summons or subpoena to any foreign bank that maintains a correspondent account in the U.S. for records related to such accounts, including records outside the U.S. relating to the deposit of funds into the foreign bank. This Section also requires U.S. banks to maintain records identifying an agent for service of legal process for its correspondent accounts.
- Section 325: Concentration Accounts at Financial Institutions
- Allows the Secretary of the Treasury to issue regulations governing maintenance of concentration accounts by financial institutions to ensure such accounts are not used to obscure the identity of the customer who is the direct or beneficial owner of the funds being moved through the account.
- Section 326: Verification of Identification
- Prescribes regulations establishing minimum standards for financial institutions and their customers regarding the identity of a customer that shall apply with the opening of an account at the financial institution.
- Section 351: Amendments Relating to Reporting of Suspicious Activities

- This Section expands immunity from liability for reporting suspicious activities and expands prohibition against notification to individuals of SAR filing. No officer or employee of federal, state, local, tribal, or territorial governments within the U.S., having knowledge that such report was made may disclose to any person involved in the transaction that it has been reported except as necessary to fulfill the official duties of such officer or employee.
- Section 352: Anti-Money Laundering Programs
- Requires financial institutions to establish anti-money laundering programs, which at a minimum must include: the development of internal policies, procedures and controls; designation of a compliance officer; an ongoing employee training program; and an independent audit function to test programs.
- Section 356: Reporting of Suspicious Activities by Securities Brokers and Dealers; Investment Company Study
- Required the Secretary to consult with the Securities Exchange Commission and the Board of Governors of the Federal Reserve to publish proposed regulations in the Federal Register before January 1, 2002, requiring brokers and dealers registered with the Securities Exchange Commission to submit suspicious activity reports under the Bank Secrecy Act.
- Section 359: Reporting of Suspicious Activities by Underground Banking Systems
- This amends the BSA definition of money transmitter to ensure that informal/ underground banking systems are defined as financial institutions and are thus subject to the BSA.
- Section 362: Establishment of Highly Secure Network
- Requires FinCEN to establish a highly secure network to facilitate and improve communication between FinCEN and financial institutions to enable financial institutions to file BSA reports electronically and permit FinCEN to provide financial institutions with alerts.

https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/468210/UK_NRA_October_2015_final_web.pdf

The consequences of criminals successfully laundering money through a particular sector

were assumed to be severe for all the areas covered. 1.19 Following the scoring of vulnerabilities and likelihoods, the matrix produces a score for the thematic area's risk, which is then categorised into Low, Medium or High risk levels. The model then considers the mitigating measures in place in terms of UK law enforcement agencies and supervisors' capability and capacity to combat money laundering in that area. The combined mitigation score has the potential to reduce the overall risk level of a thematic area where law enforcement/supervisory activity effectively mitigates the risk. Once applied, an overall risk score for the thematic area is calculated and then categorised.

Table 6.2 : National Risk Assessment on Money Laundering

National risk assessment on money laundering						
Thematic area	Total vulnerabilities score	Total likelihood score	Structural risk	Structural risk level	Risk with mitigation grading	Overall risk level
Banks	34	6	211	High	158	High
Accountancy service providers	14	9	120	High	90	High
Legal service providers	17	7	112	High	84	High
Money service businesses	18	7	119	High	71	Medium
Trust or company service providers	11	6	64	Medium	64	Medium
Estate agents	11	7	77	Medium	58	Medium
High value dealers	10	6	56	Low	42	Low
Retail betting (unregulated gambling)	10	5	48	Low	36	Low
Casinos (regulated gambling)	10	3	32	Low	24	Low
Cash	21	7	147	High	88	High
New payment methods (e-money)	10	6	60	Medium	45	Medium
Digital currencies	5	3	15	Low	11	Low

Threats and vulnerabilities

Criminals can use accountants to conceal the origins of criminal funds and/or legitimise accounts in a variety of ways, such as the creation of companies, trusts and offshore corporate

structures; providing false accounts; preparation or audit of businesses' annual accounts; insolvency malpractice; and providing advice. Many of the vulnerabilities set out below also leave accountants open to being used, wittingly or unwittingly, to assist the financing of terrorism.

The key threats and vulnerabilities within this sector identified through this assessment are:

- complicit professionals facilitating money laundering
- collusion with other elements of regulated sector
- coerced professionals targeted by criminals
- creation of structures and vehicles that enable money laundering
- the provision of false accounts
- failure to identify suspicion and submit SARs
- low barriers to entry and mixed standards of compliance with the regulators across the sector
- 4.1 This section outlines the current UK law enforcement landscape, and our law enforcement and prosecution agencies' response to money laundering.
- 4.2 The UK's law enforcement and prosecution agencies operate at the national, regional and local levels (see table 4.A). All of the law enforcement agencies described below have powers to investigate money laundering (as provided in section 7 of the Proceeds of Crime Act 2002).¹ The National Control Strategy prioritises the threats of serious and organised crime and cross-cutting issues, providing a framework that informs the response across law enforcement. Decisions on what cases to investigate, and what priority to give to particular types of crime (such as money laundering), rest with the law enforcement agencies themselves.

Table 6.3: UK law enforcement and prosecution agencies at local, regional and national level

Local		43 police forces in England and Wales (local policing in Northern Ireland and Scotland is the responsibility of the Police Service of Northern Ireland and Police Scotland).		
Regional		9 Regional Organized Crime Units (ROCUs) in England and Wales, which include Regional Assets Recovery Teams (RARTs) and Asset Confiscation Enforcement (ACE) teams.	Regional Crown Prosecution Service offices (including dedicated asset recovery resources co-located with ROCUs).	
National	HM Revenue & Customs	National Crime Agency	Serious Fraud Office	National Crown Prosecution Service Functions

Regional Organized Crime Units and Regional Asset Recovery Teams

4.6 Police forces in all nine policing regions in England and Wales have collaborated to form Regional Organized Crime Units (ROCUs). These units deliver specialist investigative and intelligence capabilities to all forces within a region, to help tackle serious and organized crime. ROCUs are the primary interface between the National Crime Agency (NCA) and forces and are accountable to their respective Police and Crime Commissioners.

4.7 They support the national coordination and tasking of the effort against serious and organized crime by providing capabilities for the police response to activity with regional impact. They also support local forces, providing specialist resources and tactical advice to support local operations to counter serious and organized crime.

4.8 Within each ROCU is a Regional Asset Recovery Team (RART). The RARTs develop financial intelligence in aid of investigation and disruption of subjects. They utilize financial investigation to conduct money laundering investigations, disrupt subjects and recover assets

through POCA legislation. There are approximately 180 staff in the RARTs, all of whom are operational.

4.9 In 2013/14 the RARTs secured approximately £1 million in cash forfeitures and 261 confiscation orders with an equivalent value of approximately £22 million. The regional ACE teams, established in September 2014, recovered approximately £7 million of assets in the first 6 months of operations.

National Crime Agency

4.10 The National Crime Agency (NCA) is the lead agency for the response to serious and organized crime in the UK.

4.11 The NCA's strategic priorities are set by the Home Secretary. The first is to identify and disrupt serious and organized crime including by investigating and enabling the prosecution of those responsible.⁵ Its principal functions are to reduce crime and to gather, analyze and disseminate criminal intelligence.⁶

4.12 Money laundering and criminal finance are important areas of work for the NCA. Money laundering, and bribery and corruption which is closely associated with it, have been identified as high priority threats in the NCA National Control Strategy, which prioritizes the threats of serious and organized crime.

4.13 The NCA has an intelligence hub which is responsible for gathering, analyzing and disseminating information and an Economic Crime Command which is responsible for leading, supporting and coordinating resources to counter economic crime across the UK – including law enforcement, regulatory bodies and the private sector. This includes law enforcement efforts on money laundering, bribery and corruption, asset recovery and asset denial. It also supports partners, for example in the SFO and FCA by providing the NCA's investigative capabilities to tackle high priority economic crime threats.

4.14 The NCA co-ordinates activity through the multi-agency Criminal Finances Threat Group, which has sub-groups on cash-based money laundering (led by HMRC); non-cash based money laundering (led by NCA); and professional enablers (led by SFO).

4.15 NCA is focusing on money laundering and international corruption to protect the UK as an international financial centre. It is proactively using financial investigation and other law

enforcement techniques in intelligence and evidence-gathering to target money laundering, as well as asset recovery tools after the event. It has various tools available to it including:

- intelligence and evidence-gathering
- cash seizure and forfeiture
- restraint and confiscation
- civil recovery and taxation

4.16 On asset recovery the NCA's priority is denying criminals their assets by every lawful means it can, not just recovering them. The focus is on the disruptive value of taking assets away, not the size of the returns. However, in its first year, NCA led and coordinated operational activity that resulted in almost £126 million (£22 million domestically and £104 million internationally) being denied to criminals impacting on the UK. In regards to assets recovered, in its first year the NCA achieved returns of £22.5 million.

4.17 The NCA also works closely with its partners to assist their efforts against criminal finances, for example by working with UK police forces to identify and seize money derived from criminal activity. Since the NCA was established in 2013, its activity has led directly to operational partners seizing over £16 million in cash and making around 150 related arrests.

4.18 NCA has also set up a dedicated Asset Confiscation Enforcement (ACE) team to coordinate activity across the Agency and with its partners, including the police, HMRC, SFO and HM Courts and Tribunal Service. Between 2 December 2013 and 31 December 2014, £40 million was collected by partners on a total of 161 priority cases across law enforcement. Through the work of ACE teams across the UK, law enforcement is tackling unenforced confiscation orders and prioritising the orders of the most serious criminals.

UK Financial Intelligence Unit

4.19 The UK Financial Intelligence Unit (UKFIU) is part of the NCA Economic Crime Command, but is operationally independent of the NCA, meaning that it has the authority and capacity to act autonomously.

4.20 The UKFIU is a law enforcement FIU which receives, analyses and distributes financial intelligence gathered from suspicious activity reports (SARs). UKFIU analyses the SARs to extract strategic and tactical intelligence, and makes all SARs available to law enforcement agencies for investigation (with the exception of SARs in certain sensitive categories). UKFIU

receives the largest number of SARs of any EU member state. In 2013/14 UKFIU received 354,000 SARs of which 14,155 were requests for consent. This is an increase of approximately 38,000 reports on the totals for 2012/13.⁷

4.21 The UKFIU works in close partnership with other key international organisations to fight money laundering and terrorist financing. The UKFIU is a fully active member of the Egmont Group (an international forum for FIUs, set up to improve cooperation in the fight against money laundering and the financing of terrorism). Membership allows the UKFIU to seek and receive financial intelligence from other members in order to support law enforcement operations and projects. It also acts as the conduit to this resource for the wider UK law enforcement community.

Serious Fraud Office (SFO)

4.22 The SFO is an independent government department that investigates and prosecutes serious or complex fraud, and corruption. It has jurisdiction in England, Wales and Northern Ireland but not in Scotland, where the responsibility rests with the Crown Office and Procurator Fiscal Service. Its expert forensic accountants and professional investigators and lawyers investigate and prosecute the most serious or complex instances of fraud and corruption.

4.23 The SFO's Proceeds of Crime Division deals with confiscation investigations, restraint proceedings, money laundering investigations and civil recovery work across the SFO's cases. The Division comprises a multi-disciplinary team of 37 lawyers and financial investigators who work closely with the criminal case teams to ensure that a financial investigation strategy is in place from the outset. The team also deals with incoming requests for mutual legal assistance involving asset freezing and the enforcement of overseas confiscation orders.

4.24 In the period 2014/15 the SFO secured 17 confiscation orders with an equivalent value of £22.7 million and one civil recovery order worth £520. They also recovered £13.8 million in net receipts. HM Revenue and Customs (HMRC)

4.25 HMRC is the UK's tax authority and was established by Act of Parliament in 2005 following the merger of the Inland Revenue and HM Customs & Excise. It is a non-ministerial department reporting to Parliament through its Treasury minister.

4.26 HMRC is responsible for investigating crime involving all of the tax and other regimes it deals with. It uses civil, as well as criminal, procedures as this allows for a greater volume of

cases to be pursued, as well as increasing the revenues secured for the Exchequer. HMRC is also a supervisor for some businesses under the regulations.

4.27 Criminals, including organized criminals, seek to attack the UK's tax and duty systems to steal taxpayer's money. To counter this HMRC has similar criminal investigation powers to those that are available to other law enforcement agencies. In addition to the predicate offences, HMRC can also investigate money laundering offences using POCA investigative powers, recover criminal cash through summary proceedings and recover the proceeds of crime through working with the independent prosecutors.

4.28 In the period 2014/14, HMRC secured £1.7 million in cash forfeitures⁸ and 171 confiscation orders with an equivalent value of £51.2 million. They also recovered £22.4 million in net receipts.⁹

Crown Prosecution Service (CPS)

4.29 The CPS is headed by the Director of Public Prosecutions (DPP), who is in turn superintended by the Attorney General (AG). It is the principal independent prosecuting authority in England and Wales and is responsible for prosecuting money laundering and other criminal cases investigated by the police, HMRC, the NCA and other government agencies. It advises law enforcement on lines of inquiry, reviews cases for possible prosecution; determines the charge in all but minor cases; prepares cases for court; and applies for restraint, receivership and confiscation orders in respect of CPS prosecutions.

4.30 The CPS also obtains restraint orders and enforces overseas confiscation orders on behalf of overseas jurisdictions pursuant to requests for Mutual Legal Assistance (MLA). Recently it has developed and implemented a comprehensive asset recovery strategy in partnership with relevant government departments and other law enforcement agencies. Central to the strategy is the identification and targeting of priority countries where UK efforts can have most impact, and the deployment of dedicated Asset Recovery Advisors (ARAs), funded by the Home Office, to Spain, UAE, and the European and Caribbean regions. These lawyers work with UK and international partners to increase enforcement of existing orders, assist ongoing operations and develop local capability to improve asset recovery and counter illicit finance.

Money service businesses 6.98 The Money Laundering Regulations 2007 ('the regulations') apply to money service businesses (MSBs) who undertake money transmission services, cheque

cashing or currency exchange. 6.99 The money service business (MSB) sector is diverse, with participants ranging from large international corporations who operate worldwide to local corner shops offering remittance services to their community. MSBs offer an important service to those who do not use, for a variety of reasons, the traditional banking sector. 6.100 Two supervisors regulate MSBs. MSBs which are not credit and financial institutions must register and be supervised by HMRC under the regulations.³⁴ The FCA is responsible for supervising money transmission, currency exchange and cheque cashing activities where they are undertaken by a regulated firm, for example a retail bank who offers currency exchange through one of its branches or as a wholesale supplier of currency to other businesses. HMRC is required by the regulations to maintain a register when they are the supervisor and conduct a 'fit and proper' test on those who apply to be registered as an MSB. Approximately 3,000 businesses are currently registered with HMRC as MSBs

Terrorist finance legislation Terrorism Act 2000 11.5 The legal definition of terrorist property is contained in the Terrorism Act 2000 (TACT) section 14. Terrorist property refers to: money or other property which is likely to be used for the purposes of terrorism, proceeds of the commission of acts of terrorism and proceeds of acts carried out for the purposes of terrorism. 11.6 Specific offences under Sections 15-18 of TACT, whether committed in the UK or overseas include:

- inviting, providing, or receiving money or other property with the intention or reasonable suspicion that it will be used for the purposes of terrorism
- using or intending to use money or other property for the purposes of terrorism
- being involved in an arrangement which makes money or other property available for the purposes of terrorism
- being involved in an arrangement which facilitates the retention or control of terrorist property by concealment; removal from the jurisdiction; transfer to nominees, or in any other way

11.8 The UK terrorist asset freezing regime meets obligations placed on the UK by Resolutions of the UN Security Council (UNSCRs) and associated EC regulations. It is implemented by the Terrorist Asset-Freezing Act 2010 (TAFA 2010). 11.9 There is a two part test to exercise the power to designate a person under TAFA found at Section 2 of the Act: in essence, there must be evidence to support a reasonable belief that the person has been involved in terrorist activity (s.2(1)(a) TAFA) and the asset freeze must be considered necessary for purposes connected with protecting members of the public from terrorism (s.2(1)(b) TAFA).

11.10 Designation under TAFE subjects a person to the following restrictions: ? a prohibition on dealing with any funds or economic resources owned, held or controlled by the designated person (in effect the funds or economic resources must be frozen) ? a prohibition on making funds, economic resources or financial services available to (or for the benefit of) the designated person 11.11 Specific offences under TAFE include: ? dealing with funds or economic resources owned, held or controlled by a designated person ? making funds, economic resources or financial services available to or for the benefit of a designated person ? circumventing the restrictions imposed by those restrictions 11.12 As well as the requirement that both parts of the statutory test are met before a designation can be made, various other safeguards are built into the asset freezing regime to ensure that it is operated fairly and proportionately:

11.13 The Treasury may grant licences to allow exceptions to the freeze for certain payments or categories of payments;

- designations expire after a year unless reviewed and renewed
- a right of appeal against designation decisions (for the designated person) and a right to challenge on judicial review grounds any other decision, for example, licensing decisions for the designated person or anyone else affected by the decision to the High Court (using specially cleared advocates to protect closed material where necessary whilst ensuring a fair hearing for the applicant) individuals are notified, as far as possible, of the reasons for their designation
- the operation of the regime is subject to an independent review by the Independent Reviewer of Terrorism Legislation
- a requirement that the Treasury report to Parliament every quarter on the exercise of its powers under TAFE during that period.
- Implementation of UNSCRs 'without delay' 11.14 FATF sets international standards, in the form of recommendations, for combatting money laundering and terrorist financing. Recommendation 6 requires freezing 'without delay' of the assets of individuals or entities designated under UNSCRs 1267 and 1373, whilst Recommendation 7 requires freezing 'without delay' of the assets of those listed by the UN under the non-proliferation regimes. The purpose of implementing a freeze without delay is to avoid asset flight in the period between identification of an individual or entity and the freeze being imposed. 11.15 UNSCR 1373 requires states to freeze the assets of terrorists and prohibit their nationals and persons within their jurisdiction

from making funds, resources or financial services available to them. It is implemented in the UK by the TFA 2010 and EU Common Position 931 and Regulation 2580/2001. The assets of individuals designated under UNSCR 1373 in the UK are frozen without delay. 11.16 UNSCR 1267 created a regime that targets individuals and entities associated with Al Qaida requiring states to freeze the assets of persons designated under that regime. The UK implements UN asset freezes by way of EU Regulation which takes direct effect in the UK. It takes three to four weeks on average for the EU to implement UN listings resulting in a delay between the adoption of designations at the UN and their implementation and a possible risk of asset flight. 11.17 The UK continues to actively raise the risks presented by the delay with the European Commission, and is also currently considering legislative options to address the delay between UN terrorism sanctions being imposed and their implementation by the EU. 11.18 On 31 December 2014, £117,000 was frozen across 80 accounts of those designated under the UK's domestic TFA regime, the UN AQ regime and the EU CP931 regime. Al Qaida (Asset Freezing) Regulations 2011 11.19 The Al Qaida (Asset Freezing) Regulations 2011 impose the criminal penalties for breaching the UN Al Qaida asset freezing regime that is given effect by EC Regulation 881/2002. This may include circumventing or assisting someone to circumvent their asset freeze, or providing false information for the purpose of obtaining a licence from HM Treasury. Charities/non-profit organisations (NPOs) 11.33 Although proven terrorist abuse of the charitable sector is rare, it is assessed that the terrorist financing risks within the charitable sector are medium-high. Terrorists and charities operate in conflict areas and therefore, determining the end destination of funds can be difficult. 11.34 Key threats within the charitable sector are: ? raising funds from the public under the guise of a charity with or without a charity licence or authority ? trustee abuse of charities to divert funds for terrorist financing ? looting of charity assets by terrorists in high-risk jurisdictions ? diversion of charitable goods at the destination by terrorists ? charities being subject to local extortion by terrorists in de facto control 11.35 Two high profile convictions have highlighted that individuals can use the name of a charity to raise funds for terrorist purposes: 11.36 Case 1: on 2 August 2012, Mohammed and Shafiq Ali were convicted of raising money to fund terrorism and were sentenced to three years imprisonment. The brothers carried out street collections supposedly for charity. They pleaded guilty to raising £3,000 which they sent to a family member in Somalia for terrorist training and fighting. 11.37 Case 2: on 21 February 2013, Ashik Ali, Ifran Khalid and

Ifraan Naseer were convicted of committing acts in preparation for terrorist attacks, included collecting money for terrorism. They raised funds by fraudulently presenting themselves as charity fundraisers using high visibility vests and collections buckets bearing the name of the charity Muslim Aid. Of the approximately £14,000 raised, only £1,500 reached the charity. A genuine Muslim Aid volunteer, Rahin Ahmed, was found guilty of assisting with the plot.

- Unregulated sectors Cash couriership
- Digital currencies
- Mobile payment systems
- Full list of AML/CFT supervisors
- A.1 This is a full list of bodies currently designated as AML/CFT supervisors under the Money Laundering Regulations 2007 (as amended): Association of Accounting Technicians (AAT) Association of Chartered Certified Accountants (ACCA) Association of International Accountants (AIA) Association of Taxation Technicians (ATT) Chartered Institute of Management Accountants (CIMA) Chartered Institute of Legal Executives (CILEX) Chartered Institute of Taxation (CIOT) Council for Licensed Conveyancers (CLC) Department of Enterprise, Trade, and Investment Northern Ireland (DETNI) Faculty of Advocates (Scottish Bar Association) (FoA) Faculty Office of the Archbishop of Canterbury (AoC) Financial Conduct Authority (FCA) Gambling Commission (GC) General Council of the Bar (England and Wales) (GCBEW) General Council of the Bar of Northern Ireland (GCBNI) HM Revenue & Customs (HMRC) Insolvency Practitioners Association (IPA) Insolvency Service (SoS) Institute of Certified Bookkeepers (ICB) Institute of Chartered Accountants in England and Wales (ICAEW) Institute of Chartered Accountants in Ireland (ICAI) Institute of Chartered Accountants of Scotland (ICAS) Institute of Financial Accountants (IFA) International Association of Book-keepers (IAB) Law Society of England and Wales (LSEW) Law Society of Northern Ireland (LSNI) Law Society of Scotland (LSS)

Summary

Hawala: Hawala is an alternative or parallel remittance system. It exists and operates outside of, or parallel to 'traditional' banking or financial channels.

Third party cheques: Utilizing counter cheques or banker's drafts drawn on different institutions and clearing them via various third-party accounts. Third party cheques and traveler's cheques are often purchased using proceeds of crime.

Credit Cards – Clearing credit and charge card balances at the counters of different banks. Such cards have a number of uses and can be used across international borders.

Money Laundering serves as an important mode of terrorism financing.

Terrorists have shown adaptability and opportunism in meeting their funding requirements.

International efforts to combat money laundering: International regimes attempt to monitor or regulate international relations and activities. By establishing norms, rules, and procedures agreed to in order to regulate money laundering, international regime reduce the disparities between divergent legal systems.

This term (usually abbreviated to MLRO) is used to describe the nominated officer appointed under Regulation 20(2)(d) of the 2007 Regulations and as referred to in Section 331 Proceeds of Crime Act 2002(POCA).

An MLRO should be of sufficient seniority to make decisions on reporting which can impact a business's relations with its clients and its exposure to criminal, civil, regulatory and disciplinary sanctions. They should also be in a position of sufficient responsibility to enable them to have access to all of the business's client files and business information to enable them to make the required decisions on the basis of all information held by the business.

An MLRO is responsible for ensuring that, when appropriate, the information or other matter leading to knowledge or suspicion, or reasonable grounds for knowledge or suspicion of money laundering is properly disclosed to the relevant authority. The decision to report, or not to report, must not be subject to the consent of anyone else. The MLRO will also liaise with SOCA or law enforcement on the issue of whether to proceed with a transaction or what information may be disclosed to clients or third parties.

Reserve Bank of India has specified Know Your Customers (KYC) standards to be followed by banks and measures to be taken in regard to Anti Money Laundering (AML) and Combating of Financing of Terrorism (CFT).

Risk Profiling of Customers: Each and every account of the Bank is classified as “Low”, “Medium” or “High” Risk accounts for the purpose of monitoring and reporting of suspect transactions.

Placement of money is called ‘smurfing’ when the money is deposited, in convenient amounts, in a number of banks and their branches. The deposits may be in the banks of the country of origin, or in some foreign banks.

In smurfing the tainted money is deposited in small amounts. The size of the amounts is determined by the government rules. Usually, it is the highest amount for which the bank is not required to notify the government agencies. The deposits are repeated a number of time, in a number of branches of a number of banks, in a number of towns or cities. The actual number of deposits, in smurfing, depends mainly upon the amount of ill gotten money involved, the notoriety of the collection of the illicit money and the government rules prevailing at the time.

Electronic Fund Transfers (EFT) shifts the money across the borders in no time and makes the black money anonymous. Any amount can be transferred to any place in the world, where EFT facility exists. The transfer does not take practically any time. It leaves no traces of the transfer. This money transfer is intermingled with thousands of legitimate transfers of which it forms only a fraction. It is difficult to identify the shady transfer. The tainted money becomes anonymous instantly. The wire transfer black money to foreign lands, in the name of a company or agency or even in the name of an individual. The mode layers any amount.

The Basel Committee on Banking Supervision provides a forum for regular cooperation on banking supervisory matters. Its objective is to enhance understanding of key supervisory issues and improve the quality of banking supervision worldwide.

The IMF was established in 1945 as a specialized agency of the United Nations system within the framework of the Bretton Woods Accords.

United Nations Office on drugs and Crime (UNODC)

Set up by the United Nations in 1997 to implement the United Nations International Drug Control Programme (UNDCP) and the Crime Prevention and Criminal Justice Programme (CPCJP), both of which were established by resolutions of the UN General Assembly in 1991.

The Commonwealth Secretariat provides guidance on policy making, technical assistance and advisory services to Commonwealth member countries. It supports governments to help achieve sustainable, inclusive and equitable development.

The European Commission and Council

The role of the EU is important in the AML arena and provides an excellent example of the manner in which national governments have been required to implement legislation to forestall AML legislation in line with the expectations of a transnational organization.

The Organisation for Economic Co-operation and Development (OECD) was officially born on 30 September 1961.

- Other countries joined in, starting with Japan in 1964. Today, 34 OECD member countries worldwide regularly turn to one another to identify problems, discuss and analyse them, and promote policies to solve them. The track record is striking.

The Financial Action Task Force on Money Laundering (FATF) was established by the G-7 Summit that was held in Paris in 1989.

The UK Financial Intelligence Unit (UKFIU) is part of the NCA Economic Crime Command, but is operationally independent of the NCA, meaning that it has the authority and capacity to act autonomously.

The SFO is an independent government department that investigates and prosecutes serious or complex fraud, and corruption.

The CPS is headed by the Director of Public Prosecutions (DPP), who is in turn superintended by the Attorney General (AG). It is the principal independent prosecuting authority in England and Wales and is responsible for prosecuting money laundering and other criminal cases investigated by the police, HMRC, the NCA and other government agencies.

Check your Answers

Explain the following

- a) Money Laundering
- b) MLRO
- c) Security Controls
- d) ML process
- e) KYC

Lesson – 7

Frauds in Insurance Sectors

Objectives

After reading this lesson you will be able to understand the following

- Fraud
- Methods of computer related fraud
- Scope of computer related fraud
- Classification of Insurance Fraud
- Maritime Fraud
- Frauds in Health Insurance
- Frauds in Fire and Allied Perils insurance
- Frauds in Finance and Account Departments

Structure

- 7.1 Fraud**
- 7.2 Methods of computer related fraud**
- 7.3 Scope of computer related fraud**
- 7.4 Classification of Insurance Fraud**
- 7.5 Maritime fraud**
- 7.6 Frauds in Health Insurance**
- 7.7 Frauds in Fire and Allied perils insurance**
- 7.8 Frauds in Finance and Account Departments**

Insurance fraud is any act committed to defraud an insurance process. This occurs when a claimant attempts to obtain some benefit or advantage they are not entitled to, or when an insurer knowingly denies some benefit that is due. Perpetrators in these schemes can be insurance company employees or claimants. **False insurance claims** are insurance claims filed with the fraudulent intention towards an insurance provider.

Insurance fraud has existed since the beginning of insurance as a commercial enterprise. Fraudulent claims account for a significant portion of all claims received by insurers, and cost billions of dollars annually. Types of insurance fraud are diverse, and occur in all areas of insurance. Insurance crimes also range in severity, from slightly exaggerating claims to deliberately causing accidents or damage. Fraudulent activities affect the lives of innocent people, both directly through accidental or intentional injury or damage, and indirectly as these crimes lead to higher insurance premiums. Insurance fraud poses a significant problem, and governments and other organizations try to deter such activity.

Types of Insurance Fraud

Life Insurance Fraud

Life insurance fraud may involve faking death to claim life insurance. Fraudsters may sometimes turn up a few years after disappearing, claiming a loss of memory.

Health Insurance Fraud

Health insurance fraud is described as an intentional act of deceiving, concealing, or misrepresenting information that results in health care benefits being paid to an individual or group. Fraud can be committed either by an insured person or by a provider. Member fraud consists of claims on behalf of ineligible members and/or dependents, alterations on enrollment forms, concealing pre-existing conditions, failure to report other coverage, prescription drug fraud, and failure to disclose claims that were a result of a work-related injury.

Automobile Insurance Fraud

Fraud rings or groups may fake traffic deaths or stage collisions to make false insurance or exaggerated claims and collect insurance money. The ring may involve insurance claims adjusters and other people who create phony police reports to process claims.

Property Insurance Fraud

Possible motivations for this can include obtaining payment that is worth more than the value of the property destroyed, or to destroy and subsequently receive payment for goods that could not otherwise be sold.

7.1 Fraud

Fraud means action or lack of action to gain benefit through intentional deception, cheating and stealing. Fraud occurs under two major categories. They include:

- Transaction Frauds
 - o This happens through fraudulent claim or conversion of an underwriting of company's assets to ones personal use such as assets misappropriation.
- Statement Frauds
 - o This involves intentional misstatement of some financial values to affect the appearance of books, profits/result or value of asset/ liability to deceive shareholders, lenders etc.
- White collar crime
 - o When a person, employed by an undertaking in a position of trust, breaches such trust, beyond delegated powers with intention to cheat it is known as white collar crime.

The figure 7.1. given below represents various factors causing the frauds.

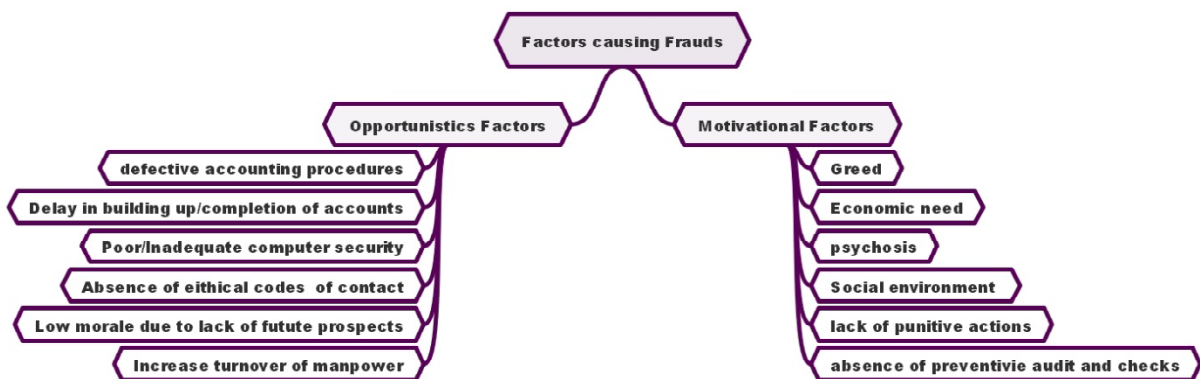


Figure 7.1: Factors causing Frauds

The reasons why frauds occur are varied. They are illustrated in figure 7.2.

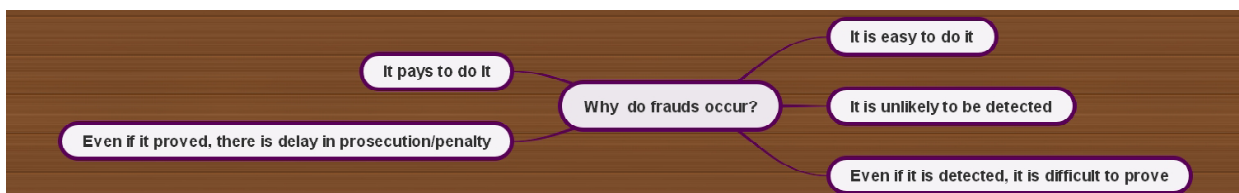


Figure 7.2: Why do Frauds Occur?

7.2 Methods of computer related frauds

- Manipulation of computer inner working by modification of programs
- Offline changing in the back office of computer output
- LAN system operated as stand alone
- Theft of business secrets from computer files
- Alteration of data already entered in the computers
- Systems have fudge facility that allows errors to be corrected
- Entering of false data into computers which would give false outputs
- Data espionage through remote access
- Theft of intellectual property rights stored in computer systems

7.3 Scope of computer related Frauds

- Variety of software/frequent upgradations
- Lack of checks on veracity of programs
- Absence of technical knowledge/understanding on computer system working
- Non-use of analysis, statistics, ratio compatibility provided by the software
- Non-compliance of systems instructions
- Non-monitoring of audit trails
- Exploitation of data stored in LAN

7.4 Steps for Reduction/elimination of cyber frauds

- Provision of security and passwords at all levels and strictly following them
- authority for changing, correcting master files, other data to be at senior levels all changes to be recorded and monitored
- Close monitoring of audit trails at senior levels
- Careful recoding and monitoring of source codes
- Provision of better computer security through antivirus software, firewalls, ID's, log files etc.

- Online automatic transfer of balances
- Acquiring special skills in computer-aided audit tools
- Automatic numbering of each input data, statements
- Establishment of totally online integrated system with centralized data centers, disaster recovery sites, servers etc.

7.5 Classification of Insurance Frauds

The following figure illustrates the classification of non life Insurance.

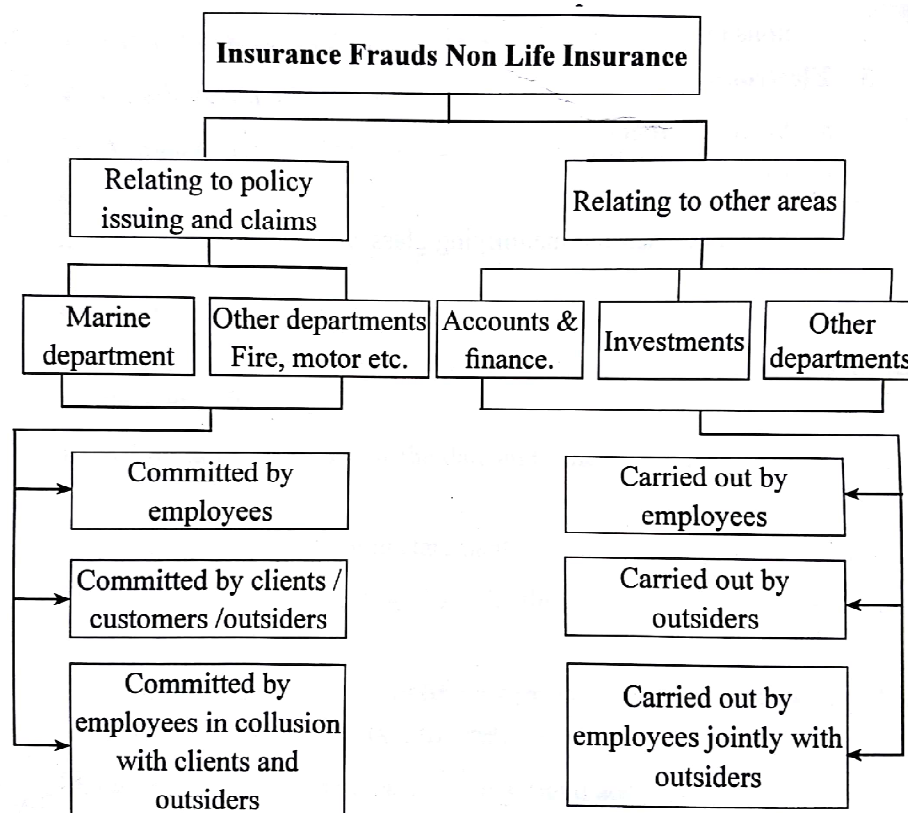


Figure 7.3: Insurance Frauds Non Life Insurance

7.6 Maritime Frauds

Maritime frauds may be classified as into two types. This is illustrated in the figure.#

- Maritime cargo insurance
 - o Cargo
 - o Goods
- Maritime Hull insurance
 - o Hull
 - o Vessel

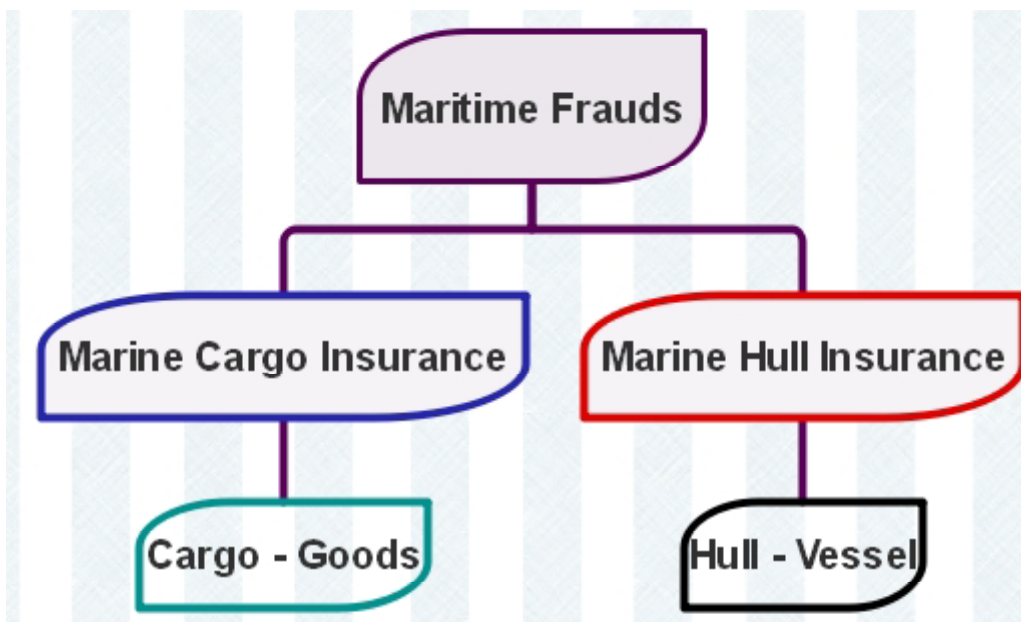


Figure 7.4: Maritime Fraud Classification

7.6.1. Marine Cargo Insurance

- Transit of goods by rail, road, air, water
- Deliberate destruction/damage to the insured cargo
- Stealing the entire cargo with the carrier resulting in non-delivery
- International theft, burglary of part of cargo resulting in short delivery
- Documentary frauds by insuring non-existing cargo

- Total loss of cargo resulting from the disappearance of vessels of carriage following their arrest, fraudulent sale
- Cost of buying back or retrieving cargo which where sold to third parties at a port other than destinations

7.6.2. Marine Hull Insurance

- Insurance of ships, streamers, boats, fishing and sailing vessels
- Scuttling of ships or deliberate sinking of ships, boats etc.
- Insurance of non-existent ships, boats by using forged/fake documents
- Frauds relating to chattering of ship known as charter party fraud by the disappearance after collecting in advance pre-paid freight

7.6.3. Prevention of Marine Frauds

The following are the prevention steps

- Establishment of the IMB in January 1981 by the ICC
- Setting up of The Far East regional investigation team (FERIT)
- Directory of approved ships, vessels containing full details is periodically updated
- Conducting of pre-transit, pre-shipment physical inspection of vessels, cargo by technically qualified authorized persons
- Buyers and sellers should take extra precautions in respect of cargo carried in chartered vessels by verifying the credentials of owners and charterers
- Maritime and port authorities should prevent voyage deviations, terminations unless the ship owners deposit adequate money for onward carriage together with handling, insurance and other expenses.

7.7 Motor Insurance Frauds

- Insurance policies, cover notes issued after happening of accidents, losses sequential cover notes are not received and antedating is done.
- Issuance of insurance policies/cover notes for non-existent (fictitious) vehicles with an intention to lodge claims on them. At the time of issue of policies higher I.D.V. is quoted which has to be settled in the event of total loss claims.

- For fleet of vehicles insurance is taken only for few vehicles.
- Number plates of uninsured vehicles substituted with register number of insured vehicles

7.7.1. Preventive Steps

- Maintenance of control registers for supplies and issues of pre-numbered motor insurance cover notes.
- Motor cover notes issuing authority to be with high ranking officers restricting issue of cover notes only to selected persons.
- In case of break of insurance for renewals, pre-insurance insurance along with engine and chassis number of the vehicle must be obtained and enclosed.
- Fully completed proposal forms duly signed by the insured should be obtained for new business and original RC to be verified in case of doubts
- IDV should commensurate with the age of vehicle.

7.7.2 Frauds in Motor on Damage Claims

There are two types of frauds in OD claims. Fraud at the time of policy issuance.

- Fake previous policy if lost before present period
- Fake NCB Certificate to get discount on premium

a) Misrepresentation of facts

- Driver implant – If the actual driver has invalid license
- Manipulation of documents overwriting on documents/fake documents
- Non-disclosure of correct facts:-
 - o Date of Loss
 - o Place of Loss
 - o Concealment of facts
 - o Fake signatures of insured in claim form if claimant is not insured.

b) Fraud in Third party claims

- Third party claims are claimed through MACT
- Third party claims can be classified under the following categories
- Substitution of vehicle
 - o Where actual vehicle is uninsured
 - o Where actual vehicle escapes from the spot of accident
- Substitution of driver:
 - o It happens when the actual driver has no DL or has an invalid DL or disqualified to hold a valid DL
 - o Conversion of non-RTA cases in RTA cases
 - o Addition of names of persons not affected by accidents either at the time of filing of FIR or while finalizing charge sheet
 - o Inflated or bogus medical bills and bogus medical certificates, death certificates etc.

c) Identification of motor third party frauds

- Date of accident is different from the date of admission in hospital
- Medical bills or admissions papers show non RTA history of the injury
- FIR against unknown vehicle and unreasonable delay in filing FIR
- Hit and run cases where police finally charge sheets some vehicle
- Contradictory statements of the police witnesses
- Postmortem report states the cause of death to be natural

d) Prevention and Detection of Motor Frauds

- Implementation of checker mechanism which checks the fake previous policy and NCB certificates from other insurance companies.
- Change of driver can be found by verification of the traffic accident report /FIR/ police report and medical details of the accident at the time of loss. This is an incident fraud hence this specific claim is repudiated.

- Manipulation of documents can be identified by verifying the documents from the respective authorities. Claim cannot be honored.

7.8 Frauds in Health Insurance – MEDICLAIM

- Fake claims which have not undergone OPD converted into IPD i.e. outpatient consultancy shown as in-house admission for a minimum of 24 hours to fit in norms of insurance
- exaggerated or inflated claims by forging and falsification of medical bills/hospital invoices/doctor's certificates and surgical reports
- Claiming payments for persons not covered by original policies by including them through additions made subsequent to illness/hospitalization by fraudulently including them with effect from prior period

7.8.1 Preventive and Detectives steps

- Introduction of cashless settlement directly with hospitals and nursing homes instead of reimbursing insured.
- Insistence of treatment only at leading registered hospitals / nursing homes where scope of fraud is minimum.
- Health ID cards with photographs and/or thumb impression of the insured using biometric technology is to be issued to arrest impersonation.
- By arranging investigations in close proximity claims and other suspicious and doubtful cases.
- Personal visits to hospitals and nursing homes for first hand verifications on real time basis when the patients are in hospital.
- Insisting and obtaining official Post-mortem reports in accidental deaths which will throw light on the exact cause of death.

7.9 Frauds in Fire and Allied Perils Insurance

- Bogus and inflated loss of stocks due to fire and floods.
- Loss of stocks in uninsured premises made up as having occurred in inside premises.
- Insurance taken after fire / flood damage but claimed under forged pre-dated cover notes.

- Loss of stocks along with books and records but claims processed without looking into backing trails etc.

7.9.1. Preventive and detective steps

- Arranging spot surveys insistence on independent fire brigade report , detailed investigation by qualified CAs, engineers, surveyors.
- Through private enquiries of those living around insured premises.
- Stocks position at time of loss to be worked out independently with the help of buyers and sellers and the insured's bankers qualified CA help to be sought in doing inspections.
- Salvage left after fire must be tested in labs to find out if fire stimulants were added to aid and cause fire.

7.10 Frauds in Finance and Account Departments

7.10.1. Methods of Frauds include:

- Not accounting full /part premium and other collections by mentioning different reduced amounts in copies of collection receipts.
- Non/short remittance or delayed remittance of daily cash collection into bank accounts.
- Deliberately making double payments for the same expenditure.
- Falsification of final accounts in order to enhance the results/financial position.
- Alterations in the payee's names/amount in words /figures after issue of checks before handing over to them and depositing them in a new bank account fraudulently opened.
- Assets sent for repairs never came back.
- Old assets undervalued deliberately and sold to interested persons for low values.

7.10.2. Preventive and Detective steps

- Introduction of real-time online integrated accounting system where alterations and corrections would be possible only at higher level office with approval from senior officers and executives. Double side carbons to be used for issuing receipts.

- Allowing only authorized persons to operate the computers during and after office hours under the direct supervision of seniors.
- Check to contain account number/bank branch name of the payees.
- Cello tape to be affixed on names of payees and on amounts in words and figures on the checks issued .
- Unused check books and receipt books to be kept under lock in safe with the responsible officials who should physically verify those at regular intervals.
- Introduction of quality audits and checks in purchase/stores department .
- Person handling treasury/cash should not handle their accounting.
- Despite all controls, identification and elimination of such leakages continues to be a challenge, it is often difficult to prove the fraud and even difficult to adduce the evidence
- Placing internal control system and taking preventive steps are only the journey and not the destination . They need to be constantly reviewed and updated at intervals within house evaluation system in action
- Computer based cyber crimes/ frauds are on the increase with the advent of computerized operating and accounting systems .
- Hence persons working in the operational and administrative areas as well as the different types of auditors/investigators must have a thorough knowledge about various control and security aspects of the computer and special computer aided audit and investigation techniques and tools.
- Like many developed countries specific laws have to be passed for tackling insurance frauds in the context of the Indian markets.
- This would give the required leverage for Indian insurance companies to combat and discourage incidents of these frauds.

7.11 Fraud Investigation

The steps involved in insurance fraud investigation are illustrated in the figure below:

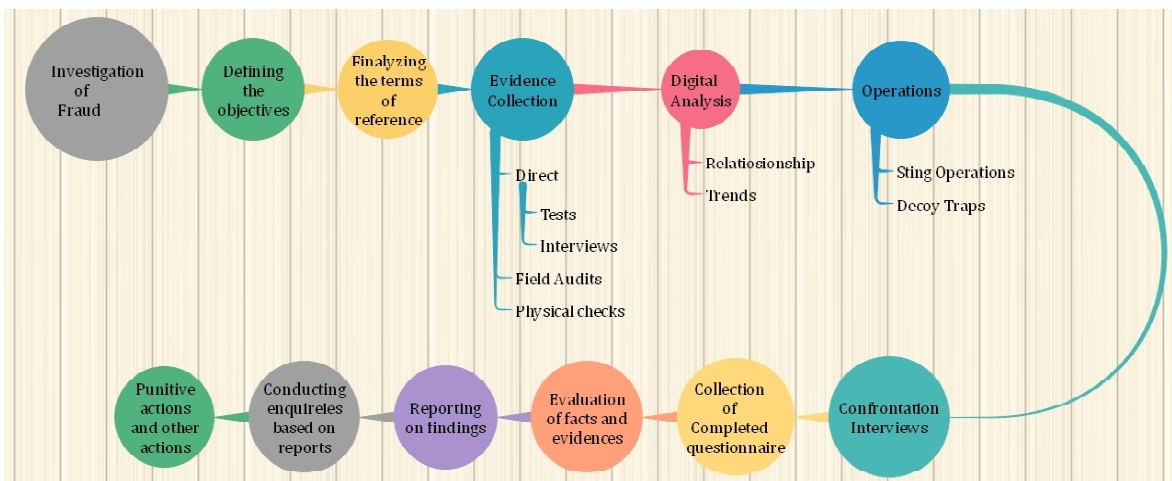


Figure 7.5: Steps involved in Fraud Investigation

Summary

Fraud means action or lack of action to gain benefit through intentional deception, cheating and stealing. Fraud occurs under two major categories. They include:

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- Maritime Hull insurance
 - o Hull
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Marine Hull Insurance

- Insurance of ships, streamers, boats, fishing and sailing vessels
- Scuttling of ships or deliberate sinking of ships, boats etc.
- Insurance of non-existent ships, boats by using forged/fake documents
- Frauds relating to chartering of ship known as charter party fraud by the disappearance after collecting in advance pre-paid freight

Motor Insurance Frauds

- Insurance policies, cover notes issued after happening of accidents, losses sequential cover notes are not received and antedating is done.
- Issuance of insurance policies/cover notes for non-existent (fictitious) vehicles with an intention to lodge claims on them. At the time of issue of policies higher I.D.V. is quoted which has to be settled in the event of total loss claims.

Frauds in Health Insurance – MEDICLAIM

- Fake claims which have not undergone OPD converted into IPD i.e. outpatient consultancy shown as in-house admission for a minimum of 24 hours to fit in norms of insurance

Frauds in Fire and Allied Perils Insurance

- Bogus and inflated loss of stocks due to fire and floods.

Check your Answers

1. What is Insurance Fraud ?
2. What is the scope of Computer related fraud ?
3. Explain classification of Insurance Fraud.

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MODEL QUESTION PAPER
M.SC CYBER FORENSICS AND INFORMATION SECURITY
FIRST YEAR- SECOND SEMESTER
ELECTIVE PAPER-II
BFSI Frauds and Counter Measures

Time:3 hours

Maximum : 80

Section-A

Answer the following in 50 words in each

(10 x 2 = 20)

1. Define Banking.
2. What is Demand Deposit?
3. What is Corporate Salary Account?
4. What is Universal Payment Interface (UPI)?
5. How is Capital Adequacy Ratio (CAR) calculated?
6. What are Cash Transaction Report (CTR) and Suspicious Transaction Report (STR)?
7. Core Banking System (CBS) - Explain
8. What is IDEA?
9. What is interface testing?
10. What is Phishing?

Section-B

Answer any five of the following in 250 words in each

(5 x 6 = 30)

1. What is Term Deposit and explain some of the schemes prevalent in the banks.
2. What is the procedure for opening NRI account and explain its salient features.
3. What are the types of advances given by the Banks? Explain them with examples.
Explain some of the cheque related frauds in banks.

4. What are the main functions of RBI?
5. How are the remittances effected through E-Wallet, NEFT and RTGS.
6. What are the main categories of risks faced by the banks?

SECTION – C

Answer the following in about 500 words each

(3 x 10 = 30)

1. Explain the three mutually reinforcing pillars under Basel II. Explain the approaches suggested by RBI to cover Credit Risk and Operation Risk.
2. What is Money Laundering and explain its stages with examples.
3. Discuss some of the frauds perpetrated in the area of lending by Banks and the precautions to be taken to prevent them.